Semiconductor Manufacturing Technology

Deleep R. Nair
Department of Electrical Engg, IIT Madras
https://home.iitm.ac.in/deleep

What is a semiconductor?

www.youtube.com › watch

[The Semiconductor Shortage: Semiconductor Shortage and ...]

Semiconductor Shortage and It's Impact: There's been a lot of talk lately about the **semiconductor shortage** and how it's affecting the ...

12:12

YouTube · TechEdKirsch · 14 Mar 2022

www.youtube.com › watch

Why Is There a Global Semiconductor Shortage?

Why Is There a Global **Semiconductor Shortage**? 8.7K views · ...

12:12

YouTube · SupplyChainBrain · 14 Apr 2021

www.youtube.com › watch

A Brief Explainer of the Great Semiconductor Shortage of ...

So you survived The Great **Semiconductor Shortage** of 2020 but how did it all start and is the shortage actually over?

This Is What It Will Take to Fix the Semiconductor Shortage

... about the problems. This Is What It Will Take to Fix the **Semiconductor Shortage**. 4.5K views · 3 years ago ...more. Bloomberg Television. 3M.

1:11

YouTube · Bloomberg Television · 24 Feb 2022

www.youtube.com › watch

Semiconductor Shortage

Paul Jacobs, XCom Chairman & CEO and former Qualcomm CEO, goes into detail on the worldwide **semiconductor shortage**, how it's invaded our ...

8:18

YouTube · Bloomberg Television · 22 Jan 2022

www.youtube.com › watch

Explained: Semiconductor Shortage & Its Impact

cnbctv18explains **#semiconductor #shortage** #automobile #manufacturing #cars #twowheelers #computers #chips #microchips #laptops #desktops ...

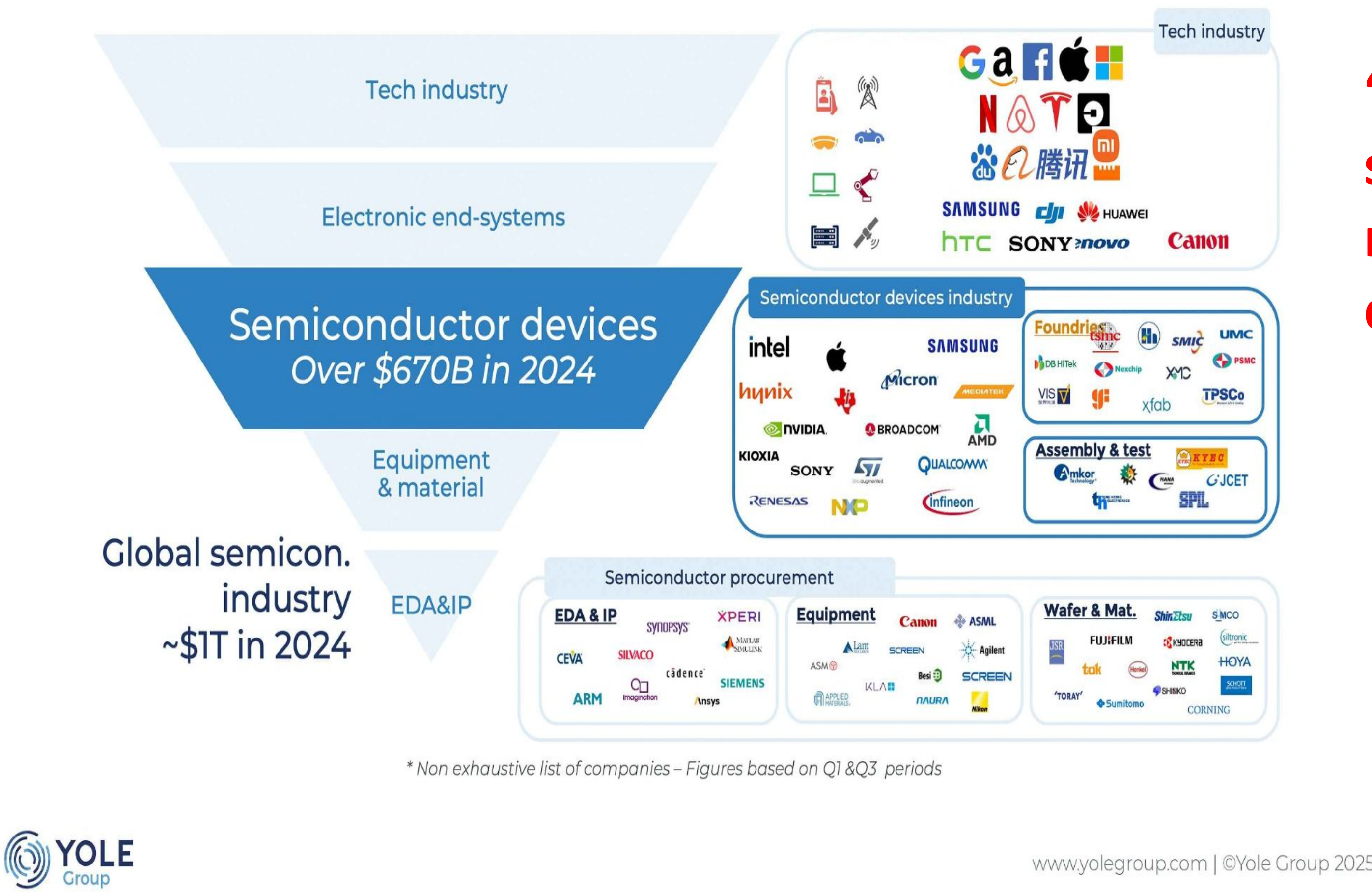
4:02

YouTube · CNBC-TV18 · 7 Sept 2021

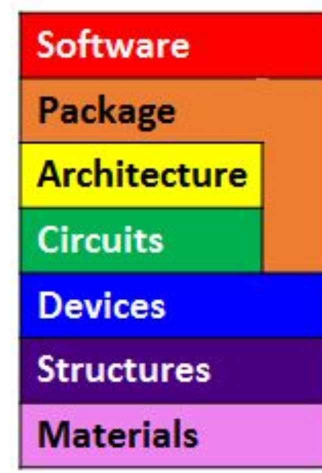
The semiconductor full stack (2024)

Semiconductors- Sophisticated products made by a collective optimization effort from software to materials and manufactured through a complex process in a highly specialized manufacturing facility

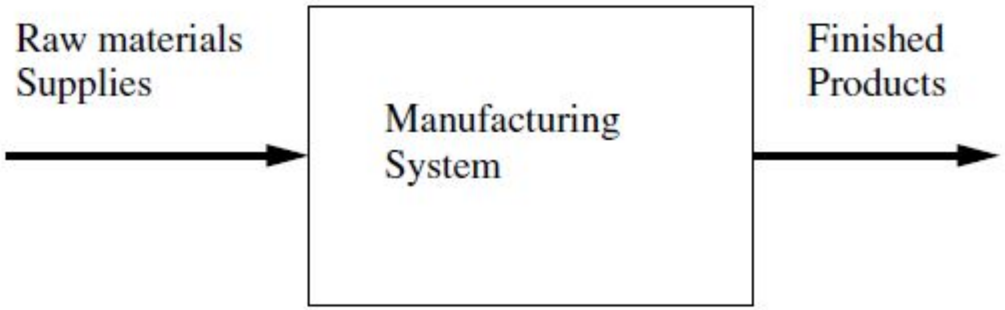
eg: Processors, DRAM, Sensors, Flash memory, Discrete power devices. ~~TFT & PV~~



"It's not rocket science—it's much more difficult"



Semiconductor production process



- Computer aided design of integrated circuits (IC)
- Front-end fabrication, in which "fabs" create ICs on silicon wafers
- Back-end testing, assembly, and packaging, in which wafers are sliced into individual ICs, encased in plastic, and put through a quality-control process

The end result is a product that meets all specified performance, quality, cost, reliability, and environmental requirements.

Semiconductor companies

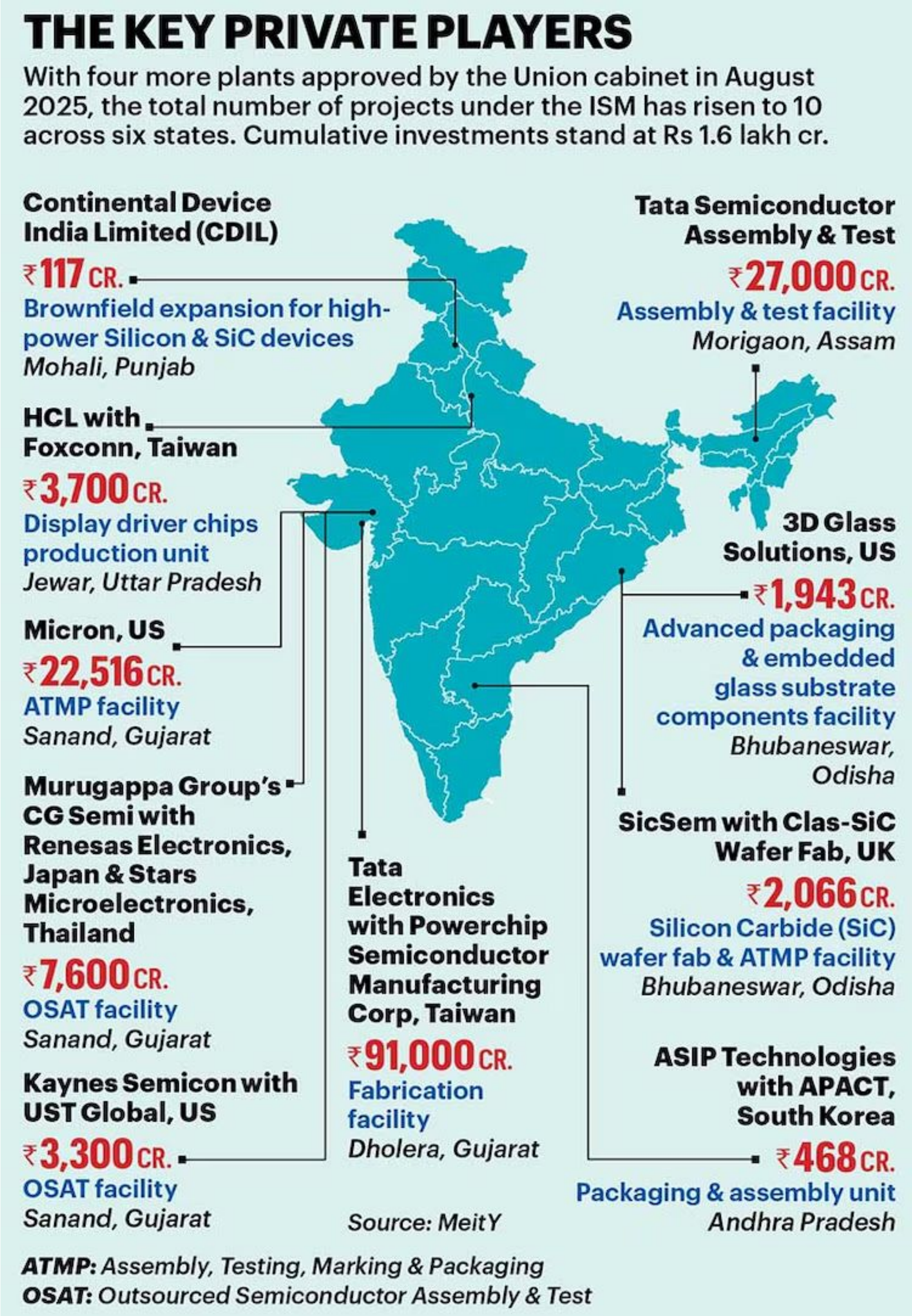
- Integrated Device Manufacturers (IDM) such as chip makers from Intel and Samsung* that design, manufacture and sell their chips/systems

- Fabless manufacturers such as Qualcomm and Nvidia that design and sell chips but outsource manufacturing to foundry companies

- Foundry companies such as TSMC*, GF and UMC that manufacture chips designed and sold by their fabless customers. TSMC also does advanced packaging (CoWoS, InFO, SoIC)

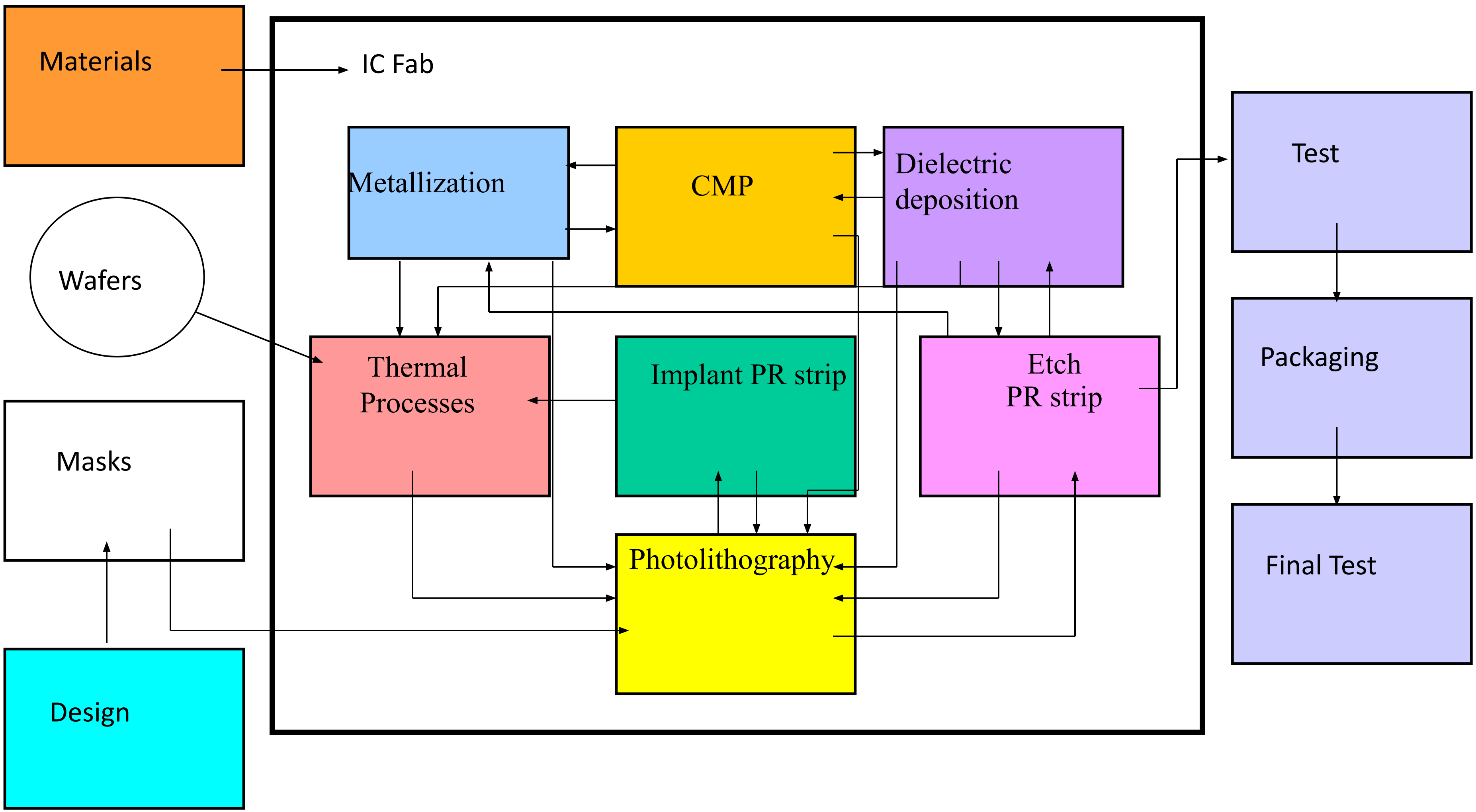
- Outsourced Semiconductor Assembly & Testing (OSAT) – Testing and Packaging of dies so that can be installed in electronic devices/systems

- Semiconductor equipment manufacturers like Applied Materials, ASML, Lam research, TEL etc which supply eqpt to IDMs and foundries



What is planned in India? (production may start in 2027)

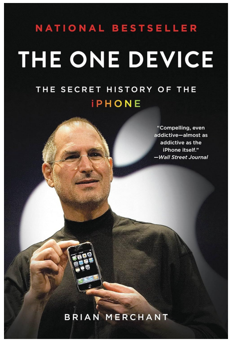
Wafer Process Flow

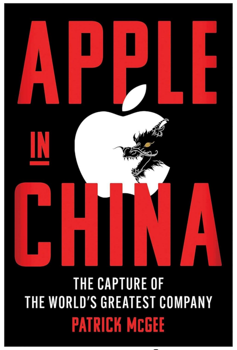


Importance of Semi Manufacturing technology -1

- To get a feel for the physical nature of the devices. Understanding of fabrication is needed to know why the device looks the way it does

- The way a device is fabricated affects key device parameters.





- Product design - Features, performance

  Manufacturing - Cost, quality and reliability

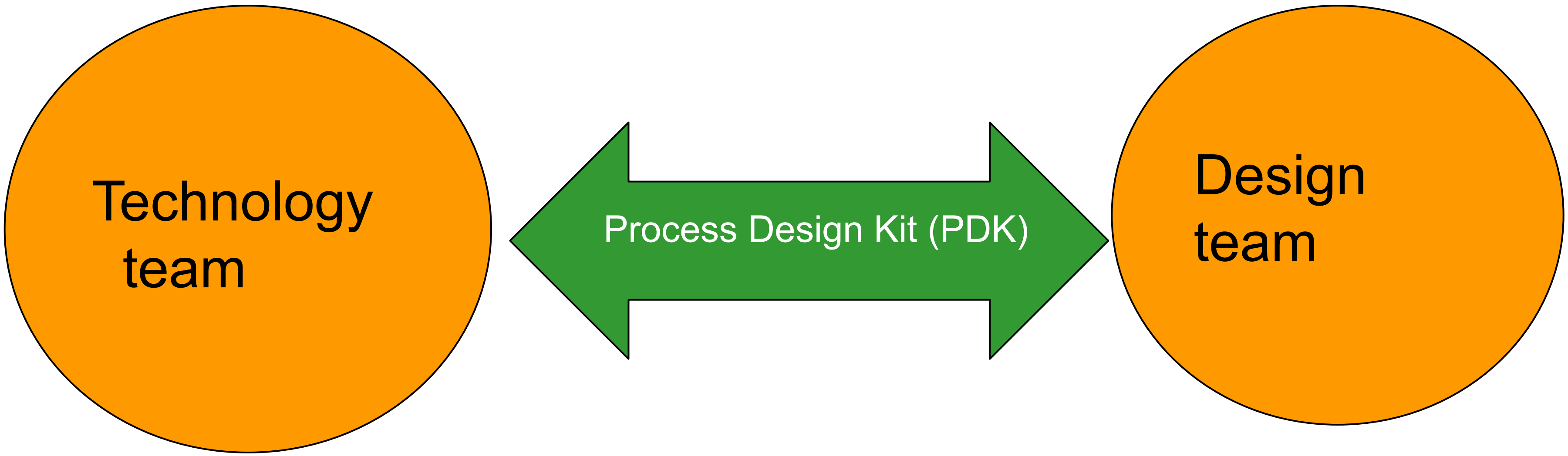
- This also affects what kind of approximations one can take in device analysis

- How to improve the performance or reduce the size of the device

- Also gives a reality check on why some ideas although logically and scientifically good doesn't see the light in engineering

- **Importance of access to EDA tools and manufacturing (eg: Huawei)**

Importance of Semi Manufacturing technology -2

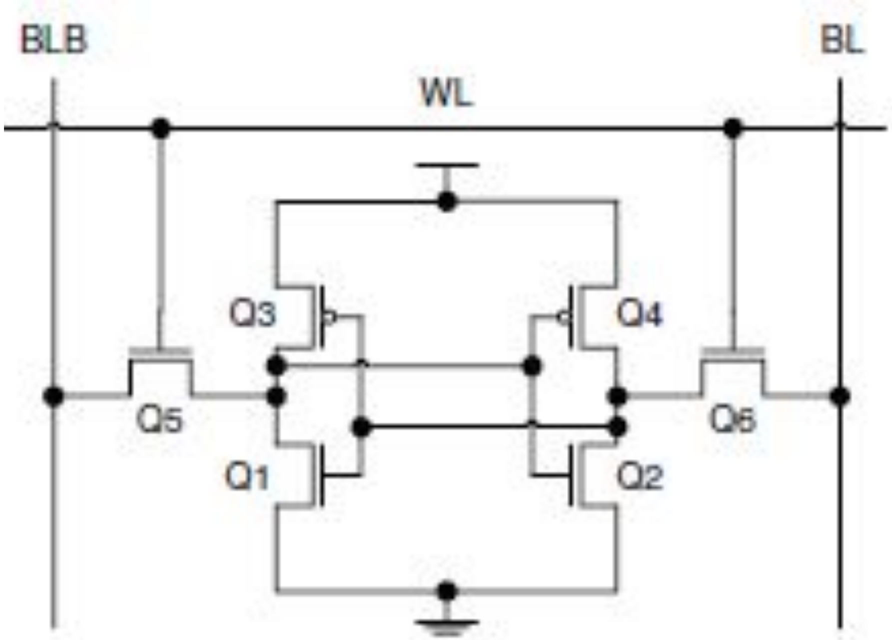


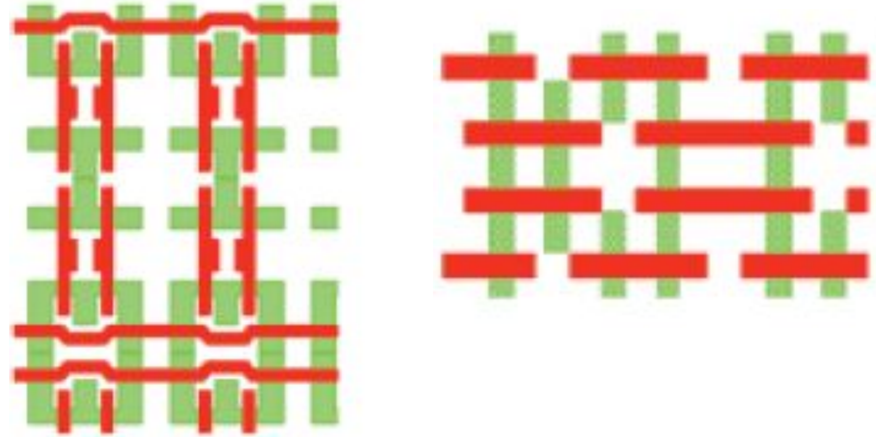
*The days when chip designers could throw tape "over the wall" to the manufacturing side are long gone**

Increasing design restrictions as DRC decks becoming complex and designs which works well in simulator aren't yielding or failing in reliability.

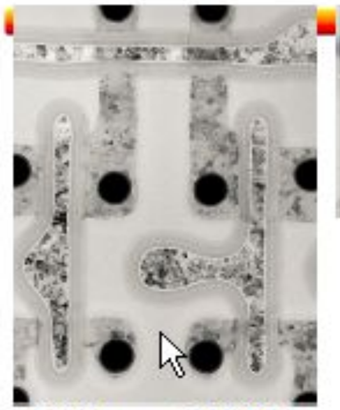
*http://semiengineering.com/litho-challenges-break-the-design-process-wall/

Evolution of 6T SRAM layout

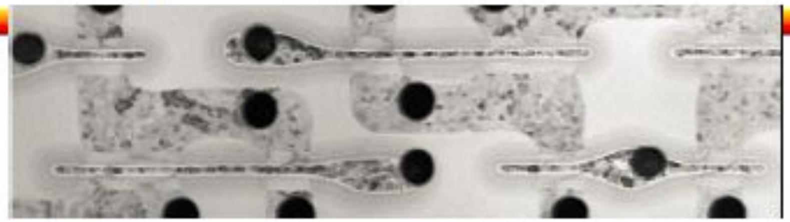




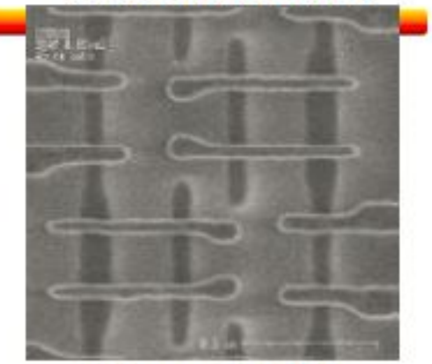
GLOBALFOUNDRIES
Dr. Jongwook Kye
Strategic Lithography
GLOBLFOUNDRIES



130nm SRAM
Traditional layout



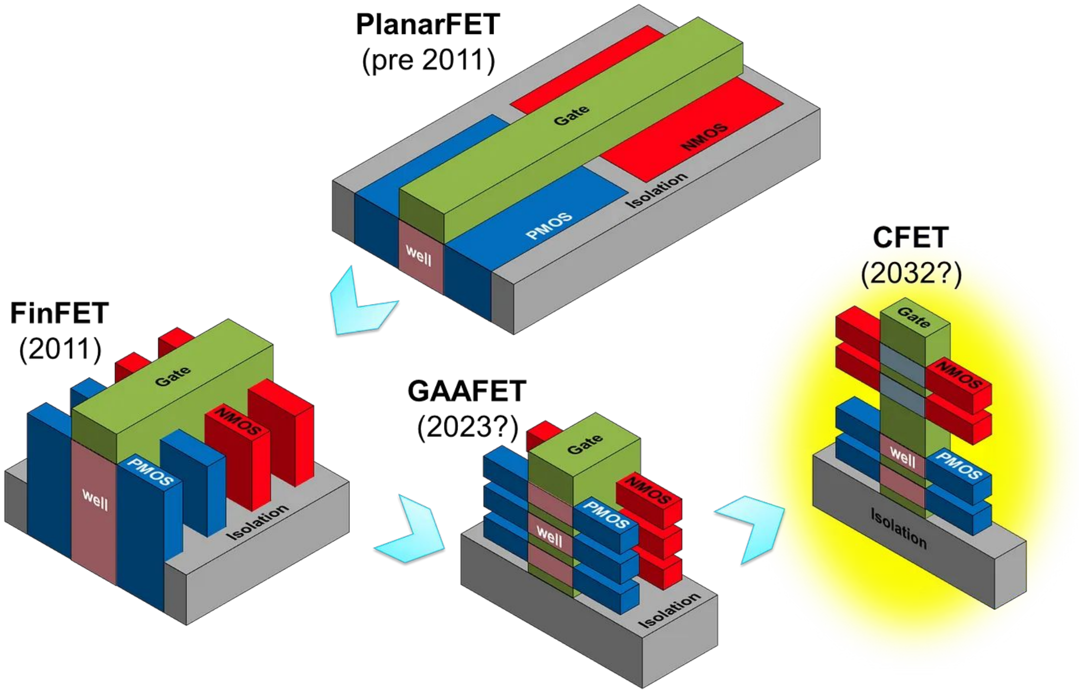
90nm SRAM
Uni-directional poly

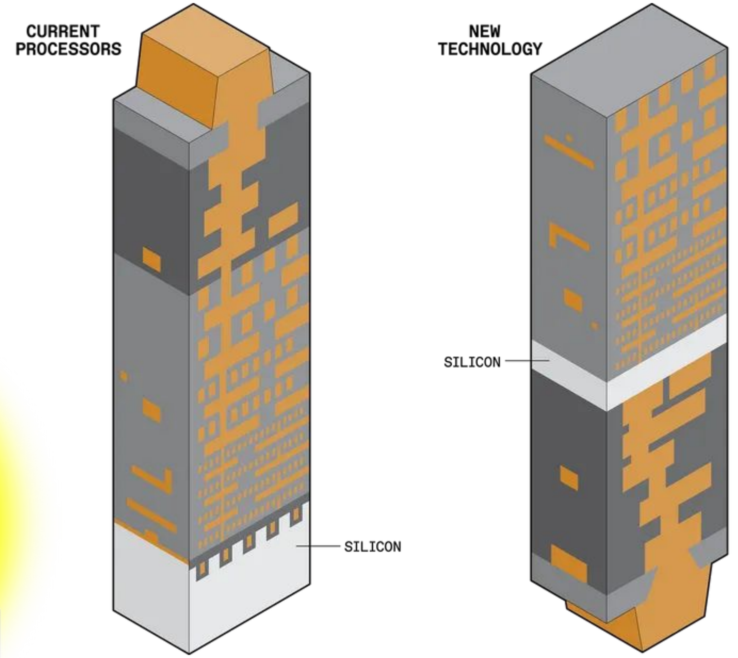


65nm SRAM
Uni-directional active

Technology driving design

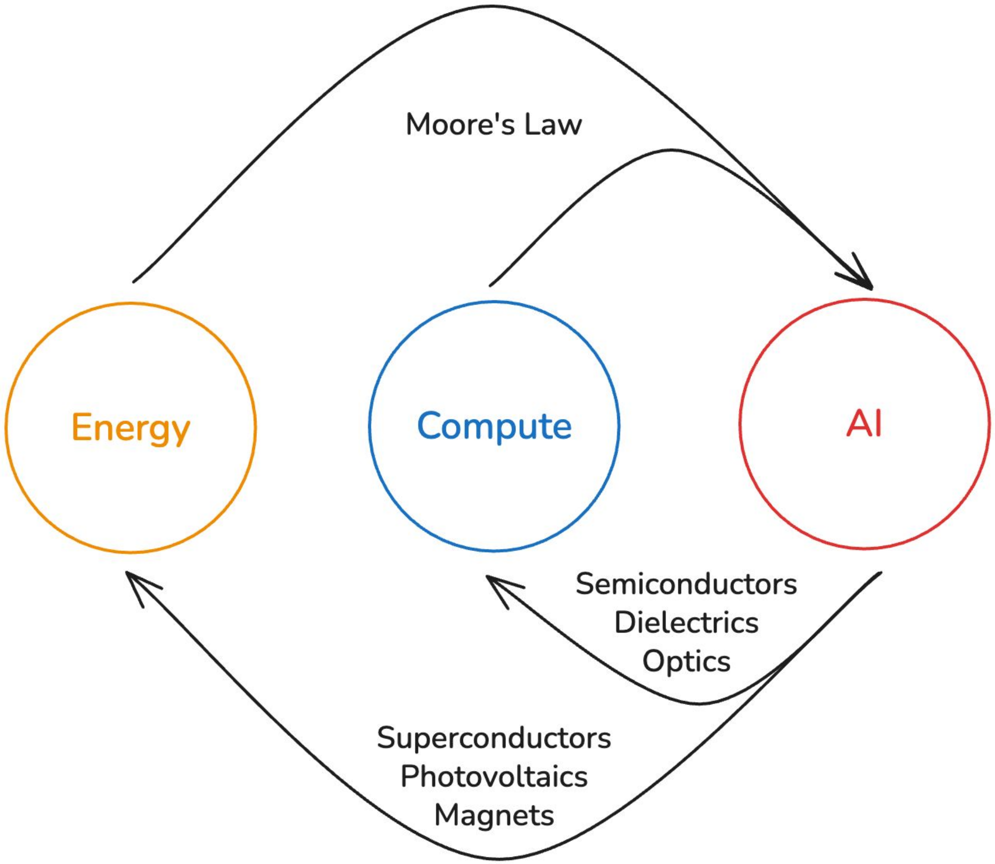
Design driving technology





Back-side power delivery network

https://medium.com/@MattTraversoPhD/how-cfet-will-revolutionize-semiconductor-morphology-and-continue-moores-law-4705bc803d49
https://spectrum.ieee.org/next-gen-chips-will-be-powered-from-below
https://semiwiki.com/eda/synopsys/294205-what-might-the-1nm-node-look-like/



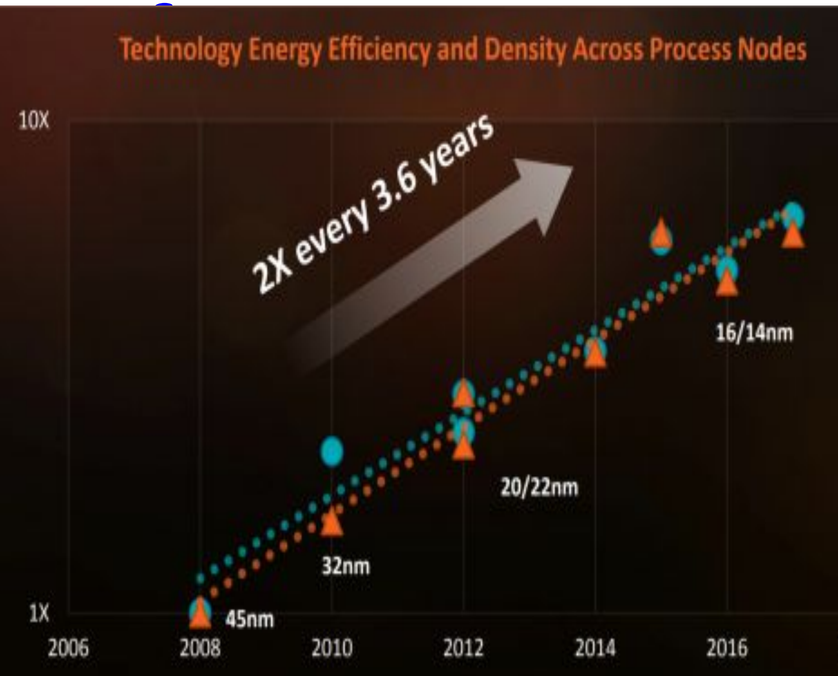
"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin. The ultimate reason for this is Moore's law, or rather its generalization of continued exponentially falling cost per unit of computation." - [Richard Sutton, *The Bitter Lesson*]

The AI industry is now operating at a scale where performance is no longer limited solely by arithmetic throughput. Power delivery, cooling, memory bandwidth efficiency, interconnect topology, data center floor space, and even electrical grid capacity have become first-order constraints.

Ironically, one of the few things slowing AI is a shortage of HVAC technicians qualified to install cooling systems in data centers.

https://www.theatlantic.com/magazine/2026/03/ai-economy-labor-market-transformation/685731/

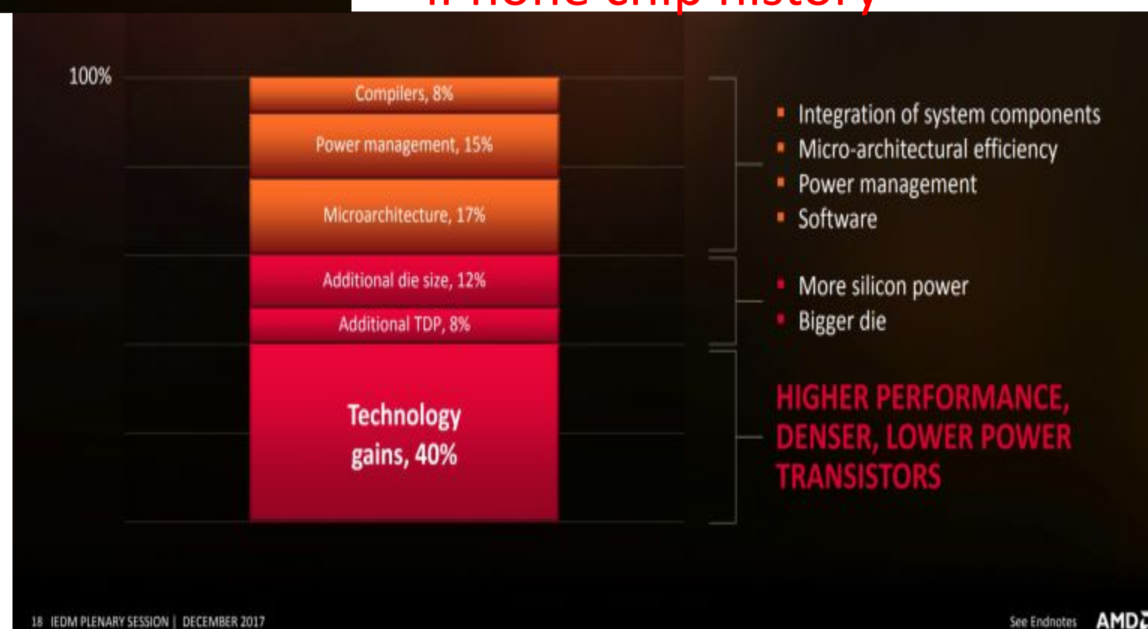
Technology for system level performance-3



2010	45nm	A4
2012	32nm	A6
2014	20nm	A8
2016	16nm	A10
2018	7nm	A12
2020	5nm	A14
2022	4nm	A16
2023	3nm	A17 Pro
2026E	2nm	A19

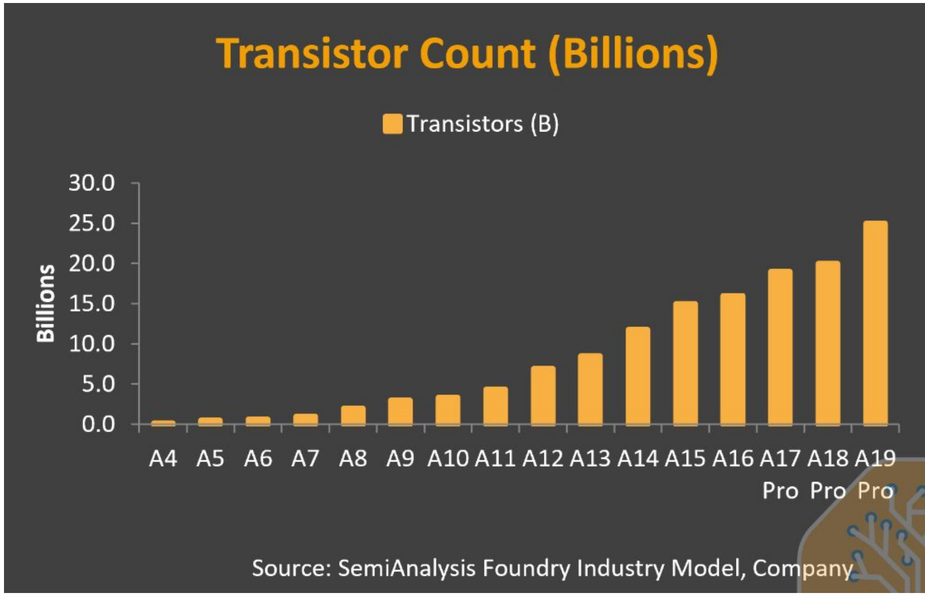
iPhone chip history

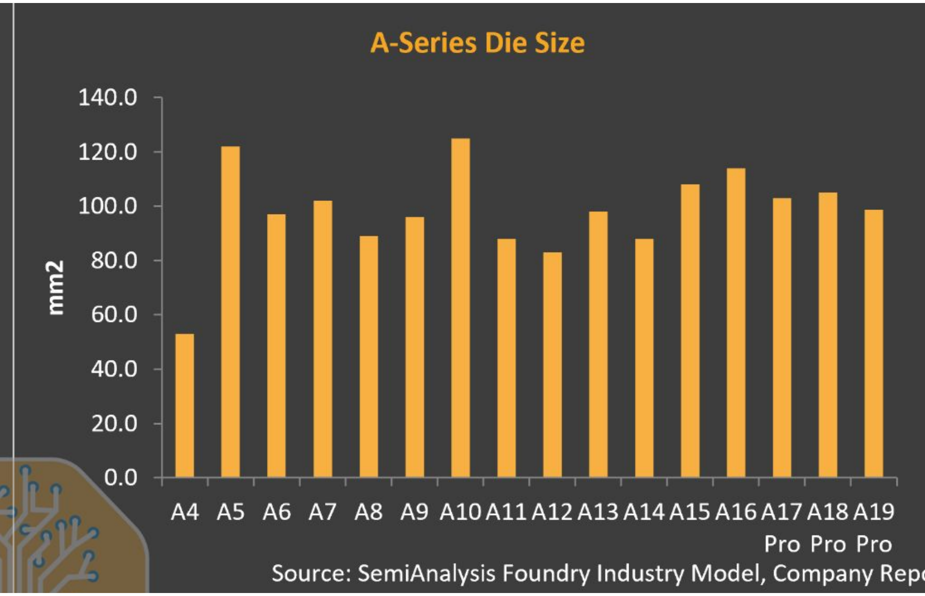
Energy/operation
Density
Functionality

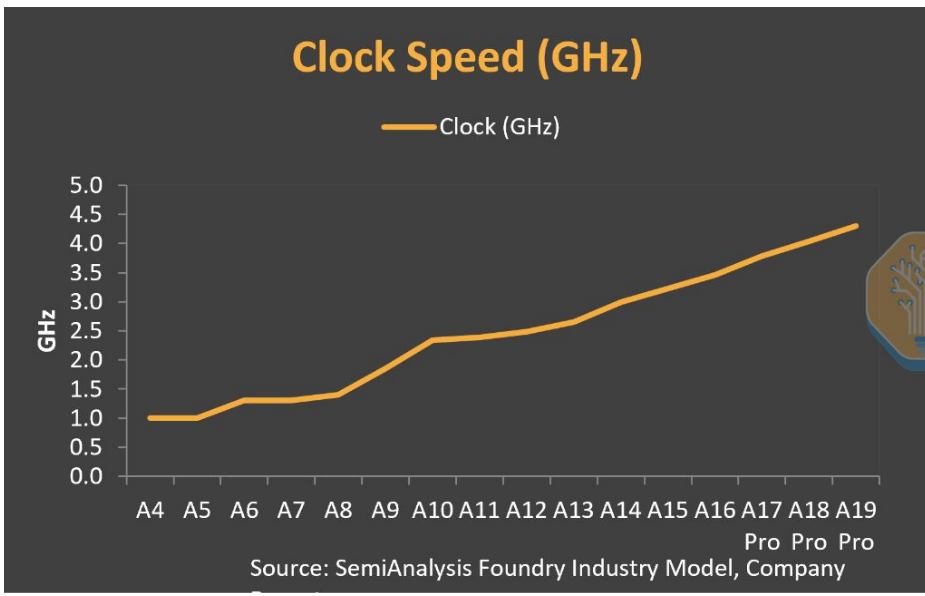


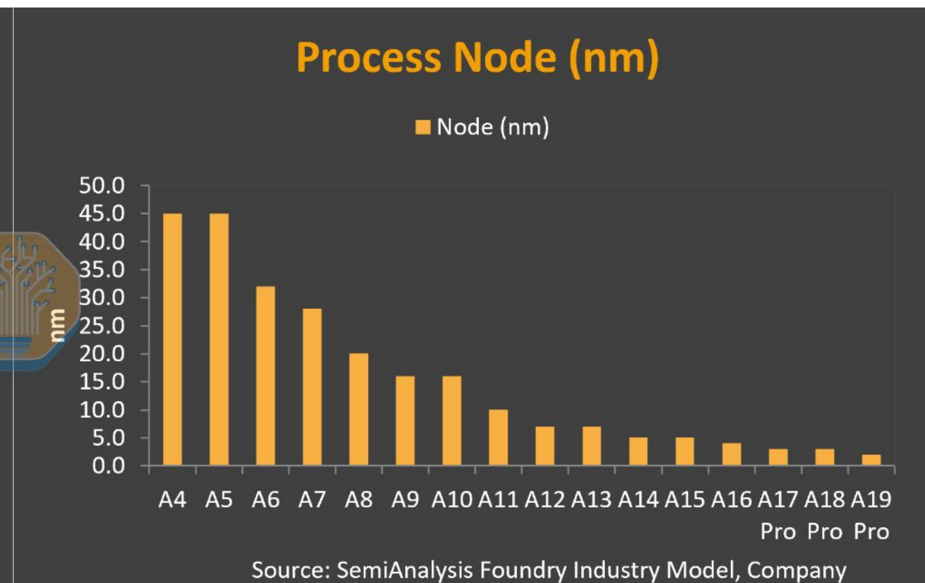
Lisa Su, IEDM 2017 Plenary talk

Apple processors



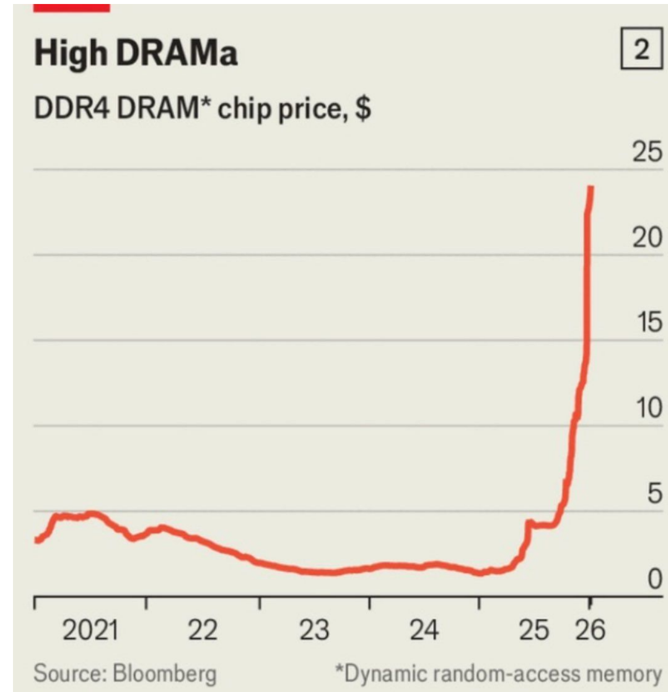




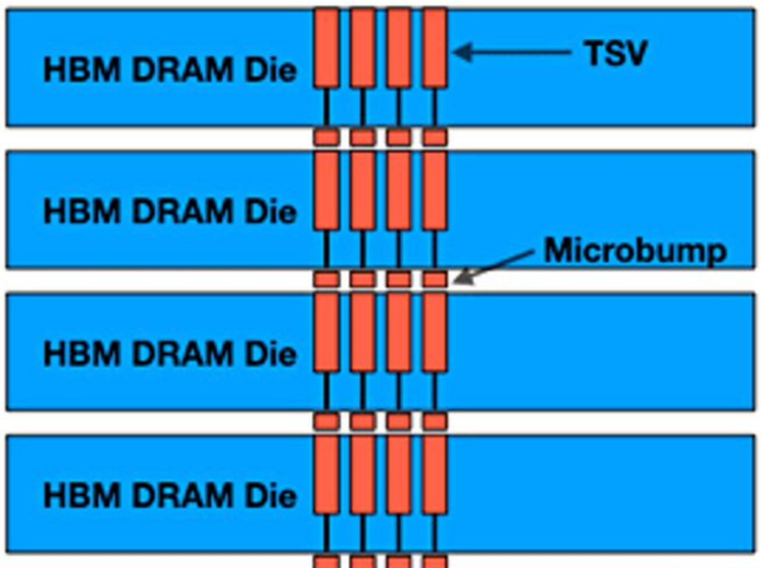


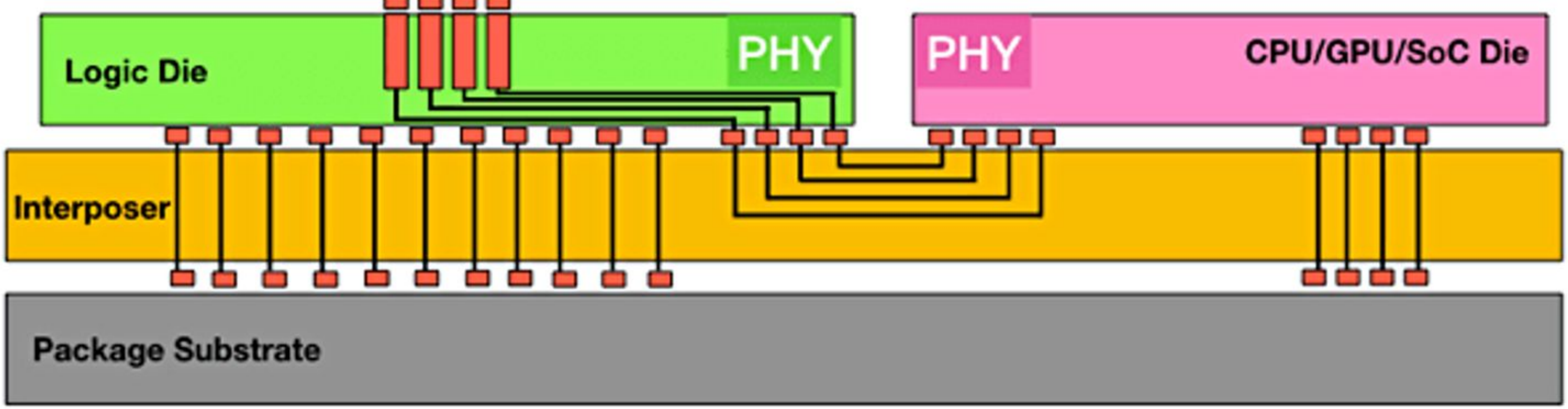
https://newsletter.semianalysis.com/p/apple-tsmc-the-partnership-that-built

High bandwidth memory



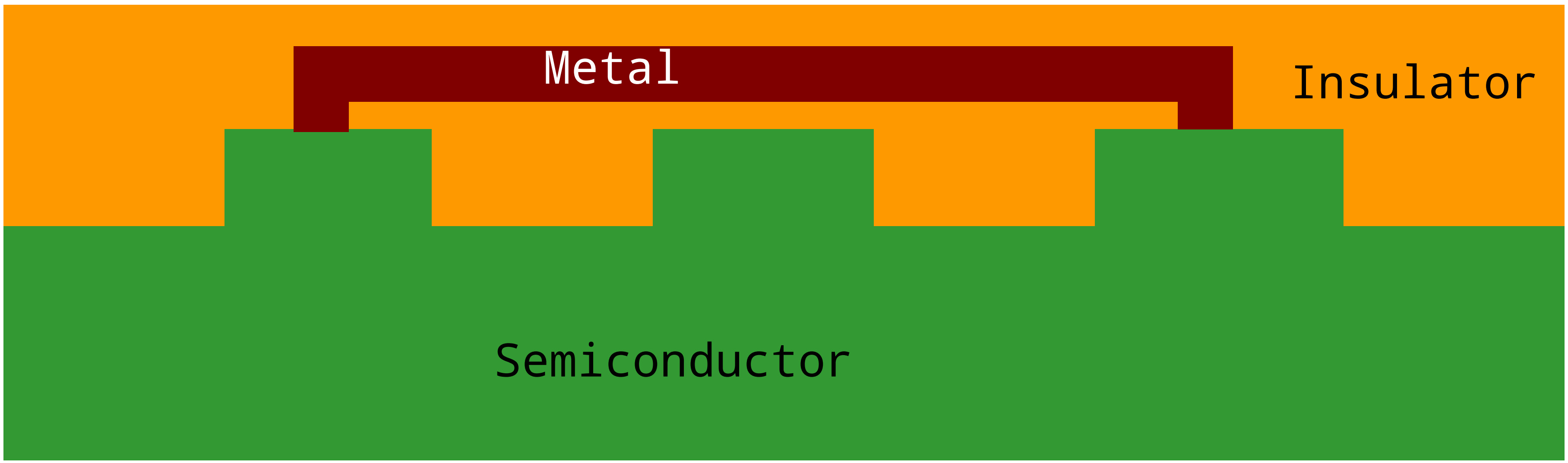
https://www.economist.com/business/2026/01/08/the-ai-frenzy-is-creating-a-big-problem-for-consumer-electronics





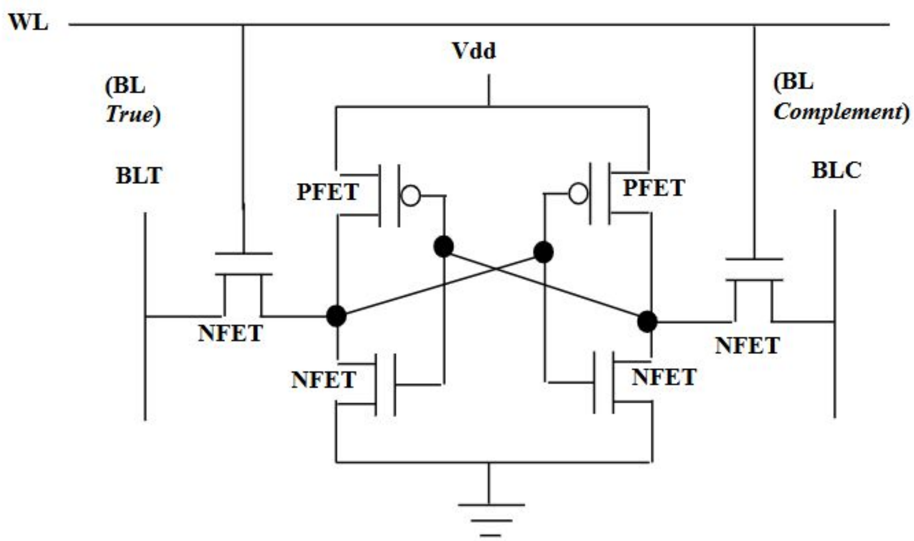
Design/architecture driving technology for system level performance

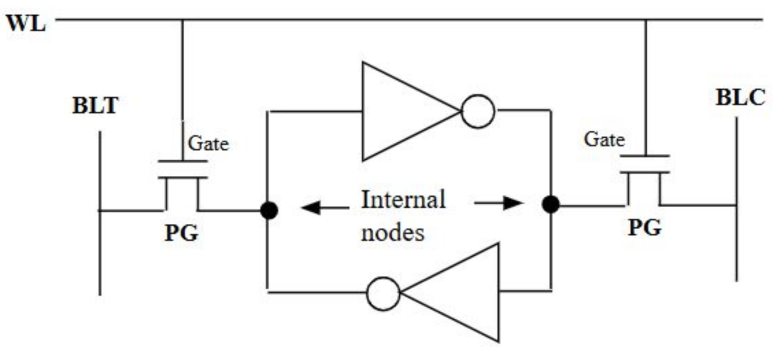
Monolithic Integrated circuit

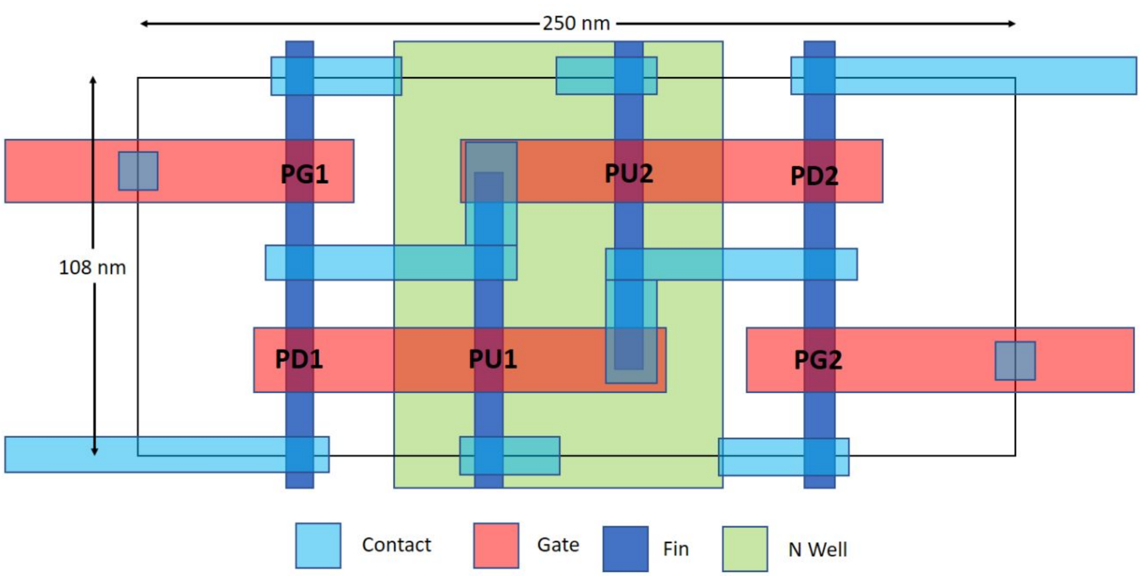


Regions of semiconducting material (devices) separated by insulators and interconnected through metals

SRAM: circuit schematic to layout







Layout is converted to different masks which allows to transfer pattern to wafer and allow selective processing

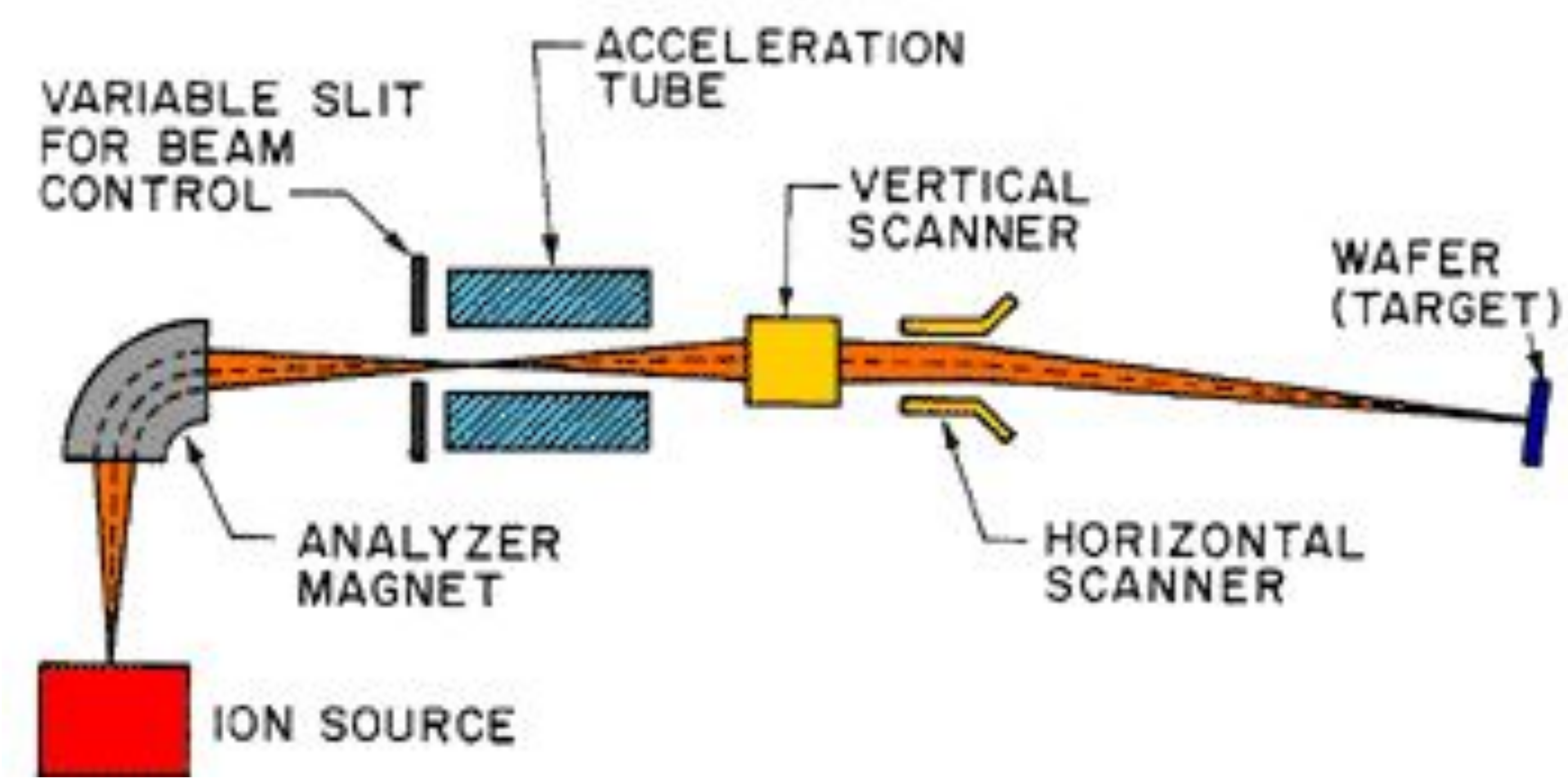
IC Fabrication Processes

Nvidia says it takes 89 masks to create the H100, and Intel cites '50+' masks used for its 14nm chips. If you add each and every steps in the entire process, it runs into 1000s. One can categorize all the process steps into these five bins

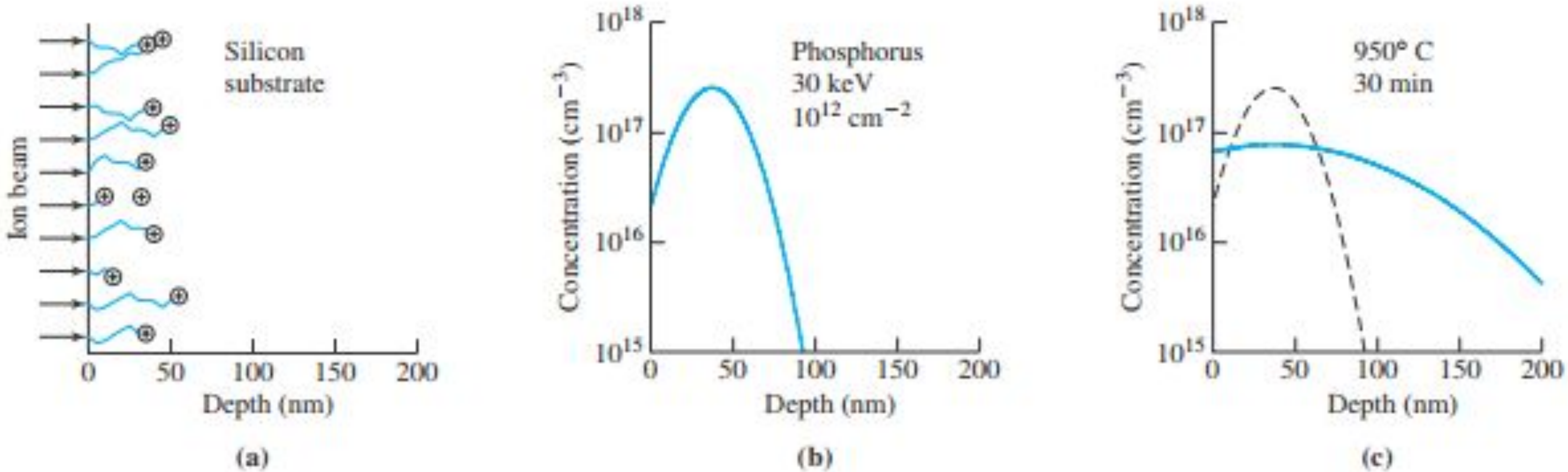
Additive processes	Subtractive processes	Patterning processes	Heating	Metrology
Doping (Diffusion/Ion implantation+annealing)	Wafer Clean	Lithography	Annealing	Inspection
Oxidation	Etching		Reflow	Electrical Test
Deposition	Chemical Mechanical Planarization		Alloying	Optical Test
Silicidation (Deposition + Annealing)				

https://www.tomshardware.com/news/nvidia-tackles-chipmaking-process-claims-40x-speed-up-with-culitho

Ion implantation



Ion implantation



(a) Accelerated ions of the doping element collide inside the semiconductor, stopping at scattered depths. The doping profile immediately after ion implantation (b) and after annealing (c).

Implant results in dopant atoms in interstitial sites and surface damage of wafer. Rapid Anneal is done to regain the crystal structure and to place dopants at substitutional sites

Laser spike anneal

Oxidation

- High quality oxide helped to establish the dominance of Si compared to other semiconductors (Ge and GaAs)
- High quality Si/SiO_2 interface (< 10^{10} defects/cm^2)
- Diffusion/ion-implantation barrier
- Not soluble in water (during cleaning)

Methods

Dry oxidation

$$Si + O_2 \rightarrow SiO_2$$

- Dense oxide (good quality, low diffusion)
- slow growth rate

Wet oxidation

$$Si + 2H_2O \rightarrow SiO_2 + 2H_2$$

- Relatively porous oxide (lower quality, species diffuse faster)
- faster growth rate

Consumes silicon volume from wafer while converting to oxide

High temperature process – 800-1100C

Thin film deposition

1. Physical Vapor Deposition (PVD) - Vapors of constituent materials created inside the chamber, and condensation occurs on wafer surface leading to the deposition of a solid film. (most commonly used for metals, TiN and Al/Cu alloy).

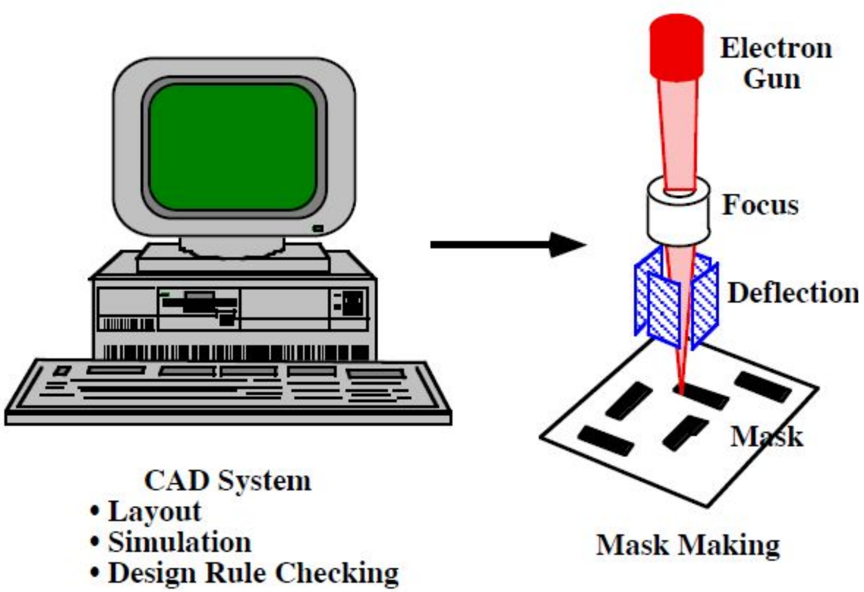
2. Chemical Vapor Deposition (CVD) - Reactant gases are introduced in the chamber, chemical reactions occur on wafer surface leading to the deposition of a solid film (most commonly used for dielectrics and semiconductors). **Epitaxy** refers to the method of depositing a monocrystalline film on a monocrystalline substrate.

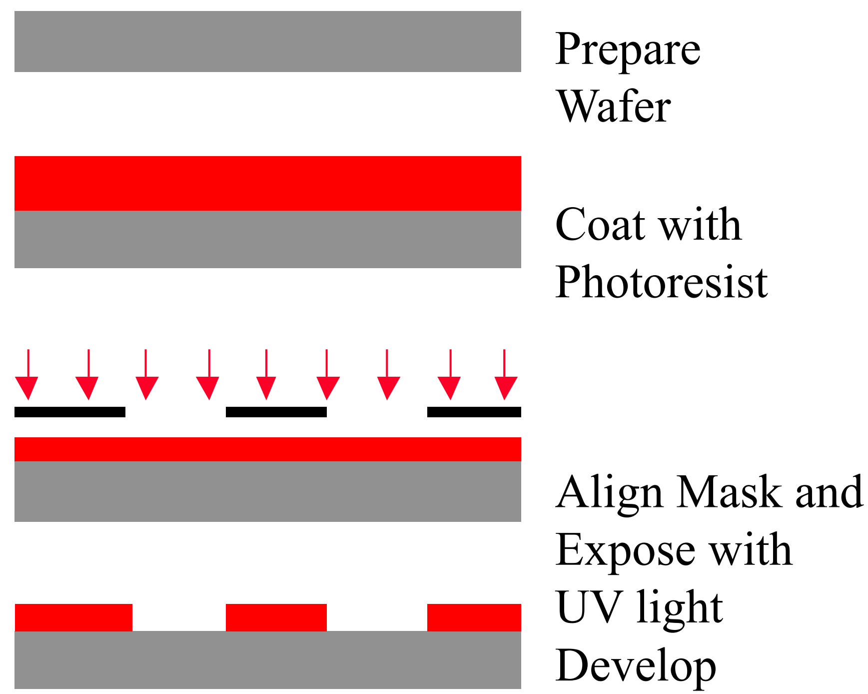
$SiH_4 + O_2 \rightarrow SiO_2 + 2H_2$
$3SiH_2Cl_2 + 4NH_3 \rightarrow Si_3N_4 + 6HCl + 6H_2$

3. Electrochemical deposition: coating from a solution that contains ions of the species to be coated. (Cu for interconnects)

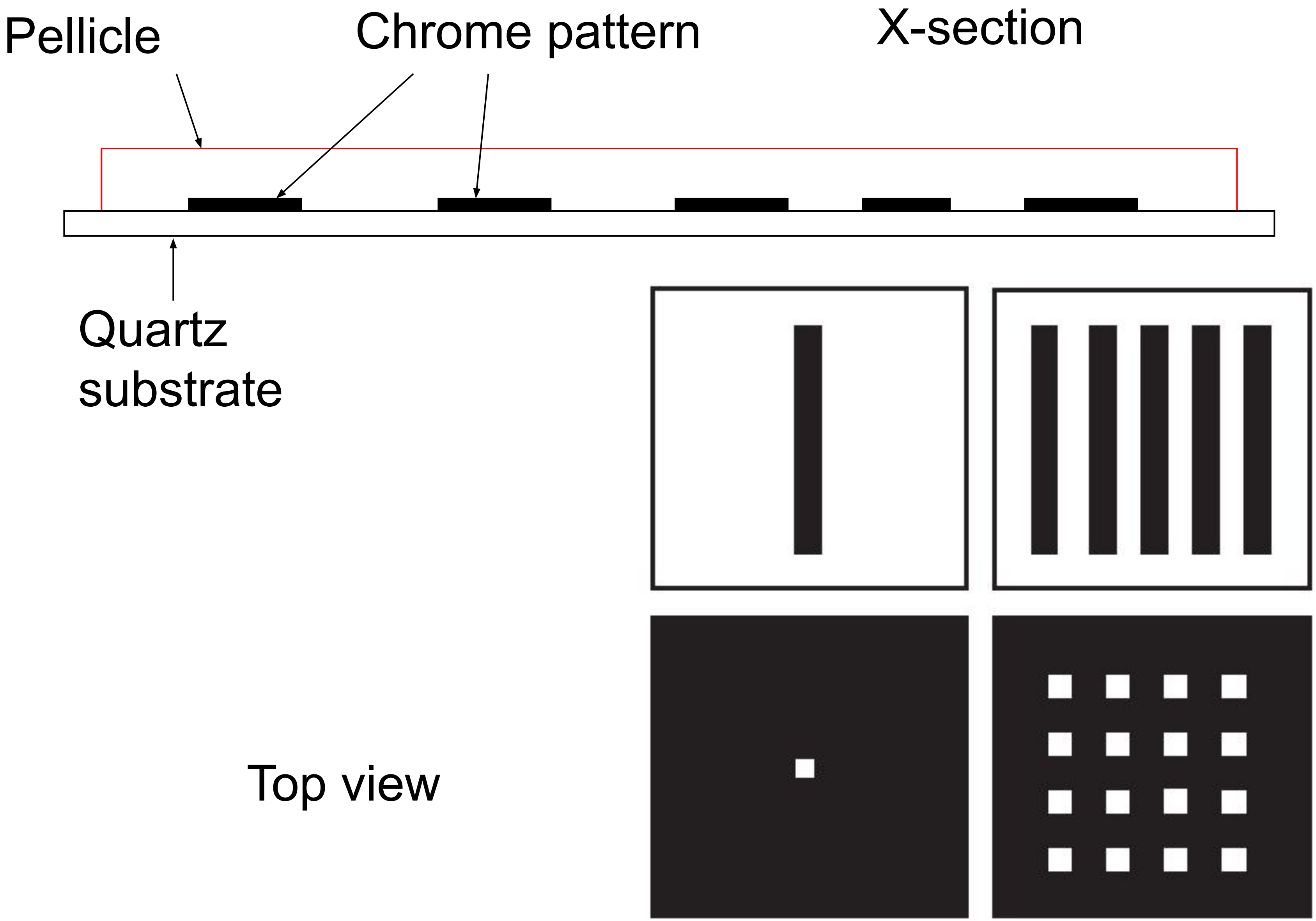
Photo lithography

- The most critical and expensive module in semiconductor manufacturing
- A process to transfer patterns from a mask to a substrate.
- ~40 % total wafer process time
- Most critical process - determines how small and closely packed the individual devices/structures can be made

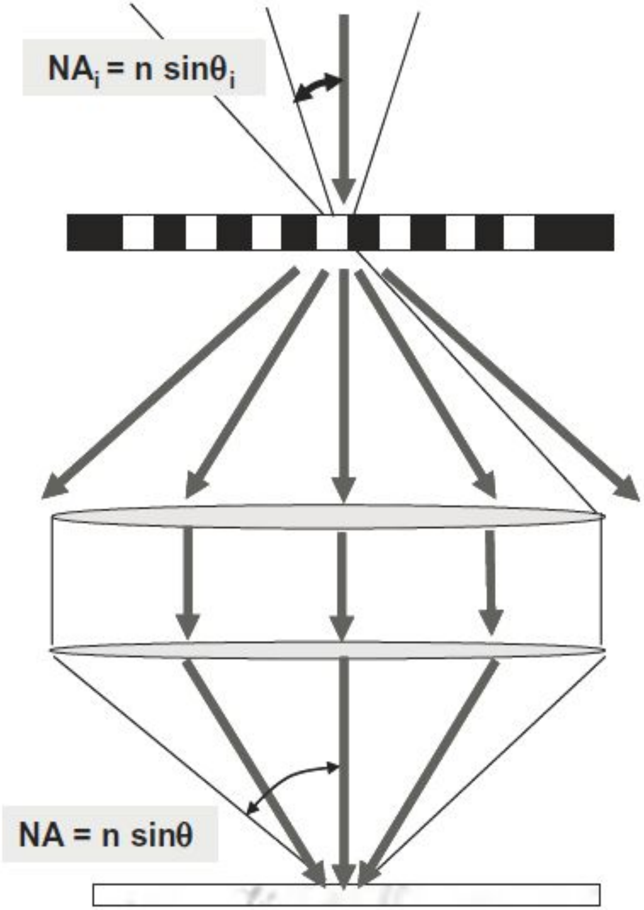




Mask/Reticle (before EUV)



Imaging process



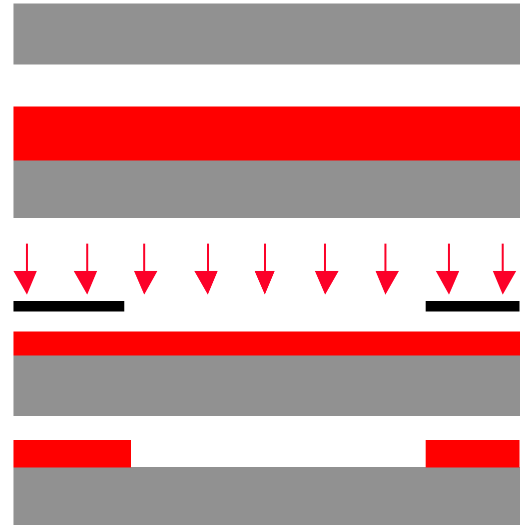
- Light of a fixed wavelength
- illuminates the mask

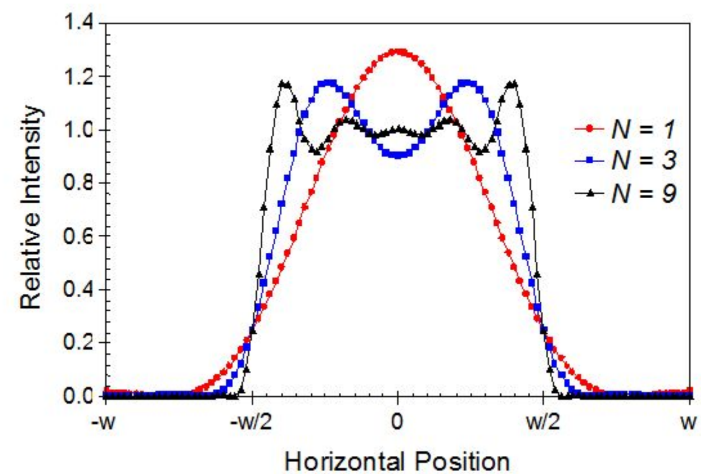
- Mask creates a diffraction pattern
- The smaller the pitch the larger the diffraction angle

- Due to its finite size, the lens can only capture some of the diffracted beams
- The key metric is NA (numerical aperture)

- Image is formed with a subset of the diffraction orders
- Loss of image fidelity
- At least 2 diffracted beams are required to obtain modulation

The larger the NA, the tighter the pitch of the pattern that can be imaged



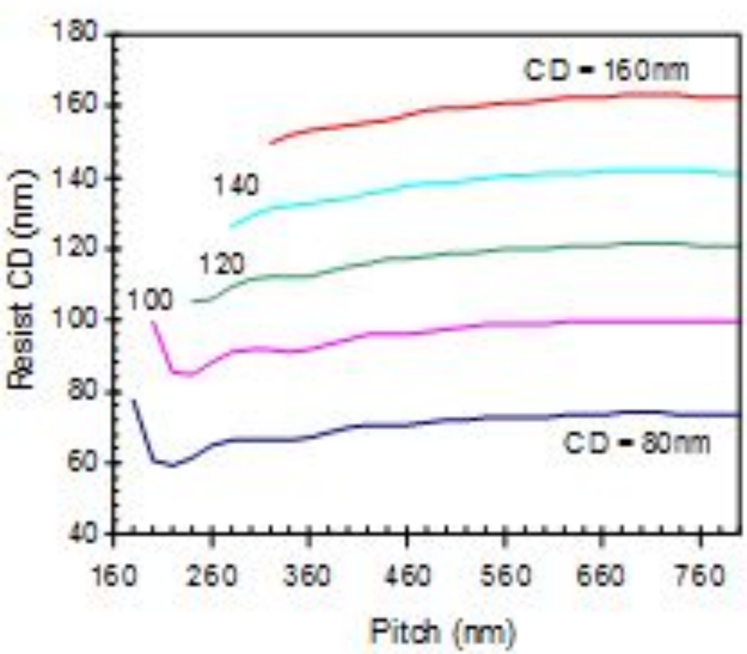


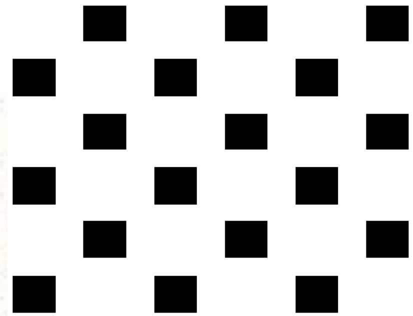
Aerial images for a pattern of equal lines and spaces of width *w* as a function of the number of diffraction orders captured by the objective lens (coherent illumination). N is the maximum diffraction order number captured by the lens.

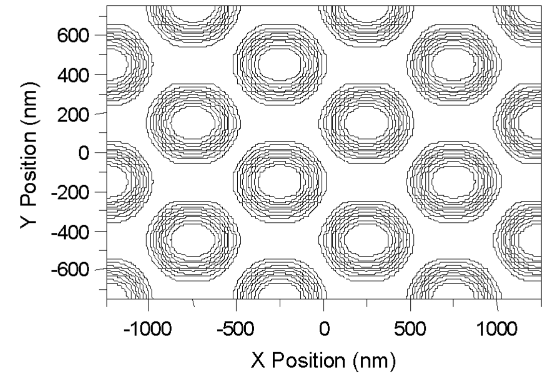
Wong et al: Nano CMOS design for DFM

Chris A. Mack, Fundamental Principles of Optical Lithography, (c) 2007

Through-pitch behavior and corner rounding

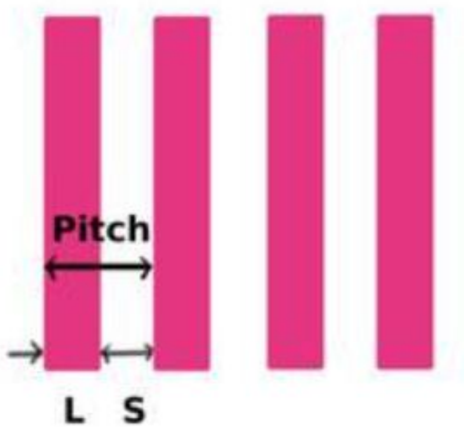


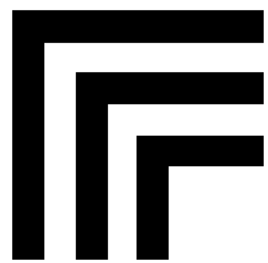


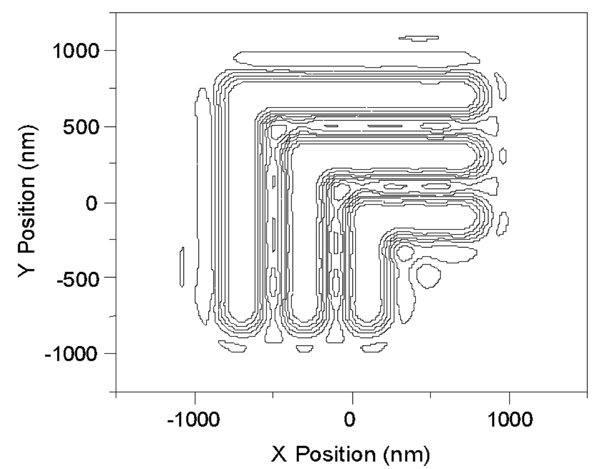


Contours of constant intensity

Resist CD through pitch (optimized for 100 nm L/S, λ = 193 nm)

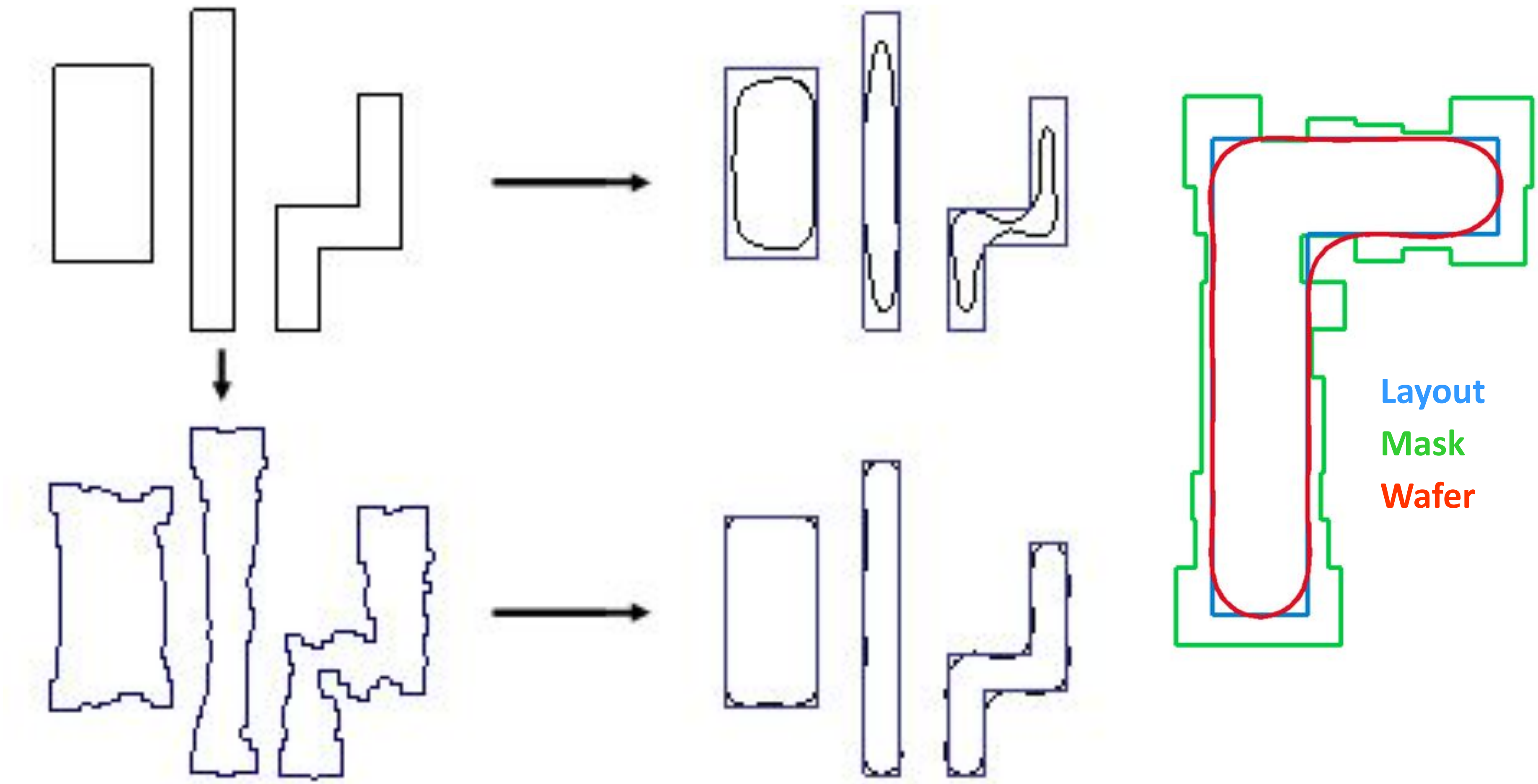




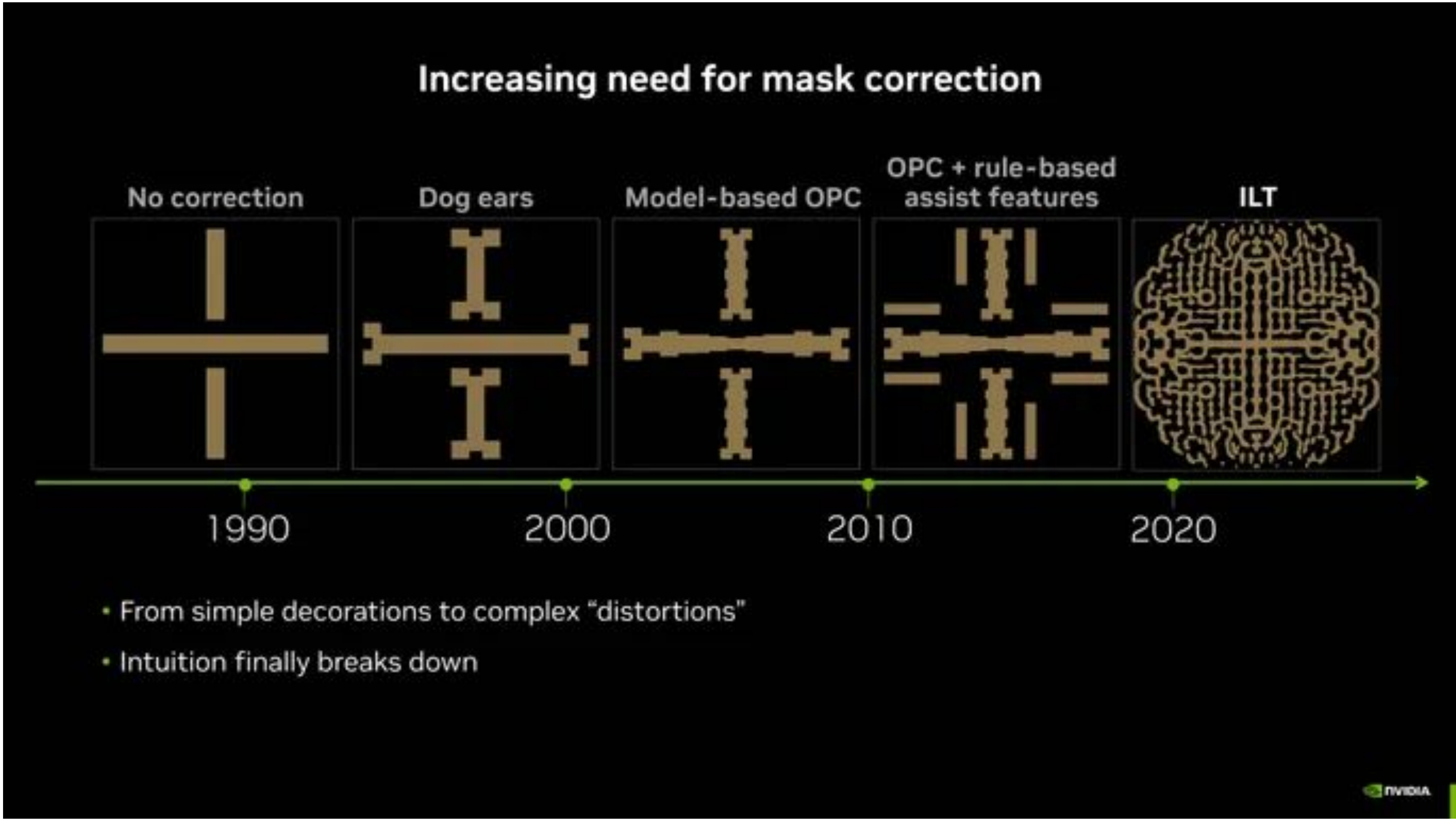


Corner rounding can be reduced by decreasing the wavelength of light or increasing the NA of lens

Dataprep: From layout to mask (for feature sizes << wavelength)



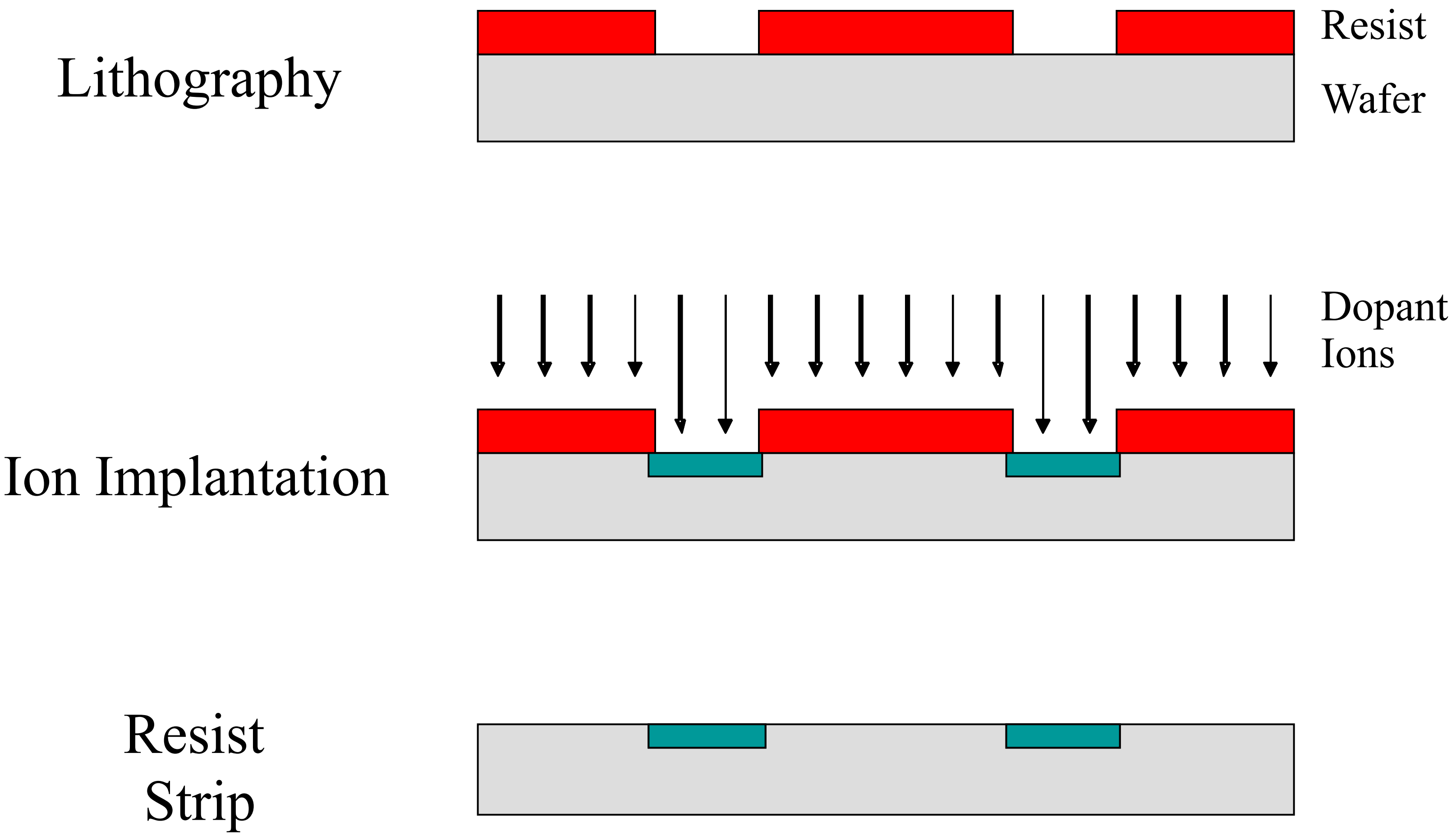
The original design (upper left) prints very poorly (upper right). After aggressive Optical Proximity Correction (OPC), the resulting design (lower left) prints very close to the desired shape (lower right).



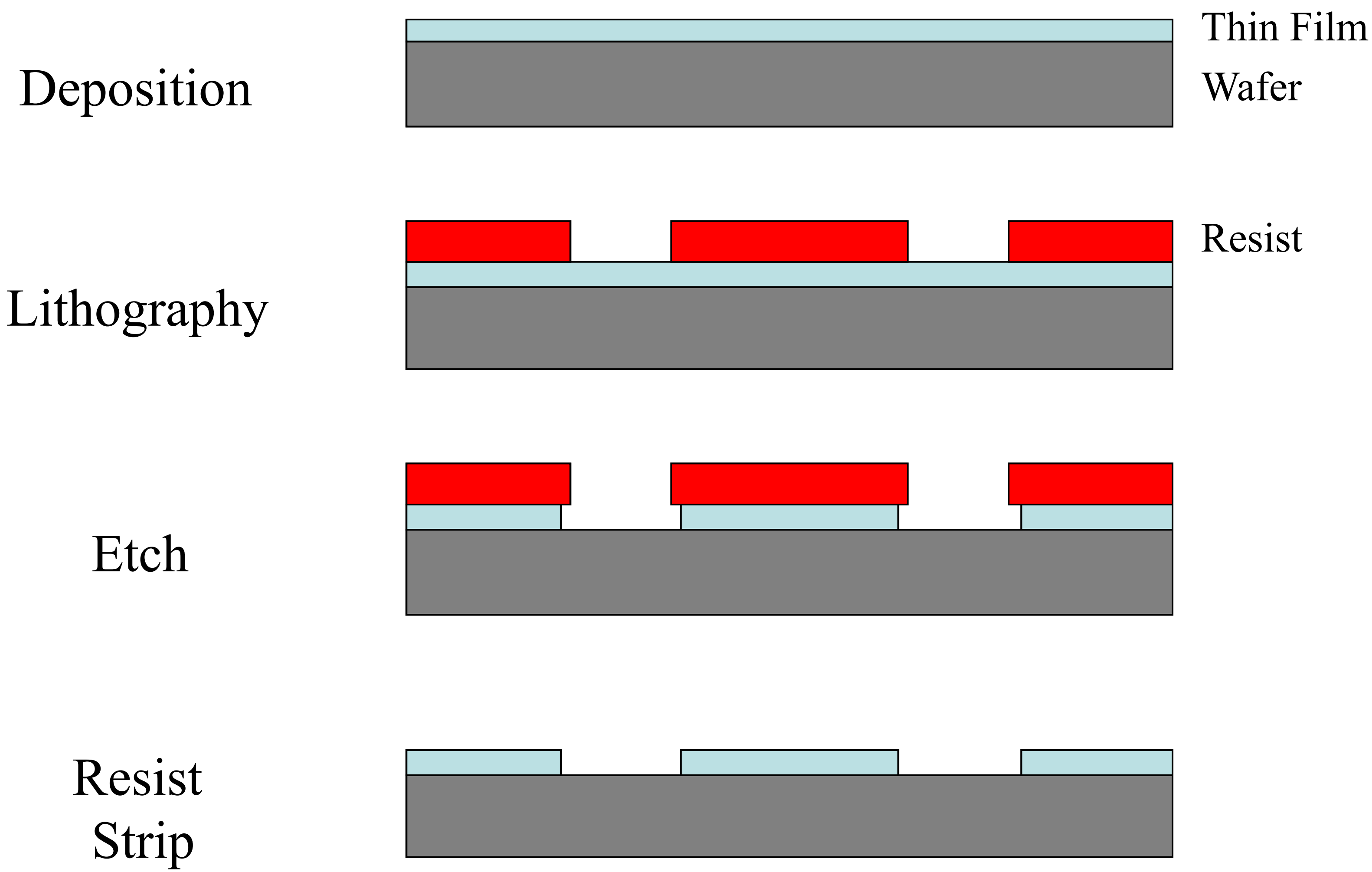
[Nvidia Speeds Key Chipmaking Computation by 40x]

[Inverse lithography technology: 30 years from concept to practical, full-chip reality]

Patterning as a means for selective doping

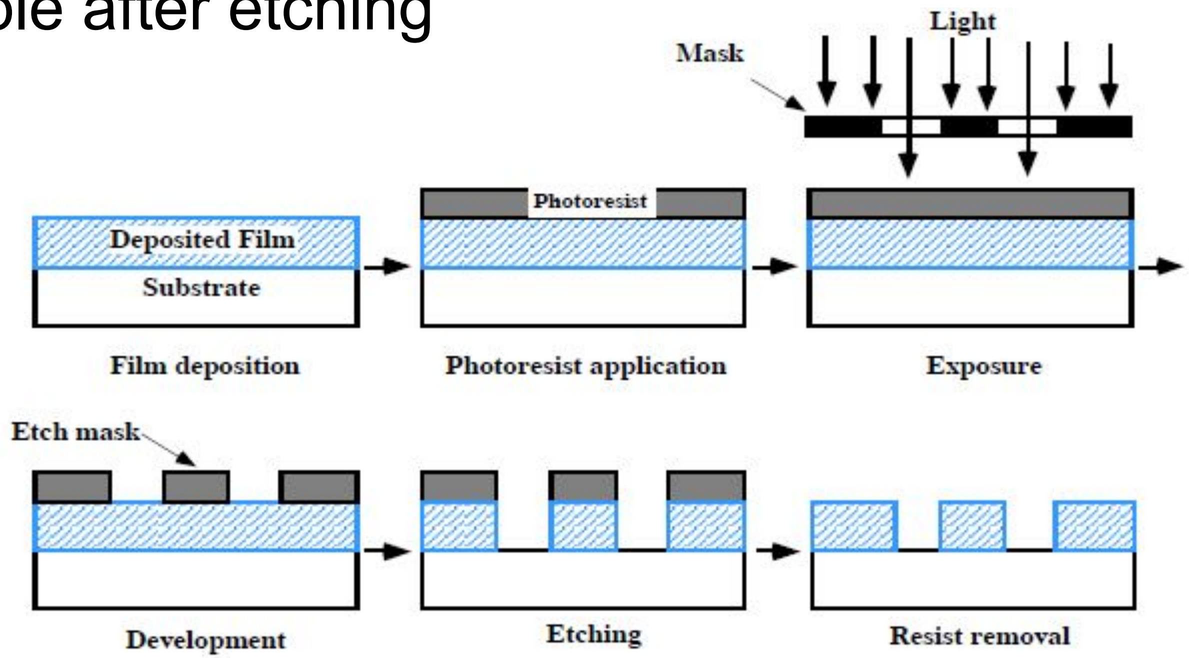


Subtractive patterning process



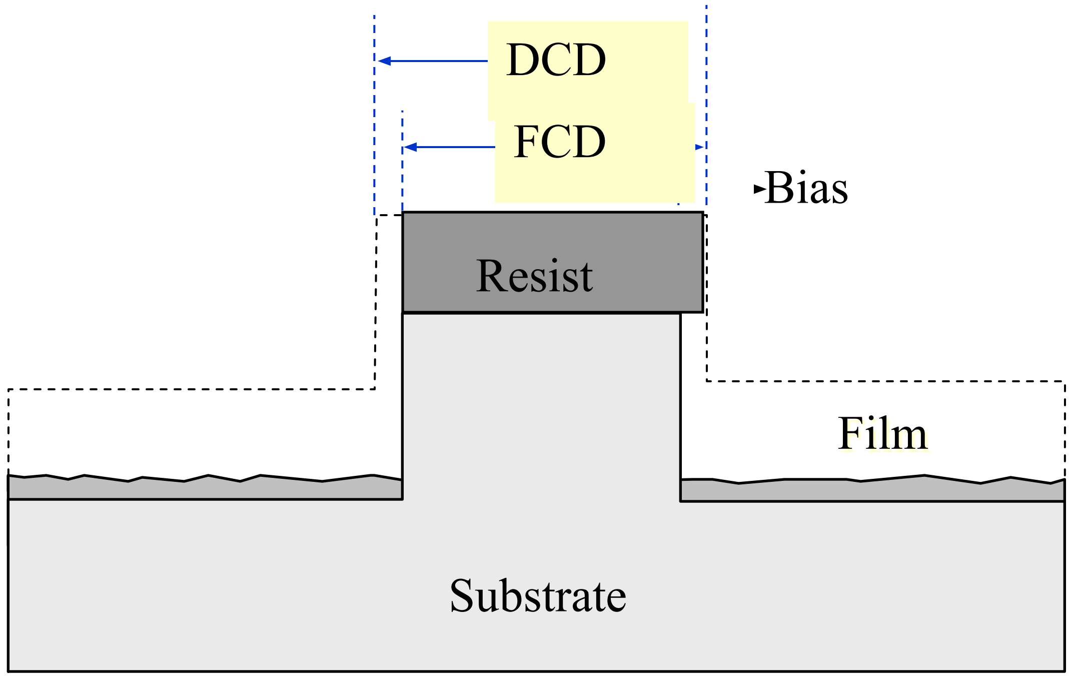
Etching

- Process by which material is removed from the substrate or from thin films on the substrate.
- Chemical, physical or combination of the two
- Selective or blanket etch
- 2nd step in the pattern transfer process. In lithography, pattern is transferred from mask to resist. In etching, it is from resist to underlying film
- Rework is not possible after etching

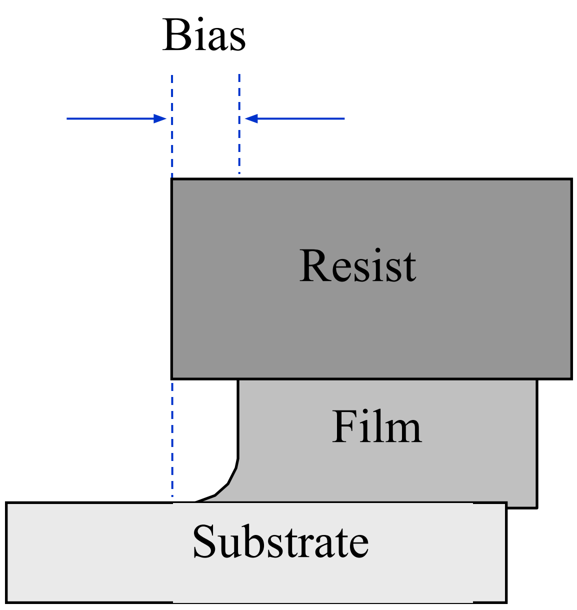


Etch bias

Measure of the amount by which the etched film undercuts



(a)



(b)

DCD = Develop CD

Feature size after lithography

FCD = Final CD

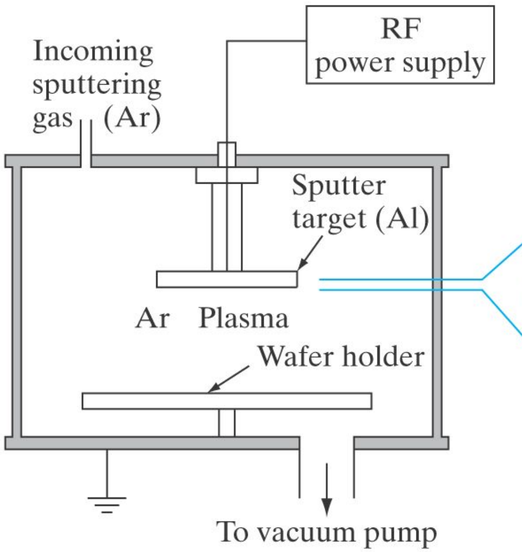
Feature size after etch

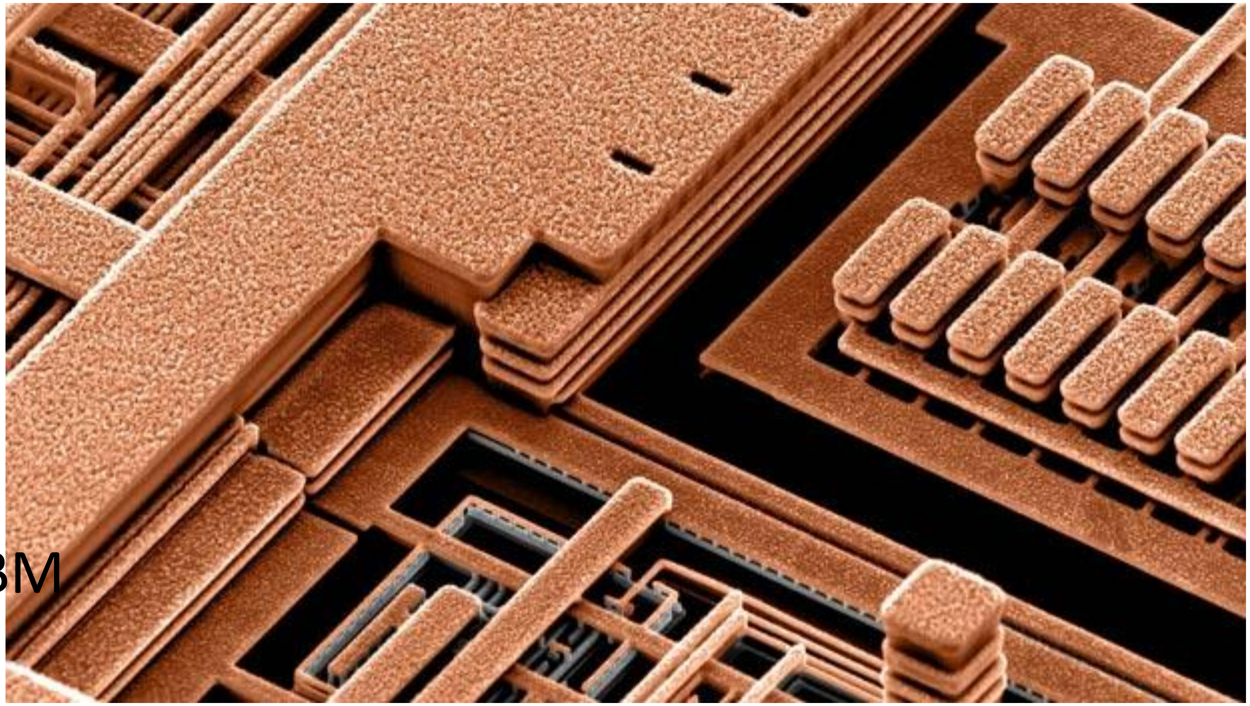
Undercut

Etch bias = (DCD-FCD)/2

Metallization

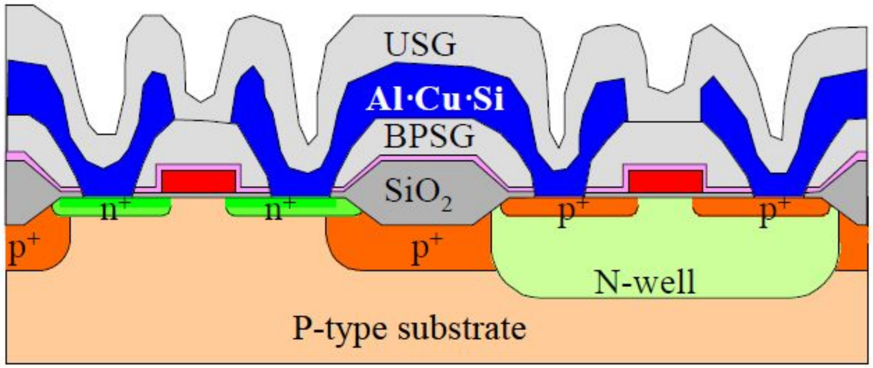
Sputtering is one way to make contacts. It is achieved by immersing an Al target in Ar plasma. Ar ions bombard target. The ejected Al atoms deposit on Si wafers

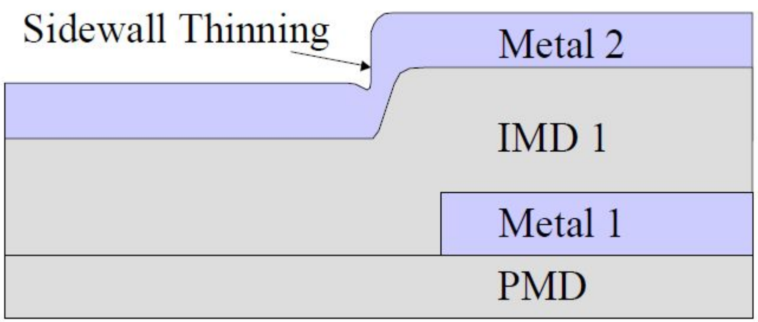


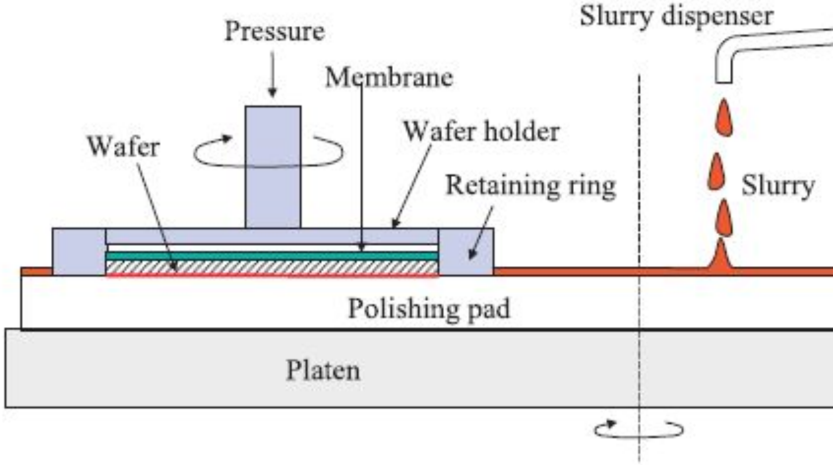


Electroplated Cu interconnects inside an IBM microprocessor

Need for chemical mechanical polishing (CMP)



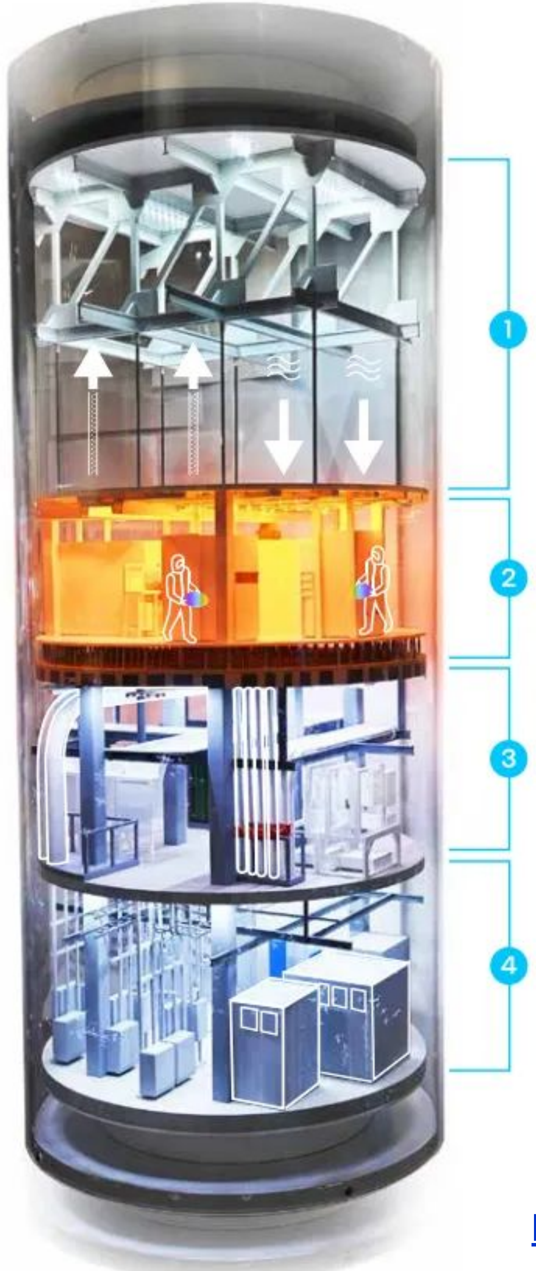




Planarized surface allows higher resolution of photolithography process

The planarized surface eliminates sidewall thinning because of poor PVD step coverage

Cleanroom Structure



1. Interstitial and fan deck (top level)

The fan deck houses systems that keep the air in the clean room particle-free and precisely maintained at the right temperature and humidity for production. The interstitial is the tallest level of the fab.

2. Clean room level

A clean room is made up of more than 1,200 factory tools that take pizza-size silicon wafers and eventually turn them into hundreds of computer chips. Clean room workers wear bunny suits to keep lint, hair and skin flakes off the wafers.

Fun Fact Clean rooms are usually lit with yellow lights. The yellow lighting is necessary in photolithography to prevent unwanted exposure of photoresist to light of shorter wavelengths.

3. Clean subfab level

The clean subfab contains thousands of pumps, transformers, power cabinets and other systems that support the clean room. Large pipes called "laterals" carry gases, liquids, waste and exhaust to and from production tools. Workers don't wear bunny suits here, but they do wear hard hats, safety glasses, gloves and shoe covers.

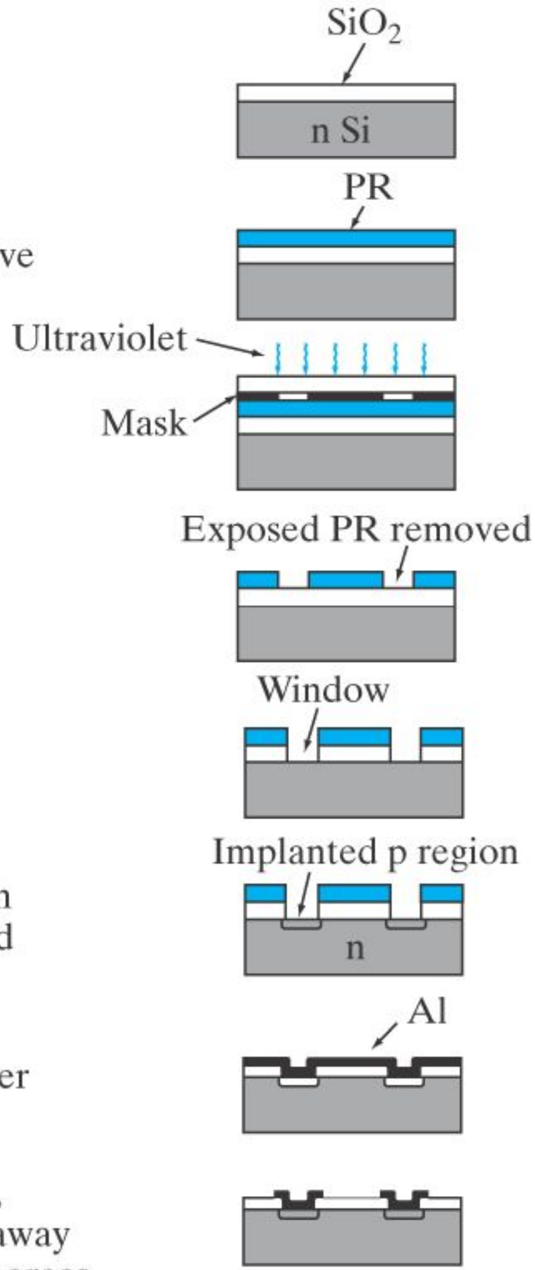
4. Utility level

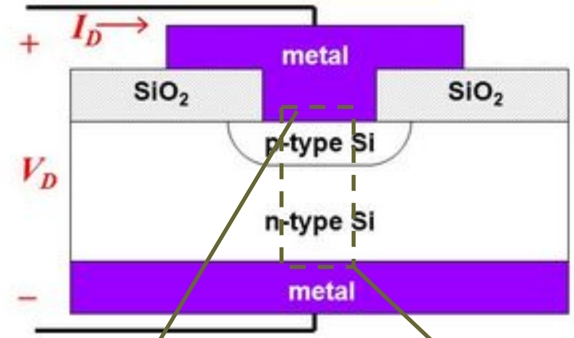
Electrical panels that support the fab are located here, along with the "mains" — large utility pipes and ductwork that feed up to the lateral pipes in the clean subfab. Also here are chiller and compressor systems. Workers who monitor the equipment on this level wear street clothes, hard hats and safety glasses.

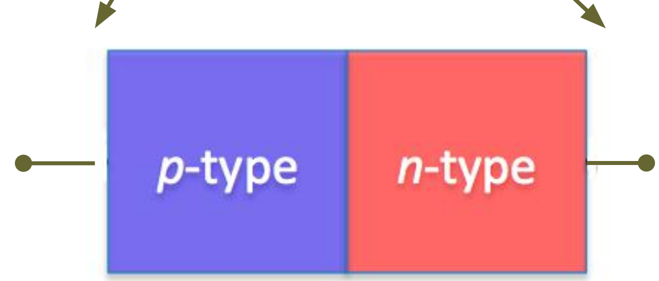
https://download.intel.com/newsroom/2022/manufacturing/fab-final-static.pdf

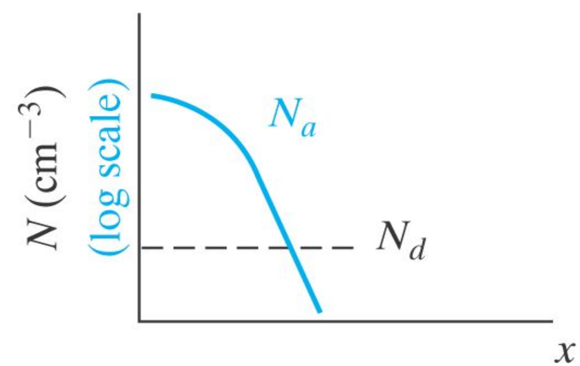
Fabrication of a pn junction

1. Oxidize the Si sample
2. Apply a layer of positive photoresist (PR)
3. Expose PR through mask A
4. Remove exposed PR
5. Use RIE to remove SiO_2 in windows
6. Implant boron through windows in the PR and SiO_2 layers
7. Remove PR and sputter Al onto the surface
8. Using PR and mask B, repeat steps 2-4; etch away Al except in p-contact areas

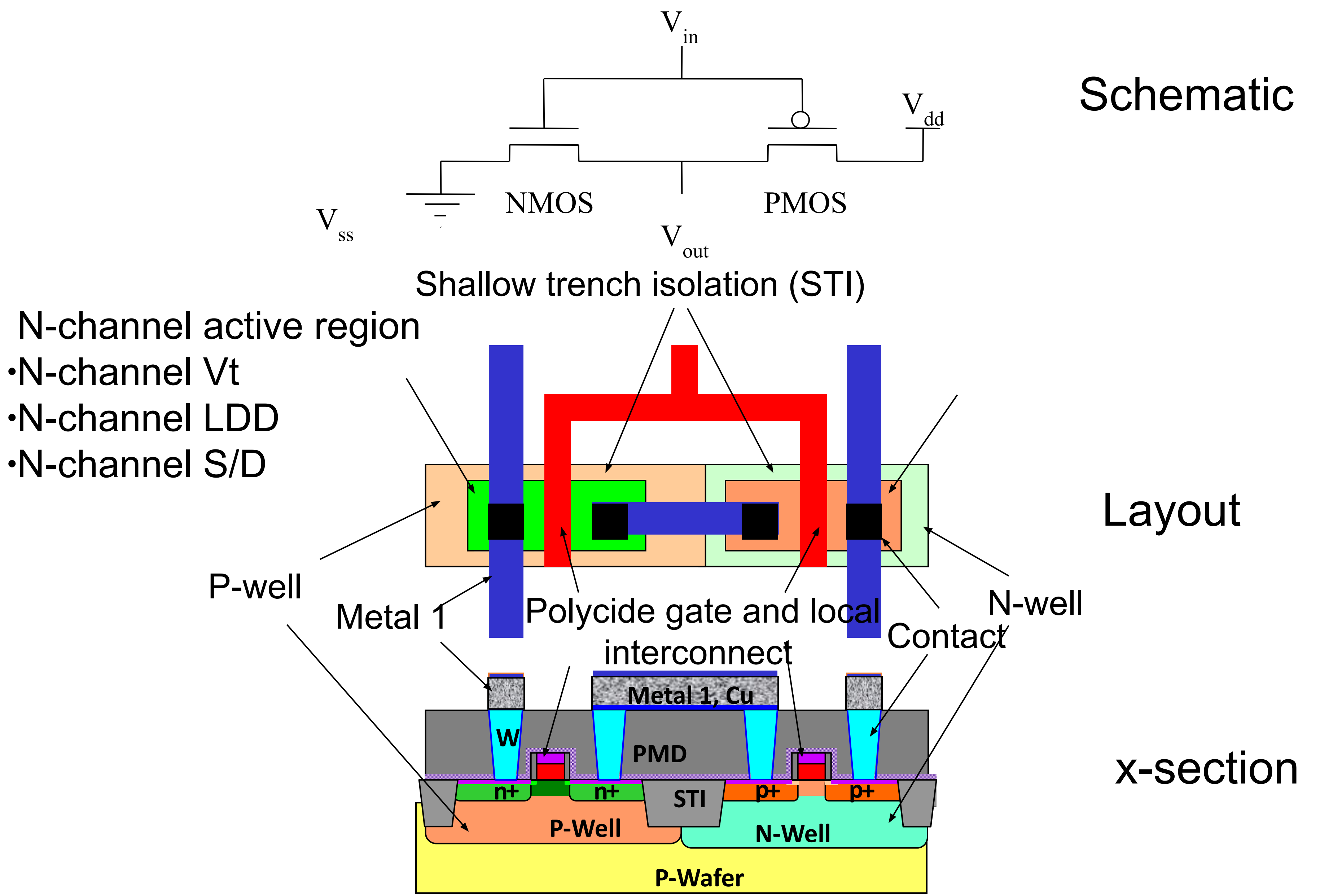




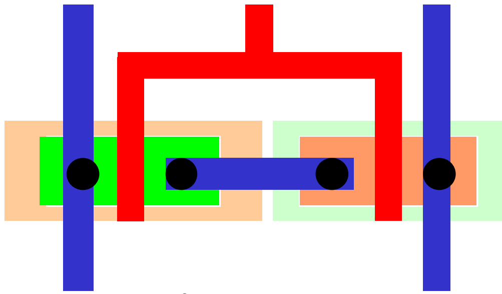




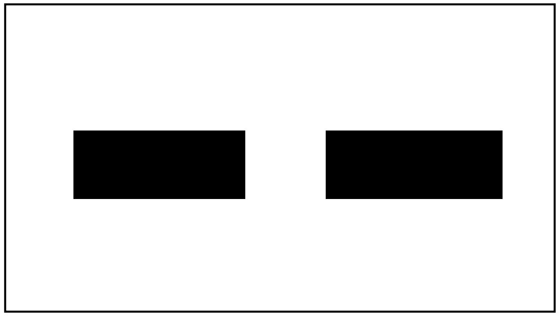
Circuit schematic to chip: CMOS Inverter



Layout and Masks of CMOS Inverter



CMOS inverter layout



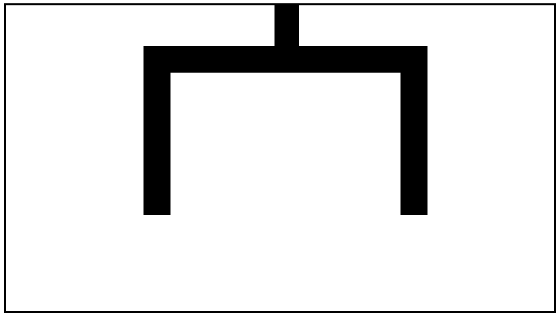
Mask 1, shallow trench isolation

Mask 2, P-well

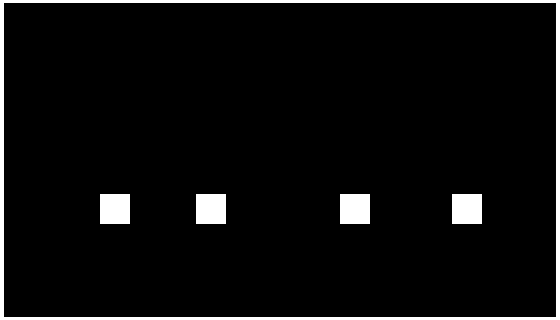
Mask 3, N-well

Mask 4, 7, 9, N-Vt, LDD, S/D

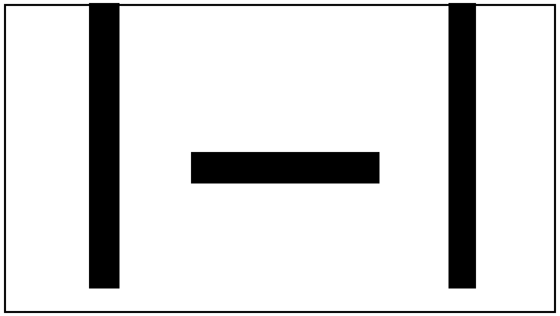
Mask 5, 8, 10, P-Vt, LDD, S/D



Mask 6, gate/local interconnection



Mask 11, contact

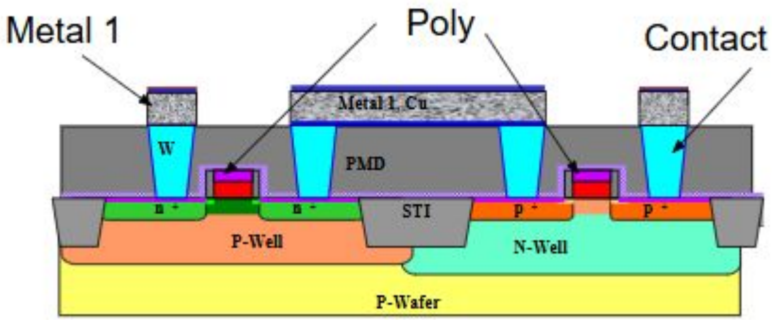


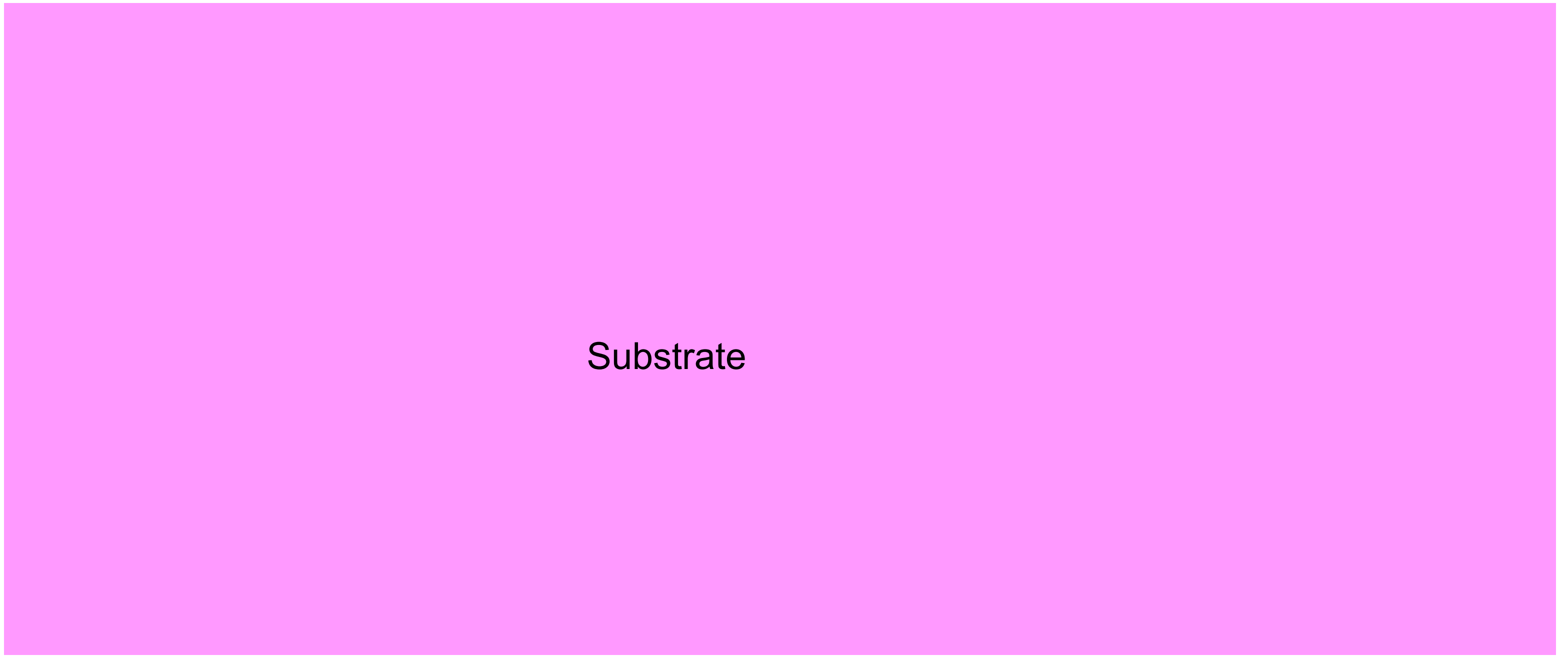
Mask 12, metal 1

CMOS process flow (130nm node)

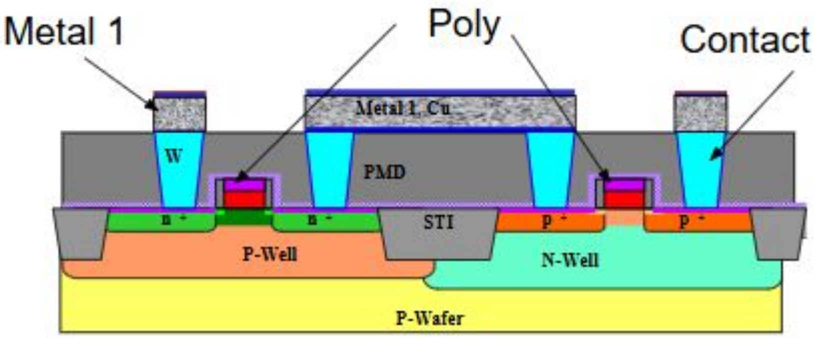
1. Active area definition/Shallow trench isolation
2. Twin-well Implants
3. Channel V_T adjust
4. Gate oxidation
5. Poly Si dep and gate stack patterning
6. Spacer1
7. Halo and source/drain extension implants
8. Second Sidewall Spacer
9. Source/Drain Implants
10. High temp rapid anneal
11. Silicidation
12. Contact etch stop liner
13. ILD deposition
14. Contact (Tungsten plug) Formation
15. Inter layer dielectric
16. First Metal layer
17. Inter layer dielectric
18. First Via
19. ….M2, V2, M3……Mx metal
20. Final passivation … open for wire bonding/bumps

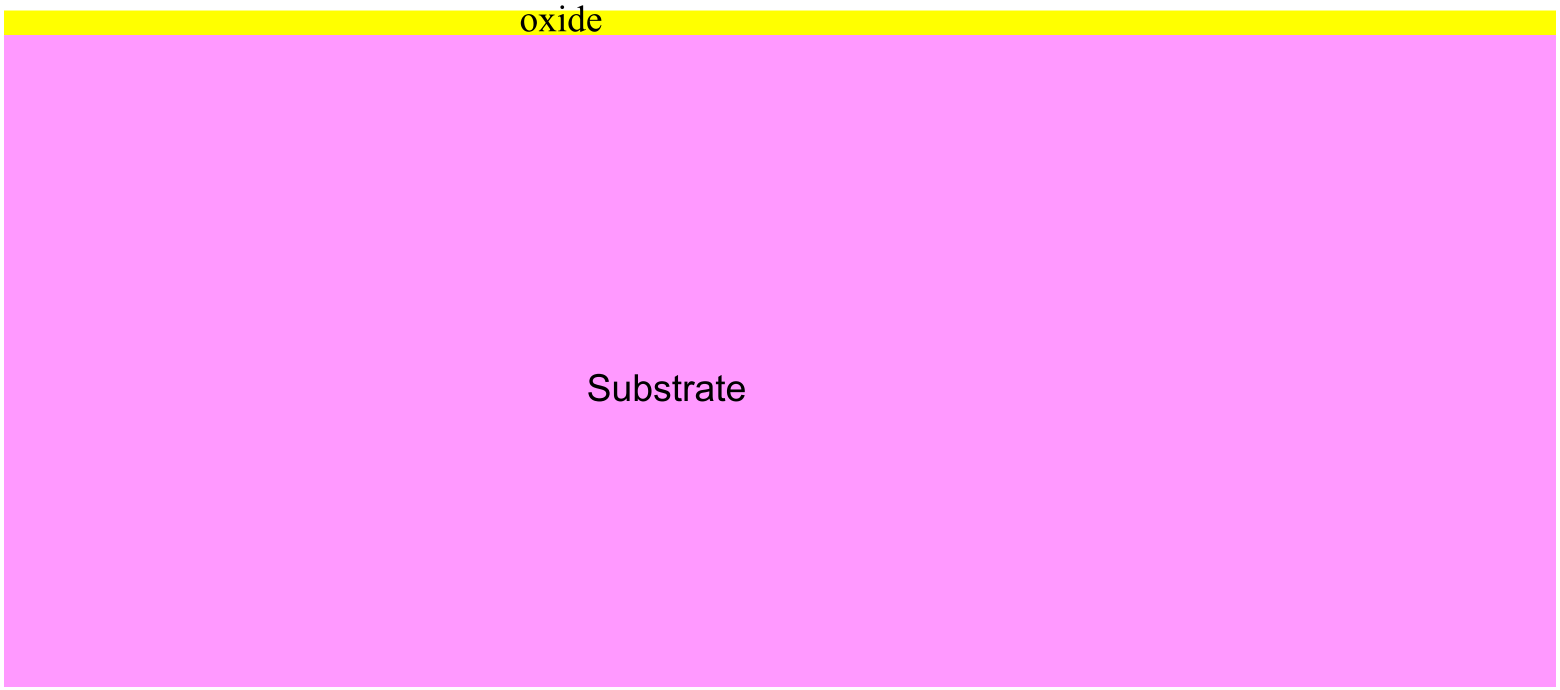
Silicon wafer



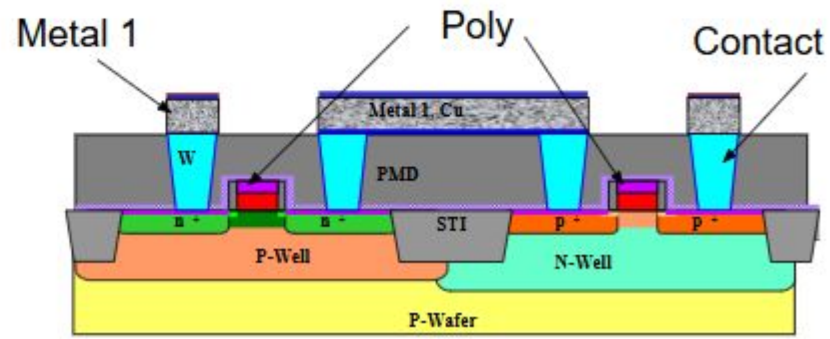


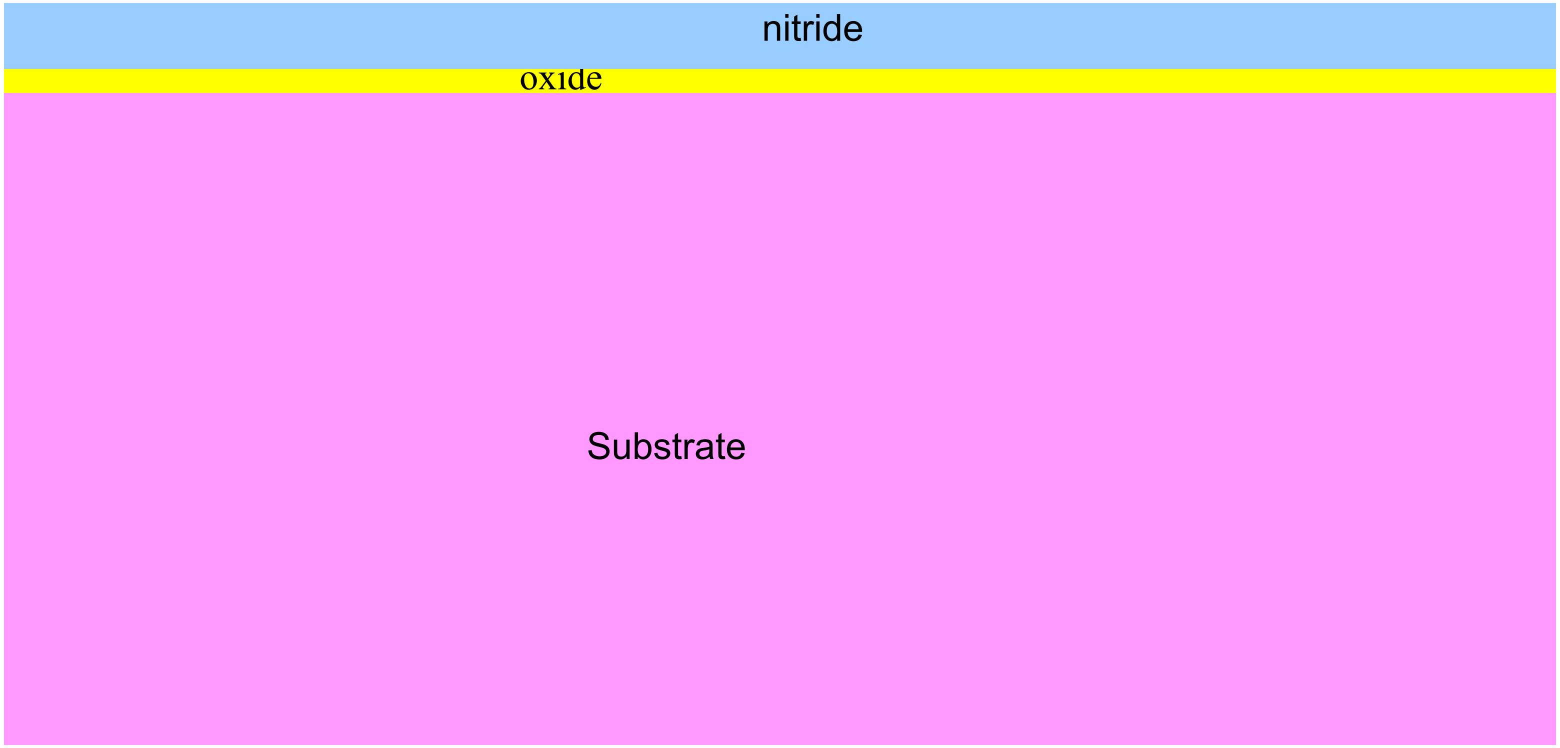
Creation of pad oxide through thermal oxidation after wafer cleaning



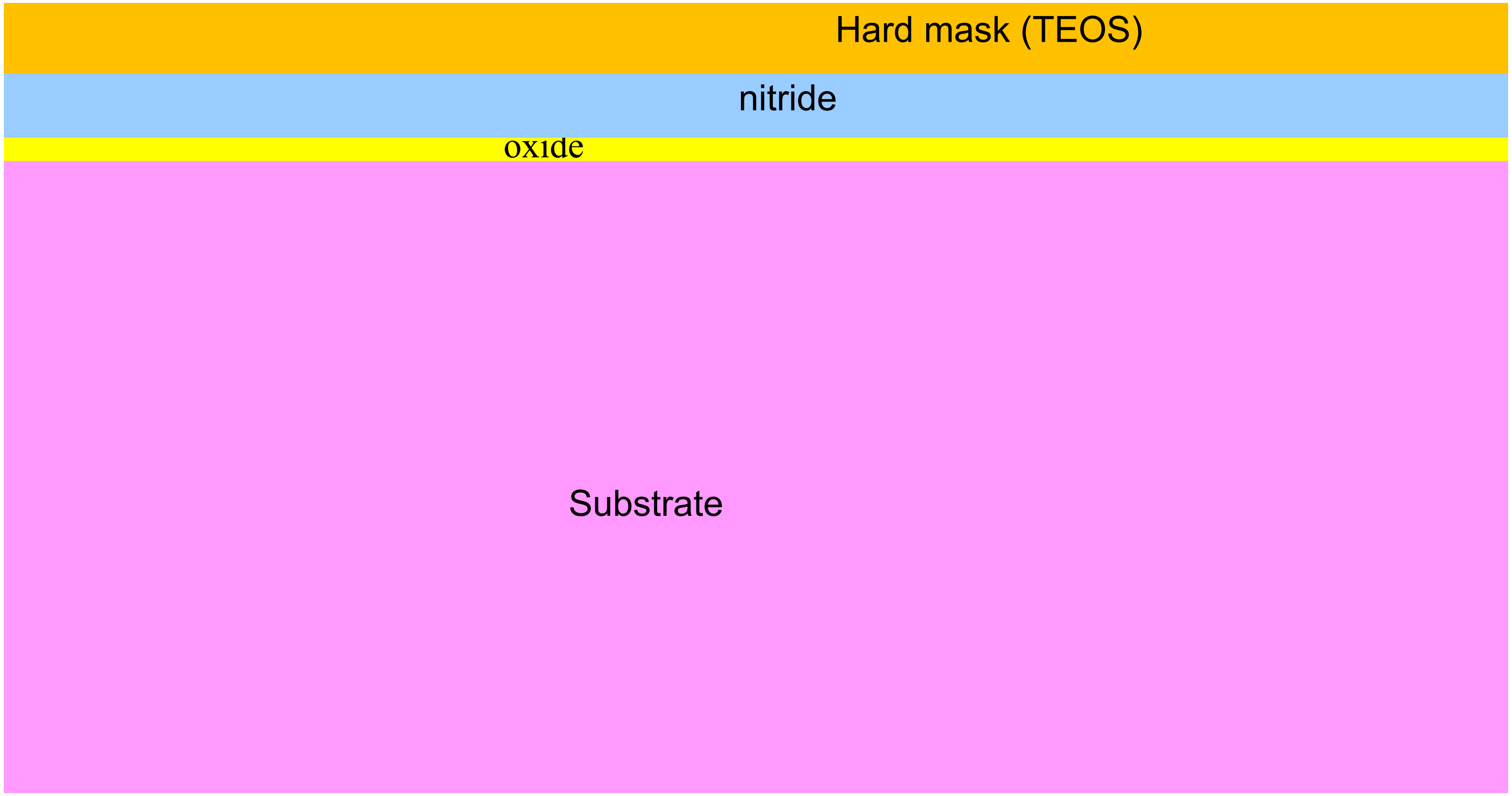


Deposition of pad nitride through LPCVD

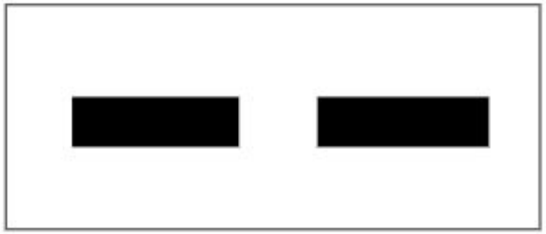




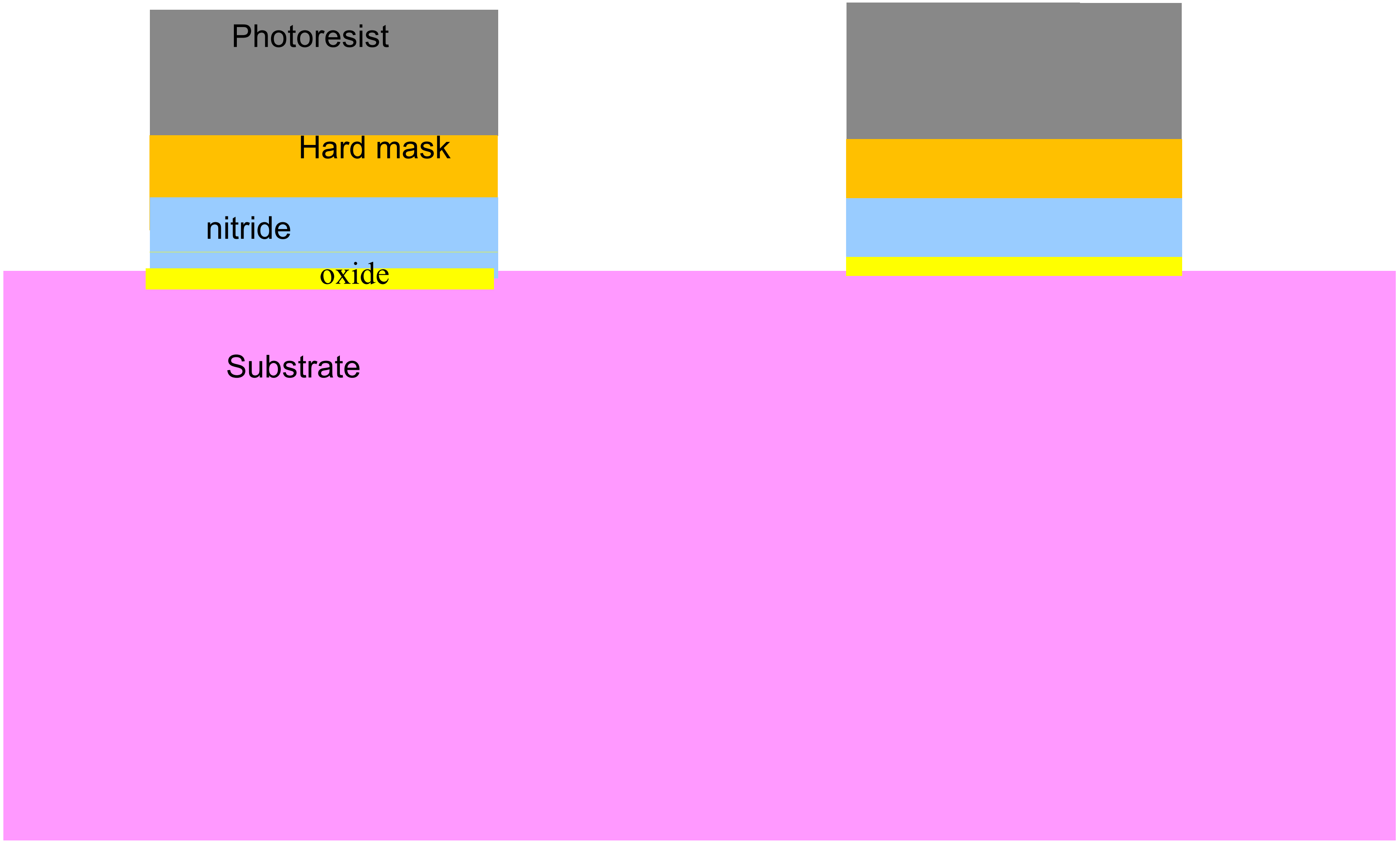
Deposition of hardmask (to pattern active area)



Pattern hardmask using litho and etch (Mask #1)

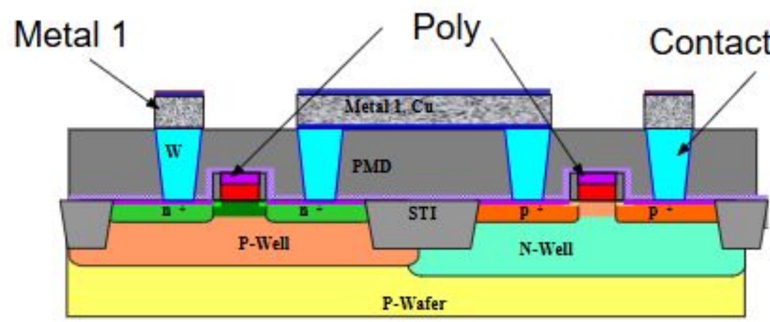


Mask 1, shallow trench isolation

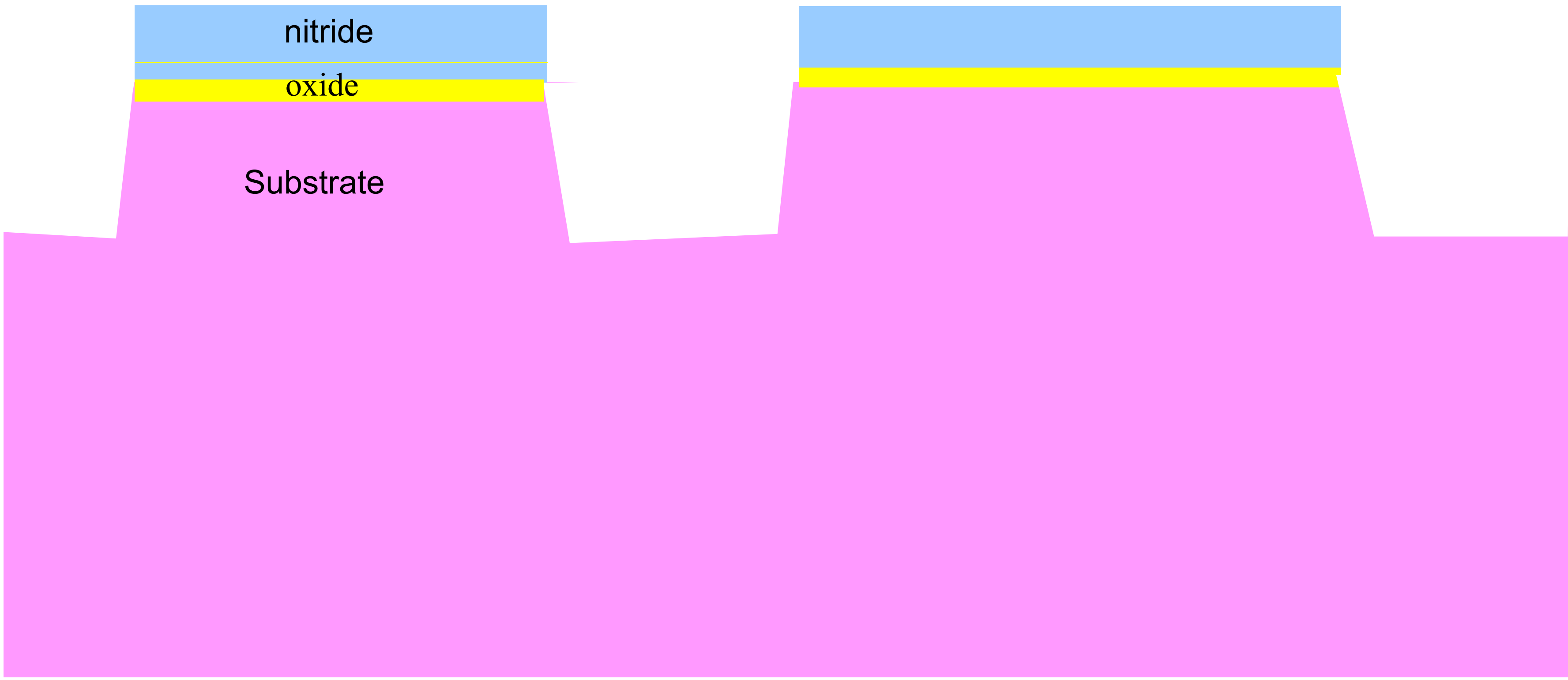


Si etch using patterned hardmask

Softmask using 1-2 um photoresist cannot withstand this aggressive etch

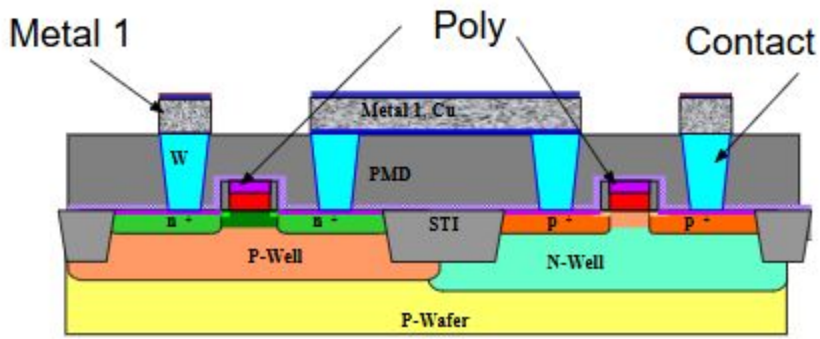


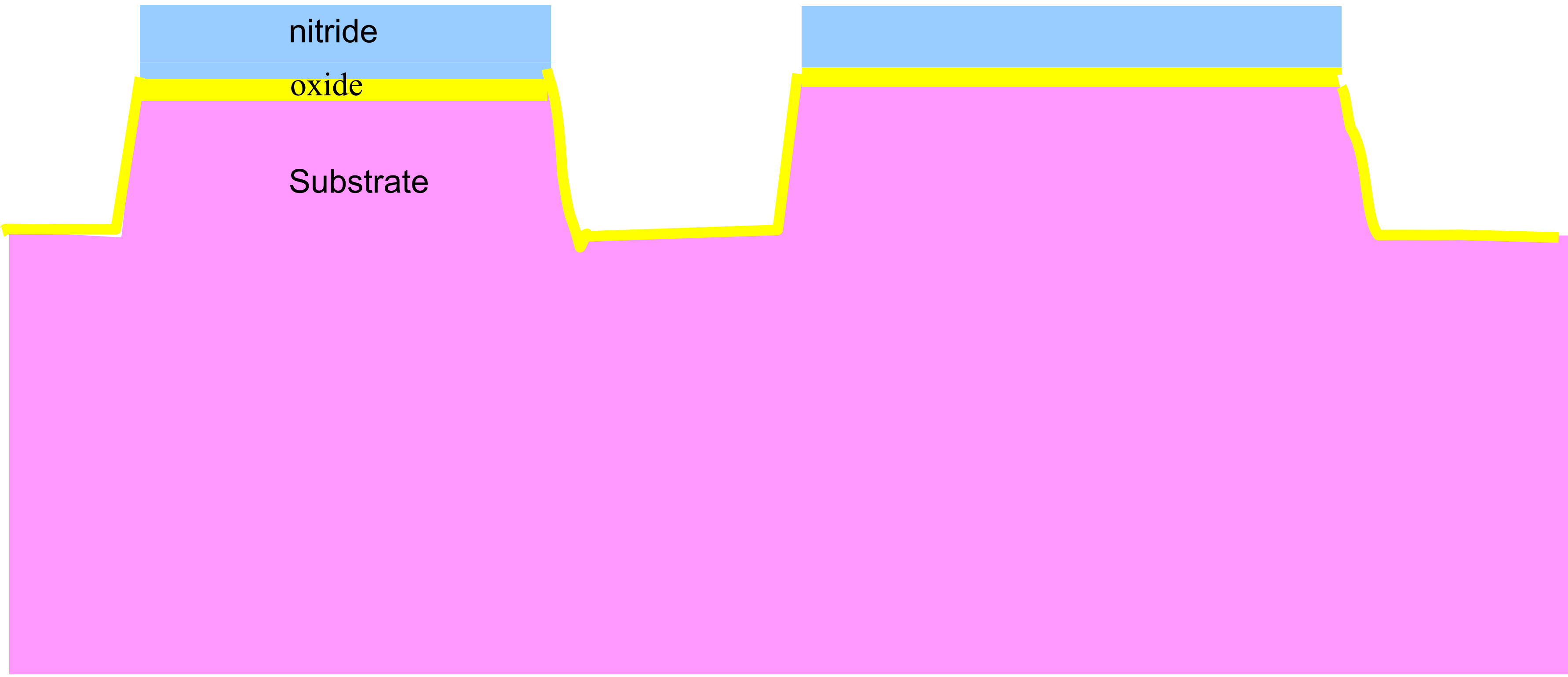
STI depth not to scale. They are < 0.5 µm deep



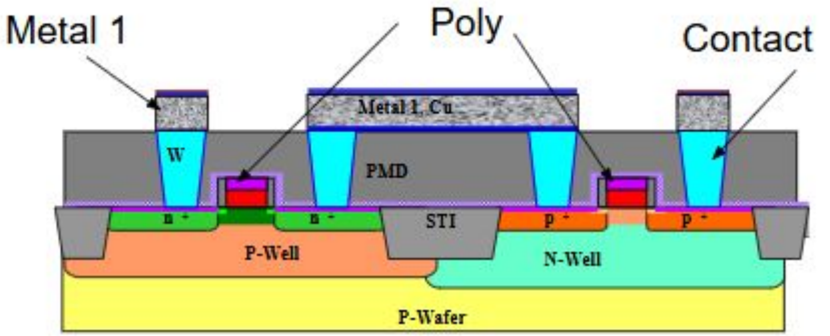
Growth of STI liner

Thermal oxidation is done to create STI liner

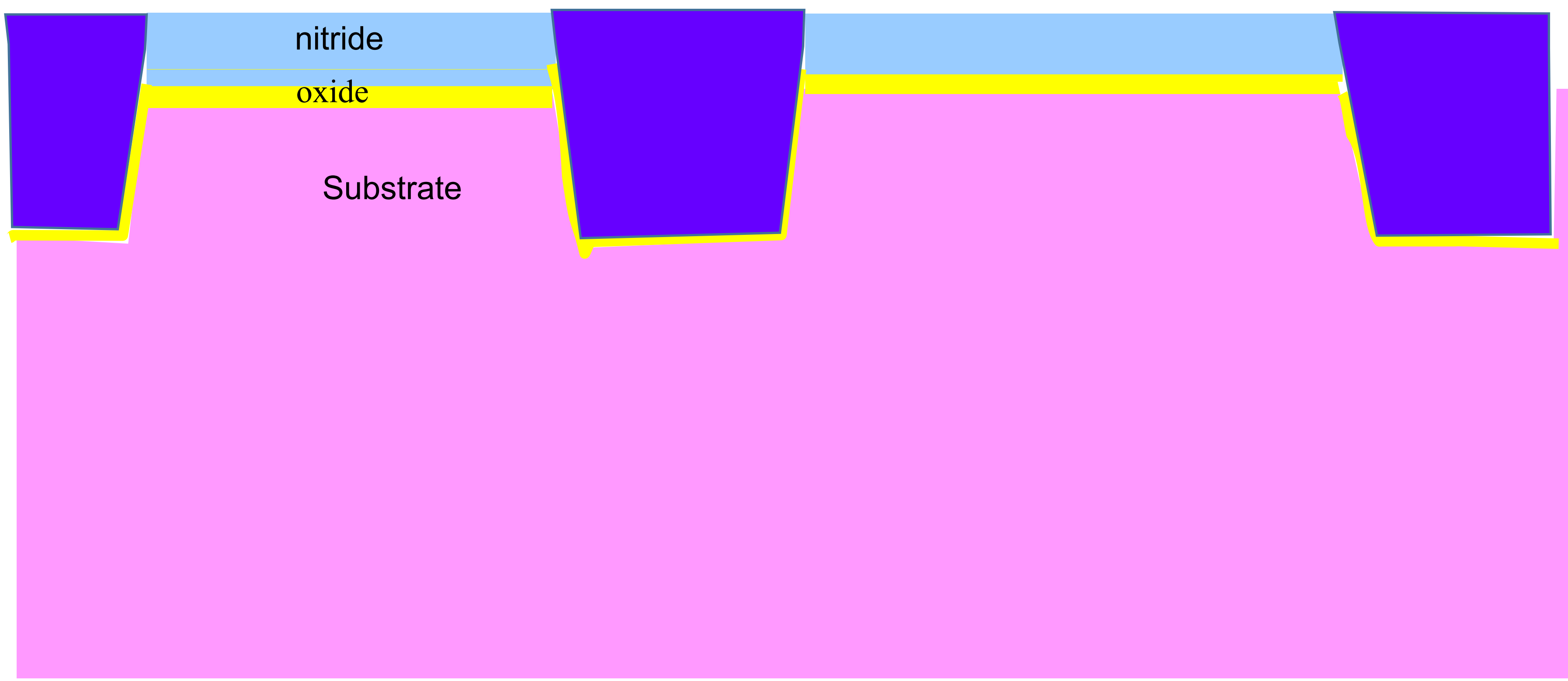




Fill the trench with deposited oxide and do CMP

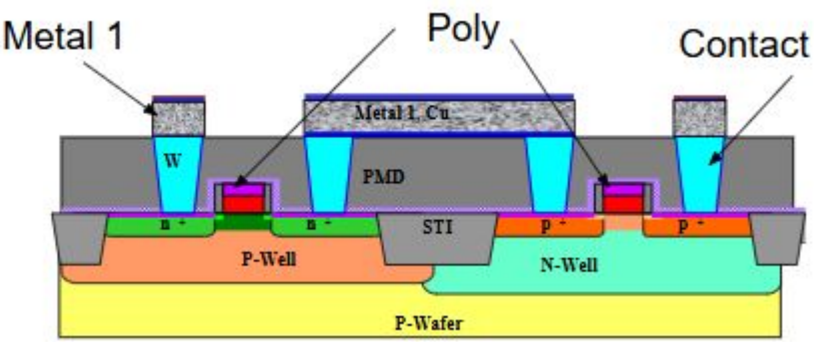


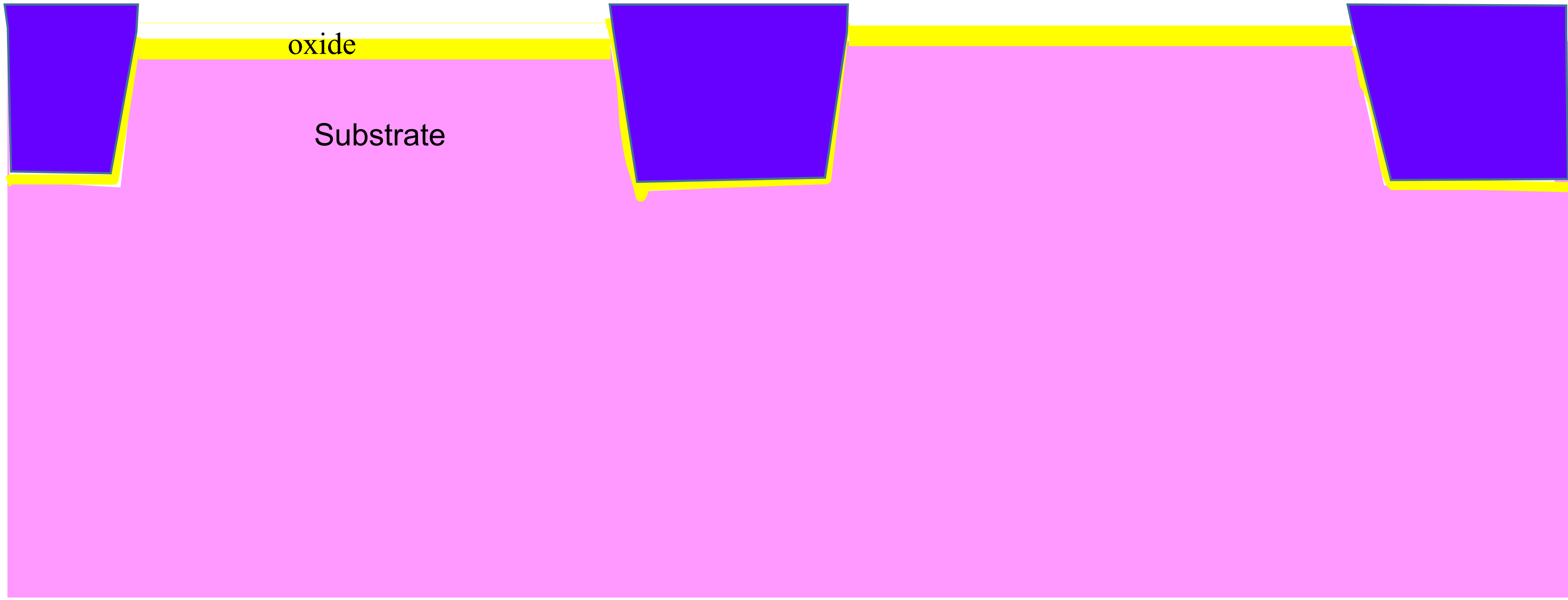
CVD is used to deposit oxide followed by Chemical Mechanical Polishing (CMP) which stops on pad nitride



Strip pad nitride

Using wet etching (Hot Phosphoric acid)



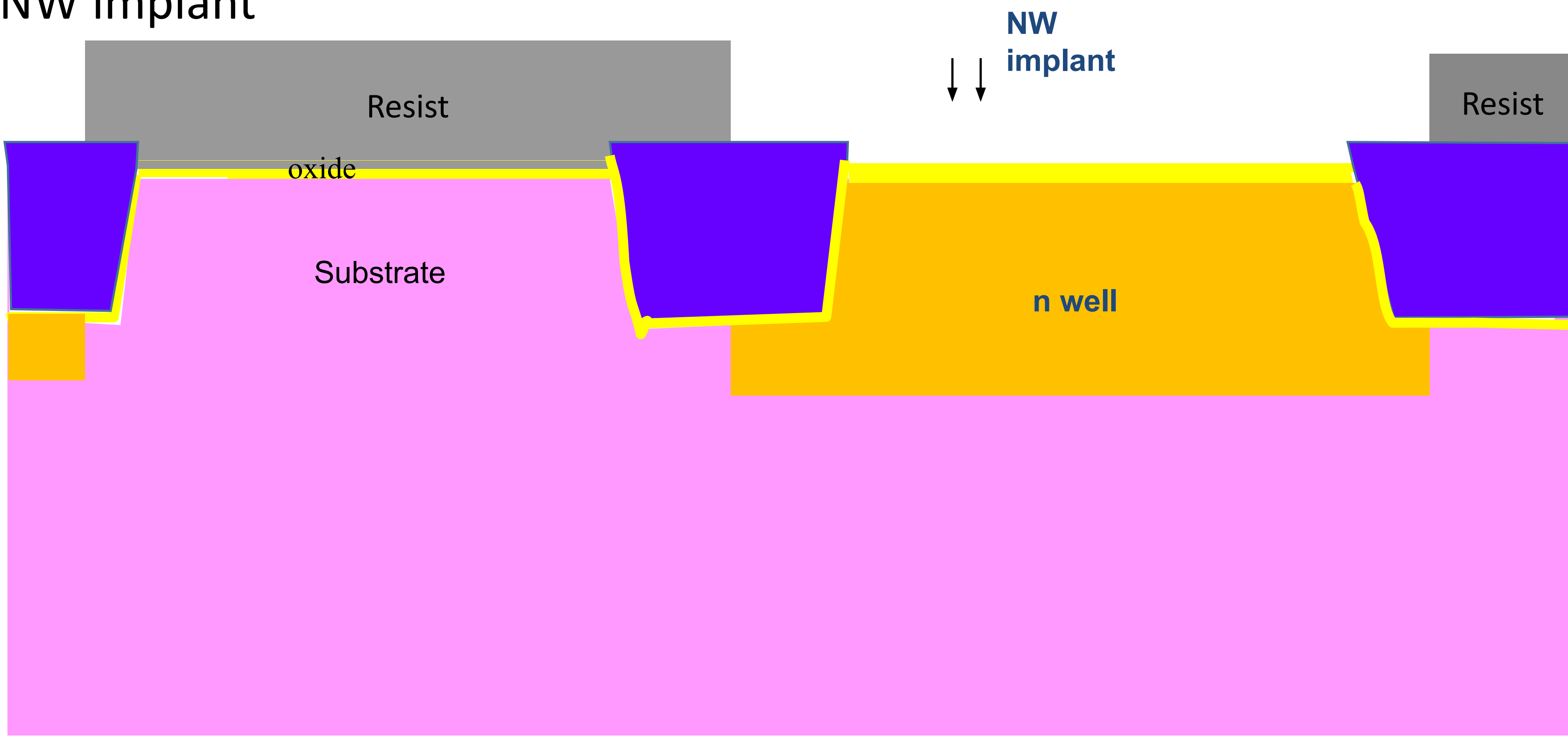


Define NWELL region (#2)

Mask 2, N-well

Using lithography and ion implantation (high energy). It may consist of multiple implants

Opens all PFETs and passive areas which require NW implant

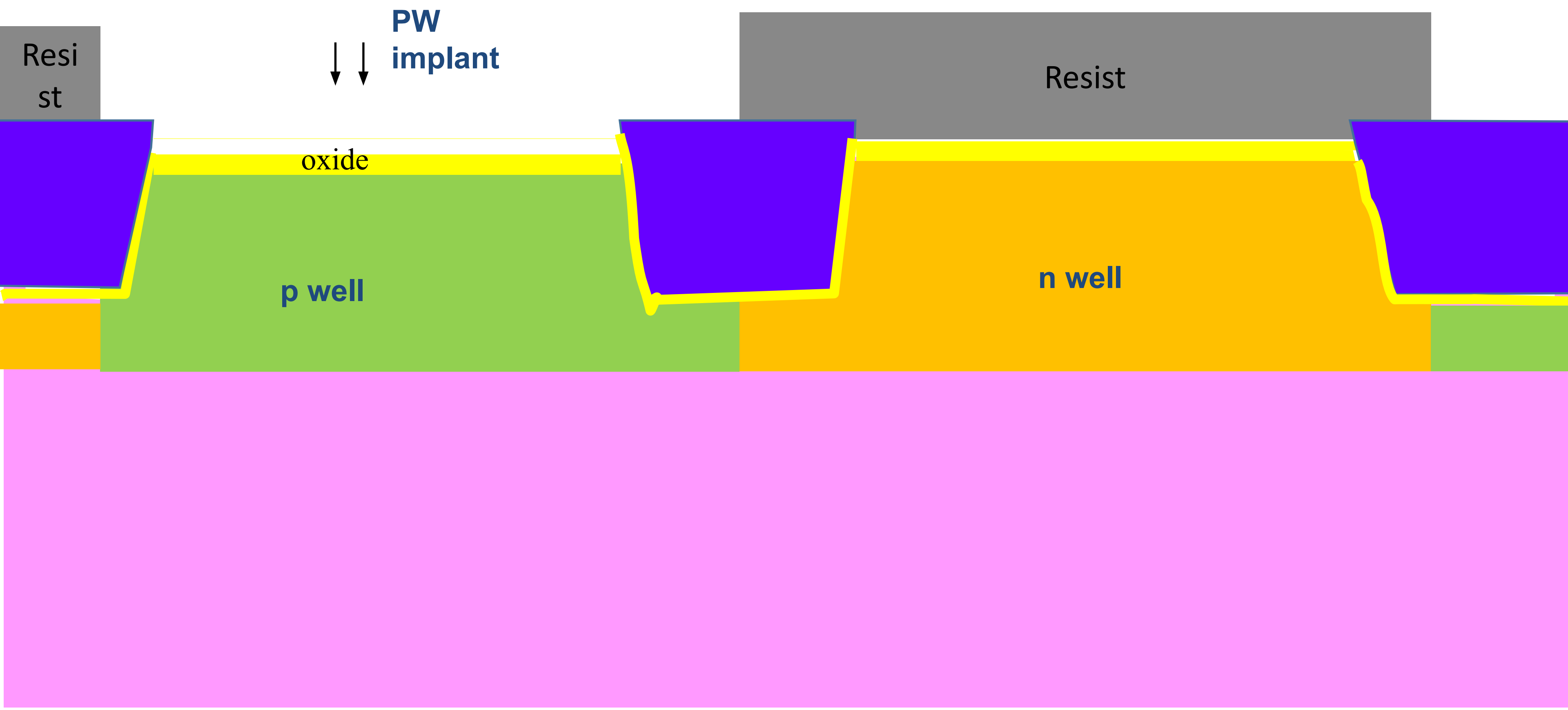


Define PWELL region (#3)

Using lithography and ion implantation (high energy). It may consist of multiple implants

Opens all NFETs and passive areas which require PW implant

Mask 3, P-well



Channel V_T adjust implants Mask #4 and #5

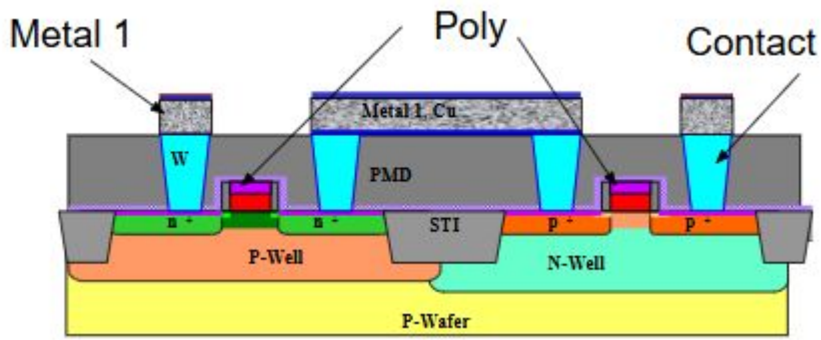
Using lithography and ion implantation (medium energy).
Opens only that NFET/PFET area (low/regular/high V_T)

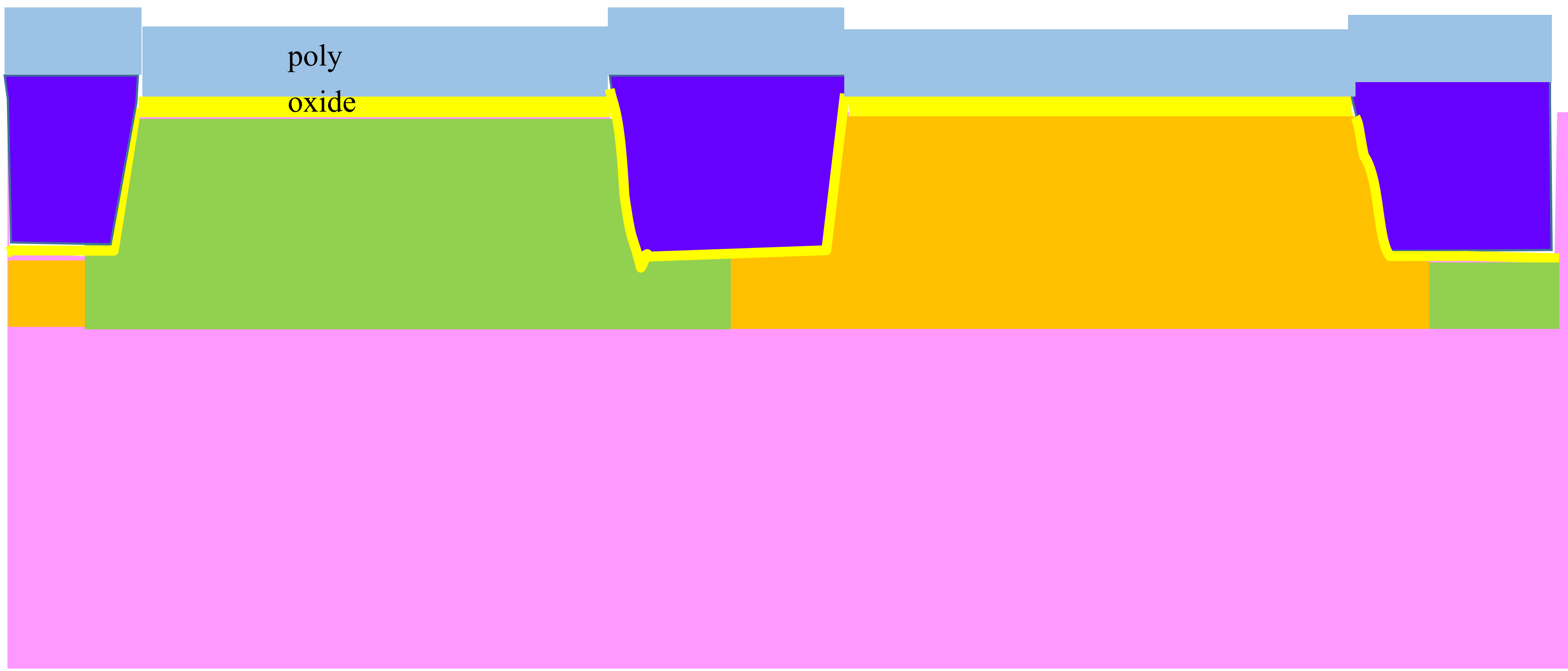
There could be a well anneal to activate the well/channel dopants

Strip pad oxide (as it has seen all implants) and grow fresh gate oxide (dry oxidation)

Polysilicon deposition

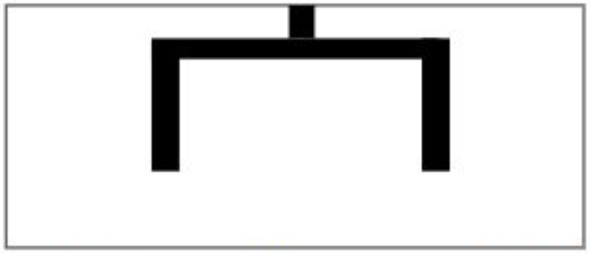
Using LPCVD



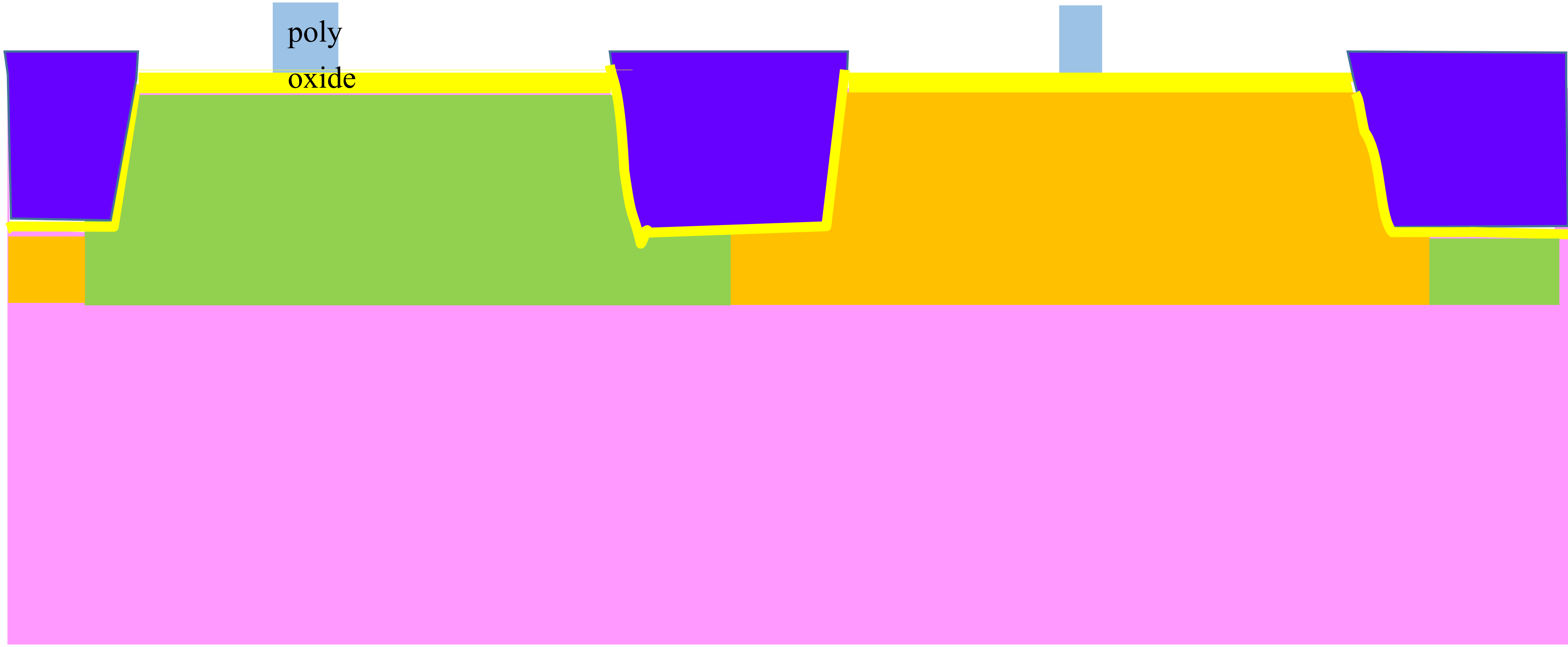


Gate patterning (#6)

Dry etching using hardmask. There could be FETs with different L_G



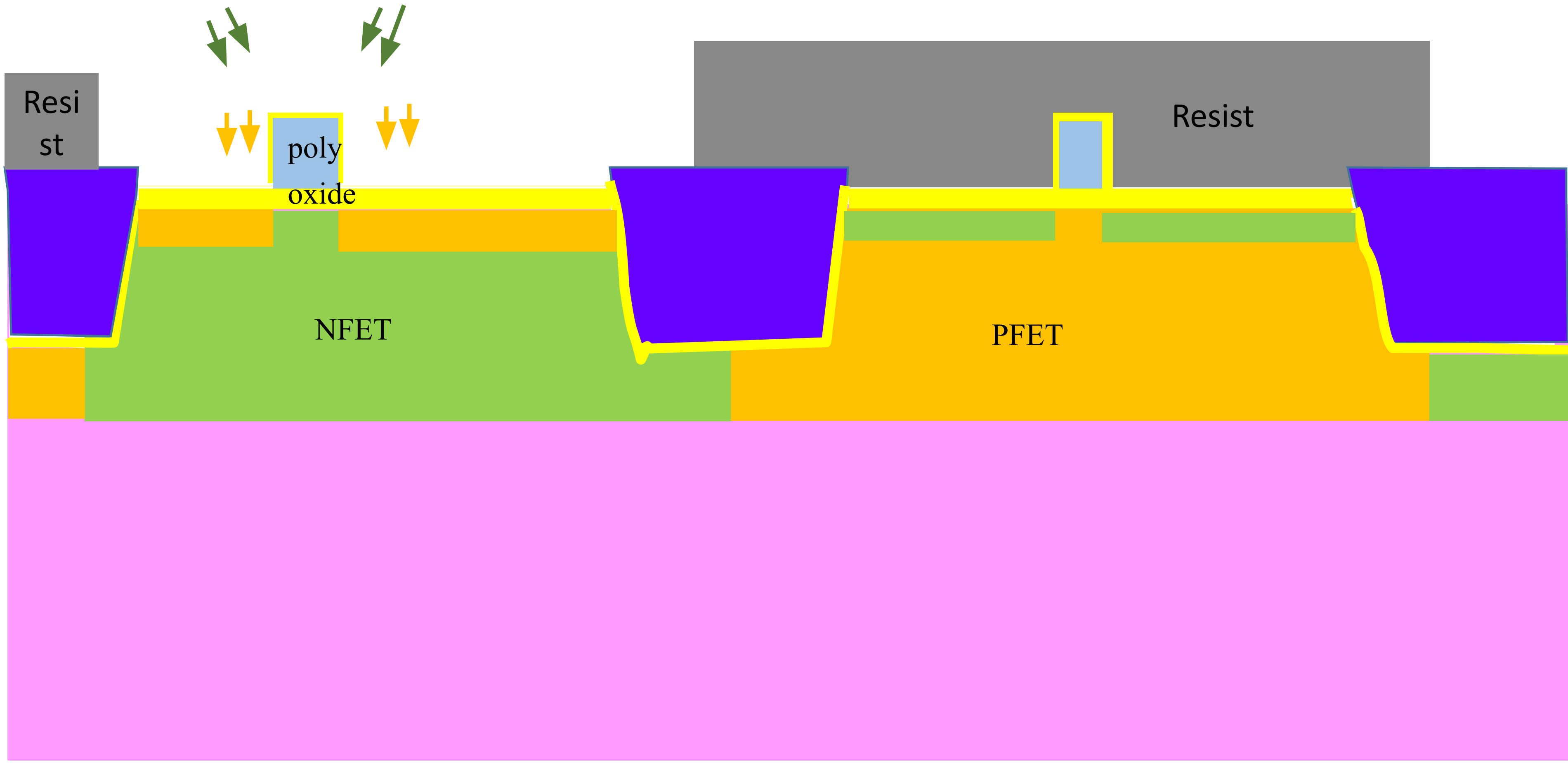
Mask 6, gate/local interconnection



Next step is poly reoxidation. It oxidizes the exposed surface of poly and also increases gate oxide thickness at the source/drain sides

Source/Drain extension (LDD) and Halo (#7 & #8)

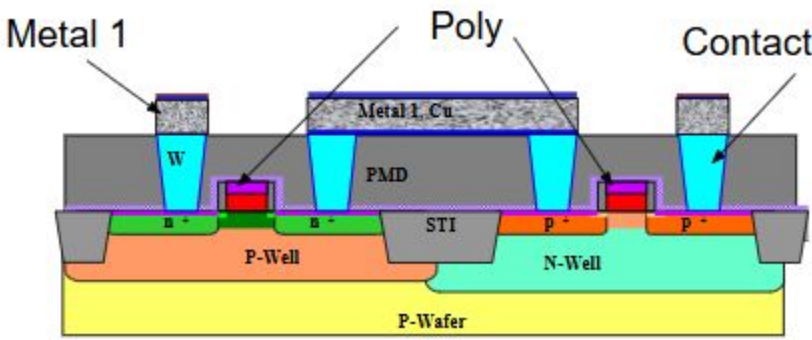
Using lithography and ion implantation (medium energy). Halo is a tilted implant to create a pocket implant under gate. LDD is for same polarity as source/drain. Opens only that NFET/PFET area (low/regular/high V_T)

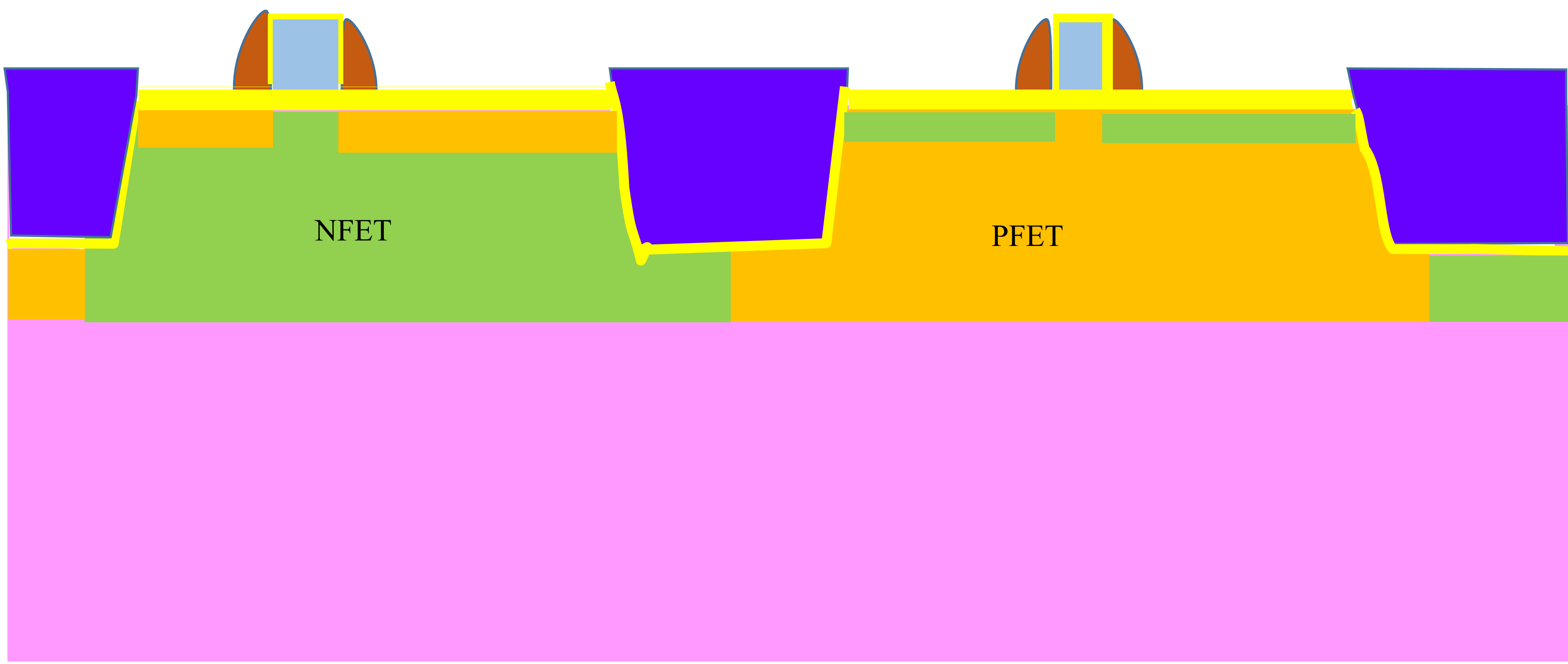


Nitride Spacer

Spacer is used to separate shallow & deep source drain regions. It also limits the shorting of gate-source/drain through silicide pipes

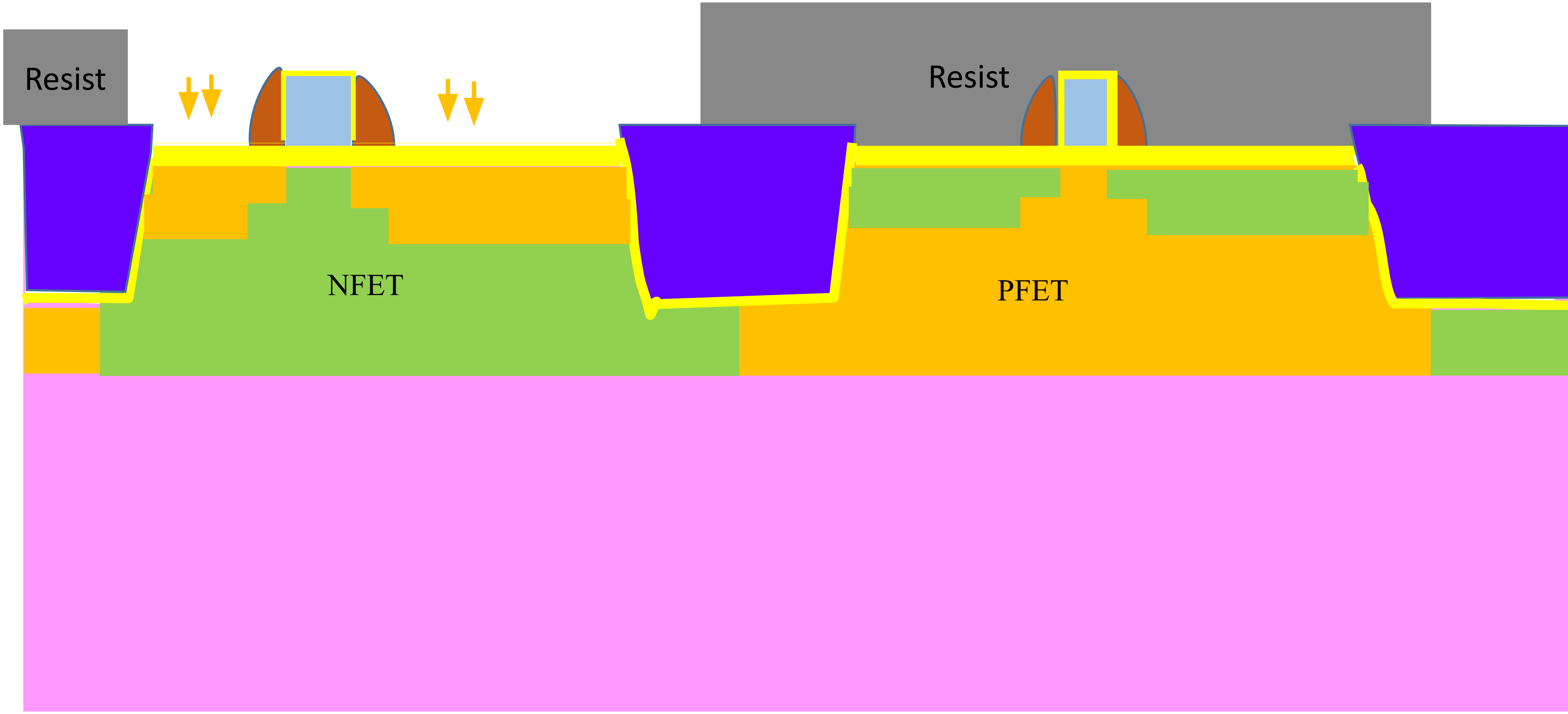
Deposition followed by anisotropic etching





Deep source/drain implant (#9 and #10)

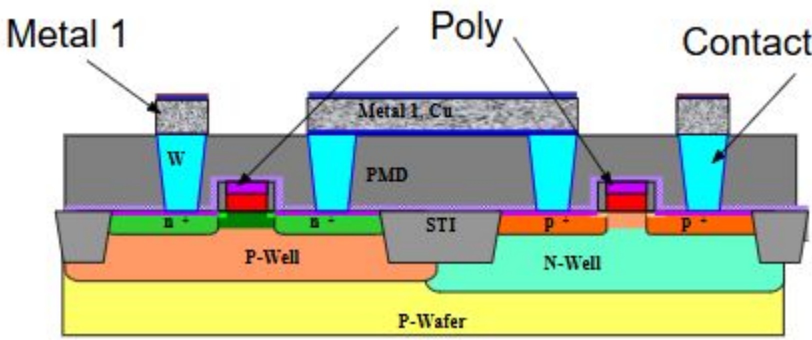
Using lithography and ion implantation (heavy dose). It may consist of multiple implants. Opens all NFETs/NW contact or PFET/PW contact. ESD areas are blocked



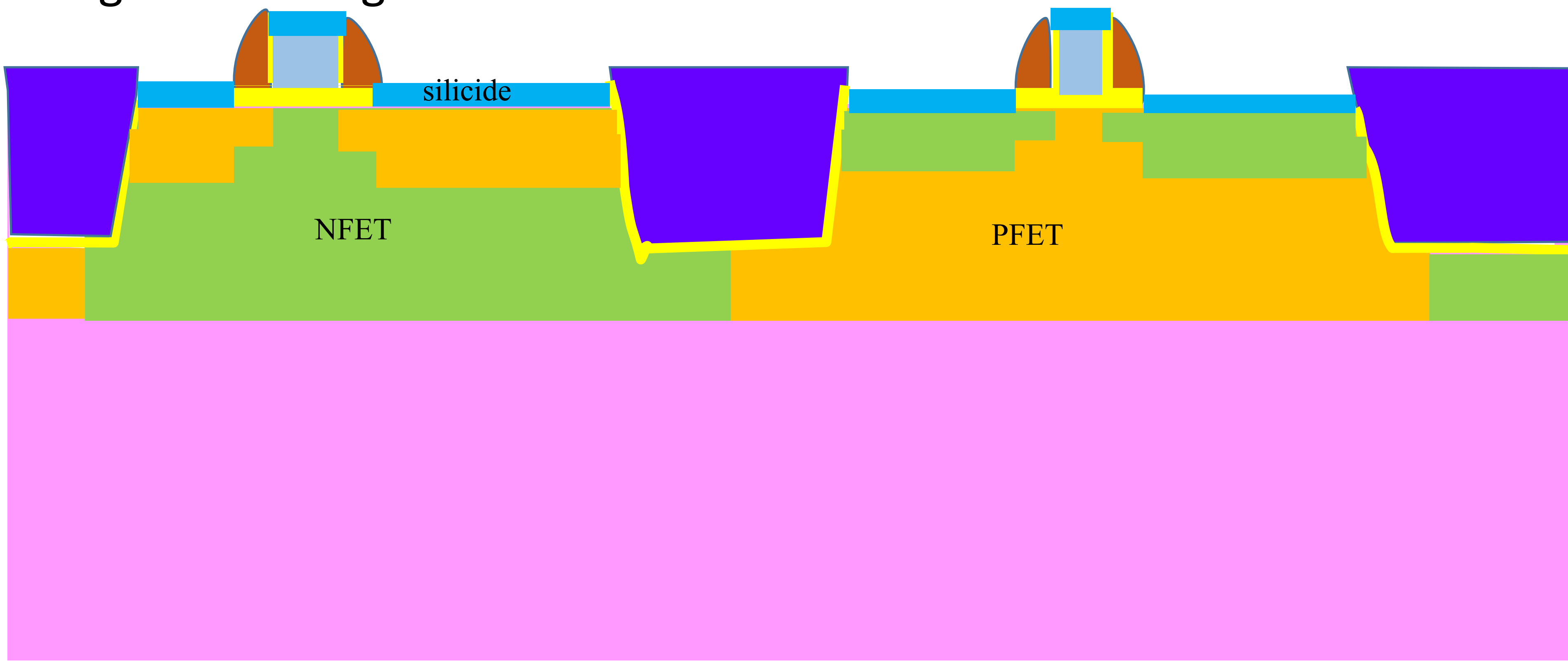
Rapid Thermal Anneal to activate dopants

Strip oxide over source/drain and top of poly using wet etch

Deposit Co or NiPt and anneal at around 500^{0}C



Co or NiPt reacts with exposed Si and form Cobalt/Nickel silicide. The unreacted Nickel over Spacer and STI are removed using wet etching



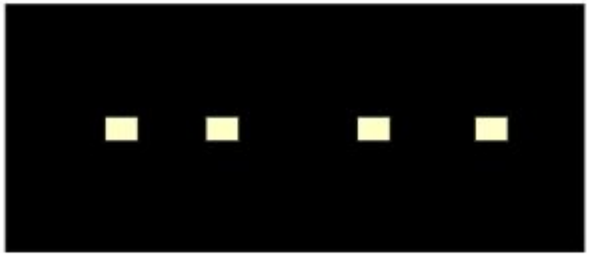
Contact etch stop liner (PECVD nitride) deposition

Inter-level dielectric deposition (CVD oxide)

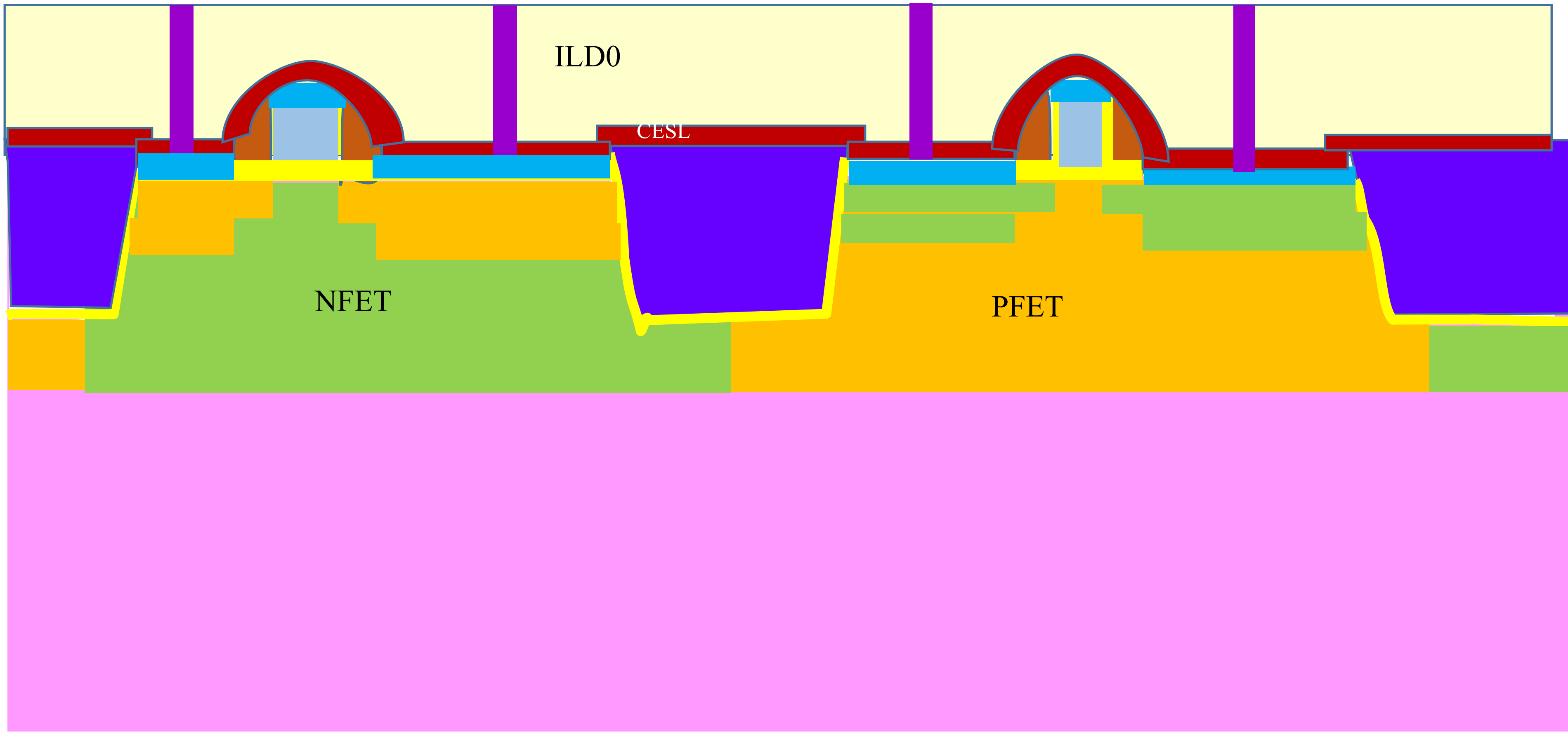
CMP

Contact patterning (#11)

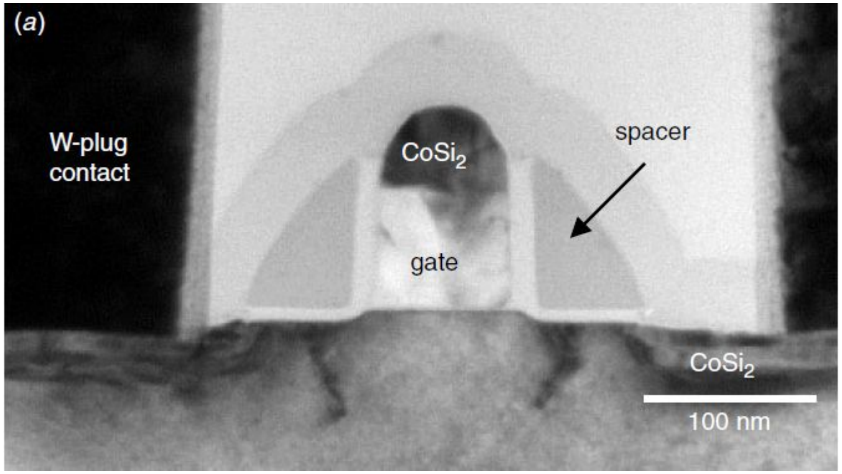
Using dry etch to open contact holes
Fill it with Tungsten (W) using CVD
Followed by W CMP

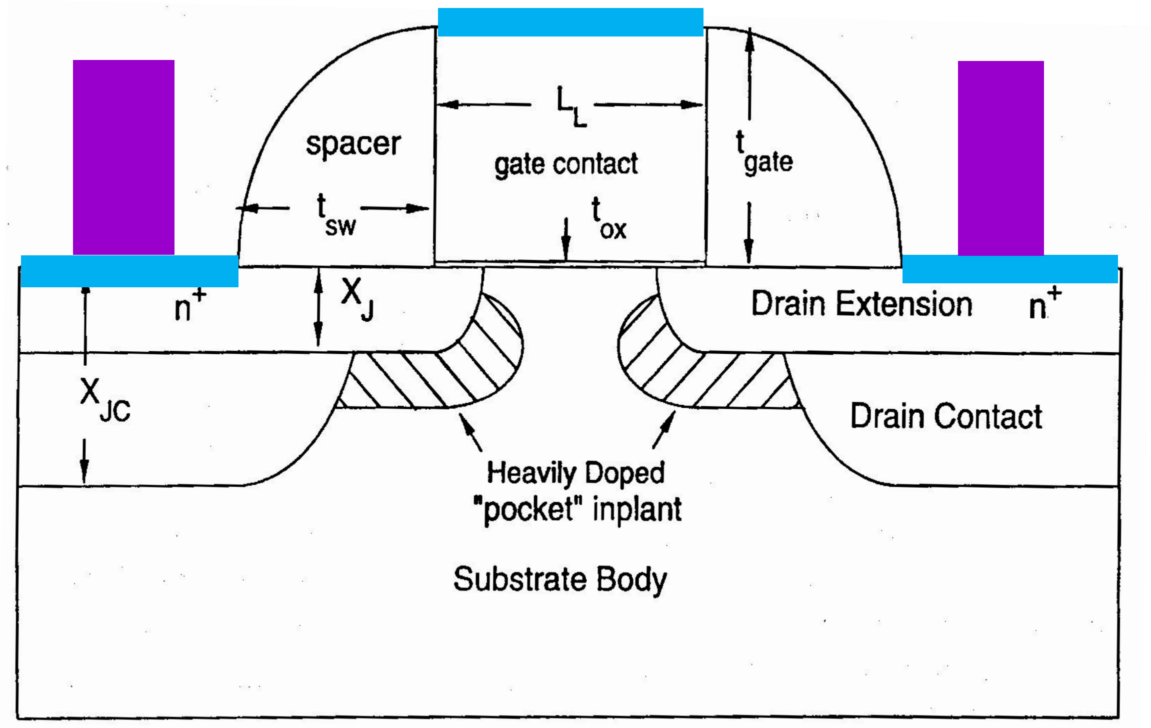


Mask 11, contact



FET cross section

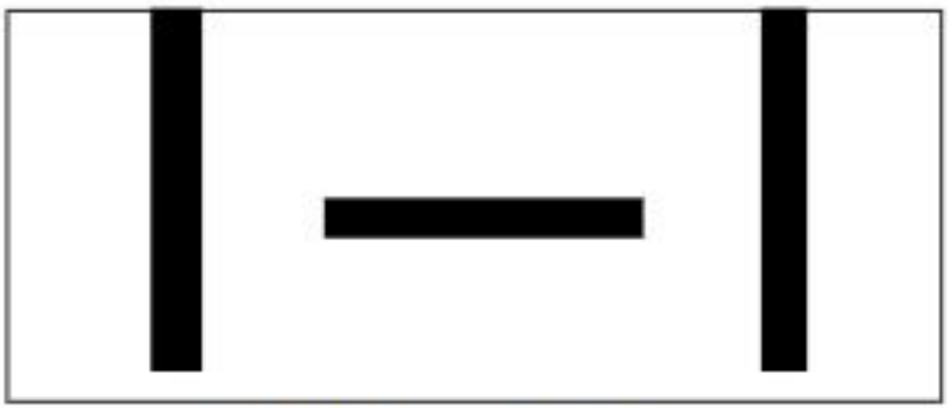




Chih-Hang Tung, George T. T. Sheng, and Chih-Yuan Lu , *ULSI Semiconductor Technology Atlas*, 2003

Inter-level dielectric (ILD1) deposition (CVD oxide)

M1 using damascene process (Cu electroplating) (#12)



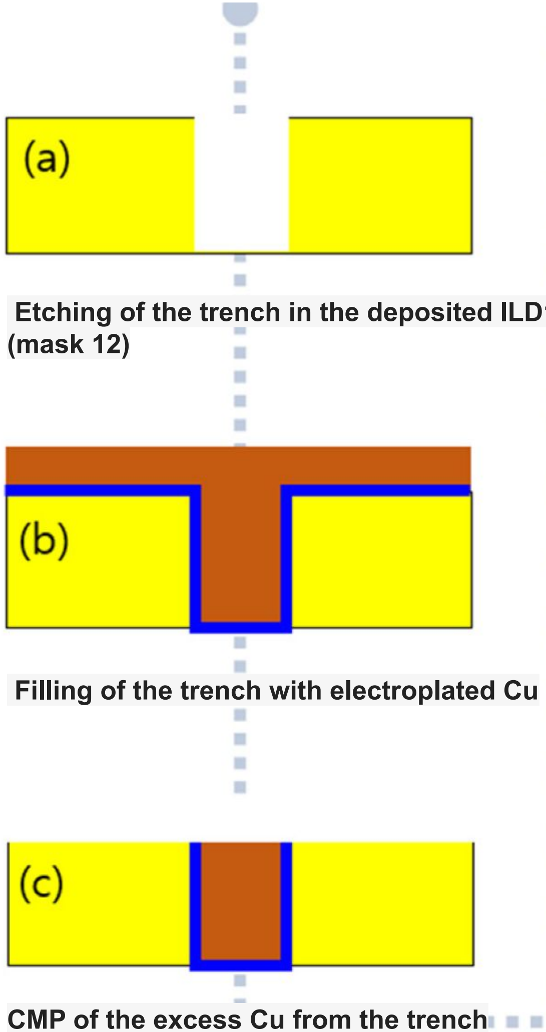
Mask 12, metal 1

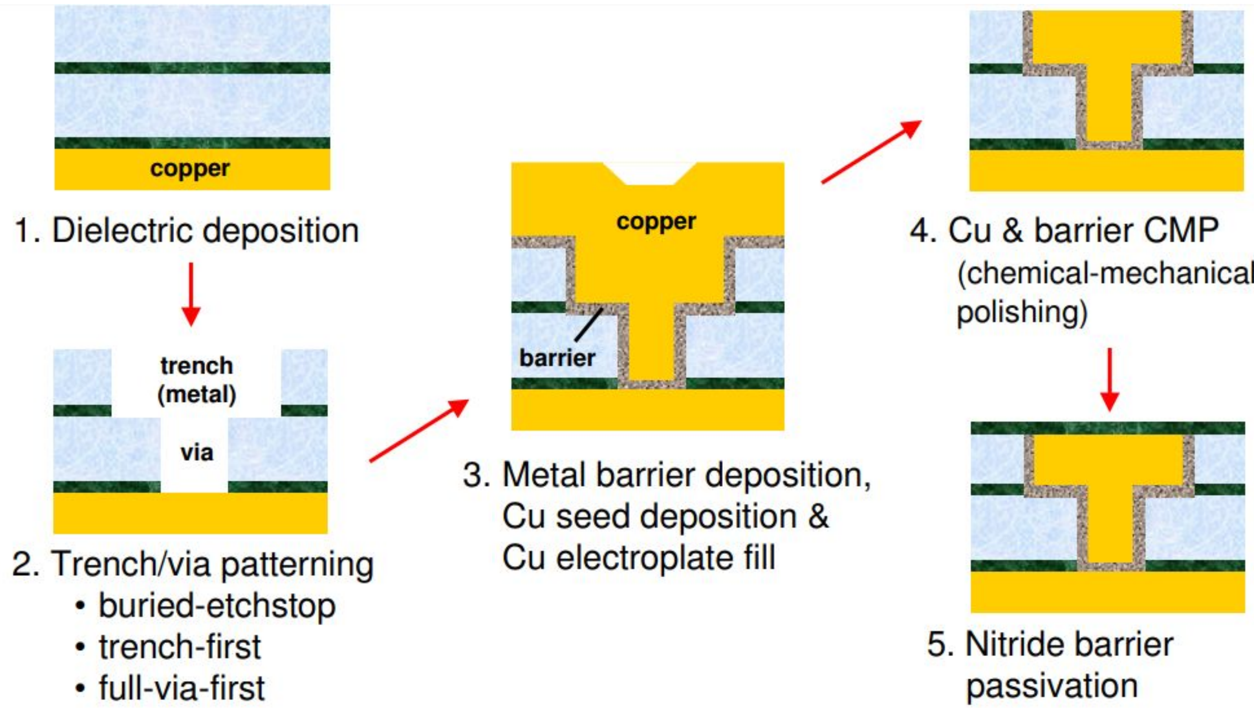
Damascene process

Blanket deposit material on a surface which contains trenches. Polish away the material from flat areas, but allow to remain in the trenches

Word Damascene is derived from the decorative practice of creating metal inlays as developed by artisans of Damascus in middle ages.

Single and Dual Damascene process

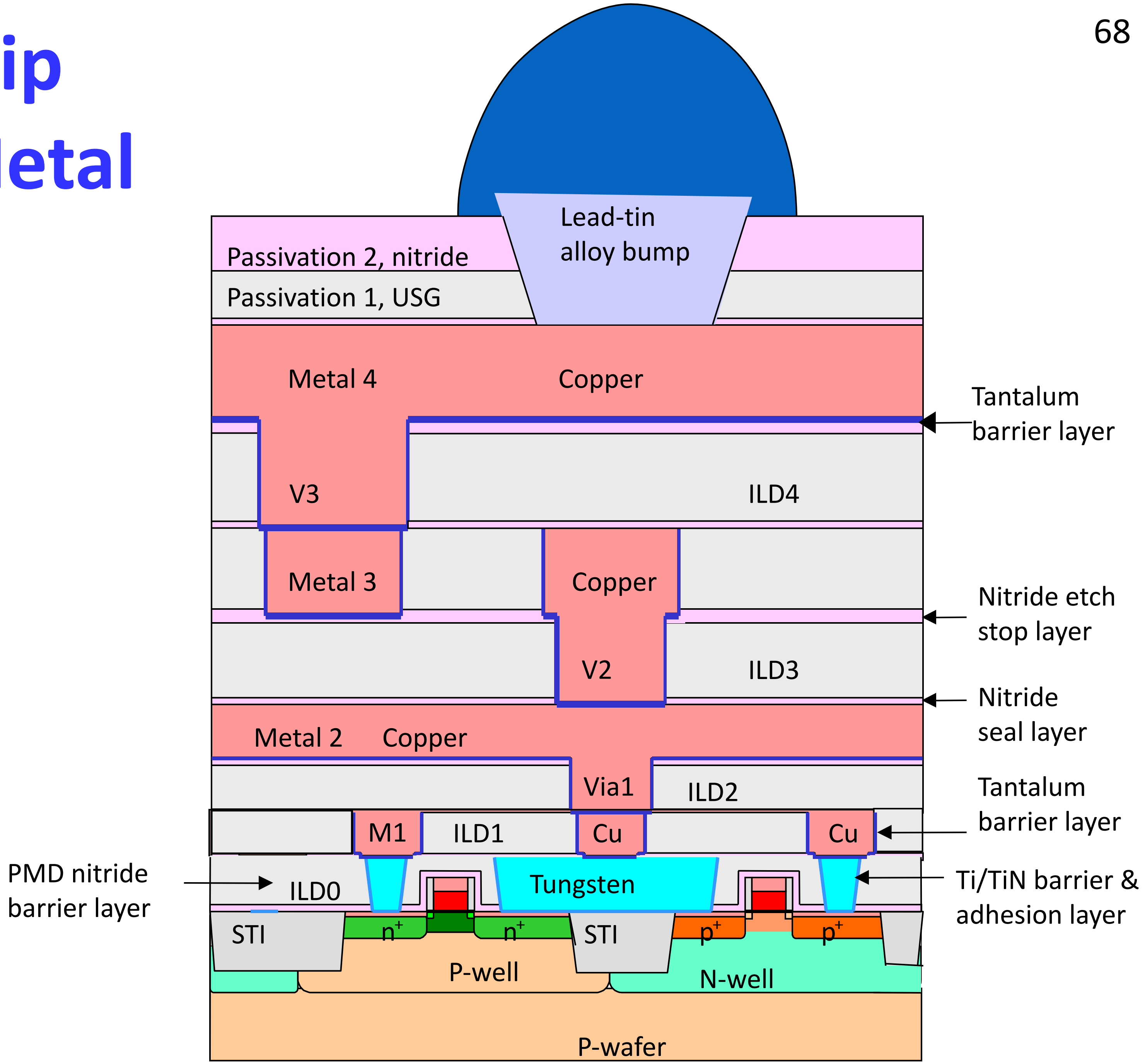


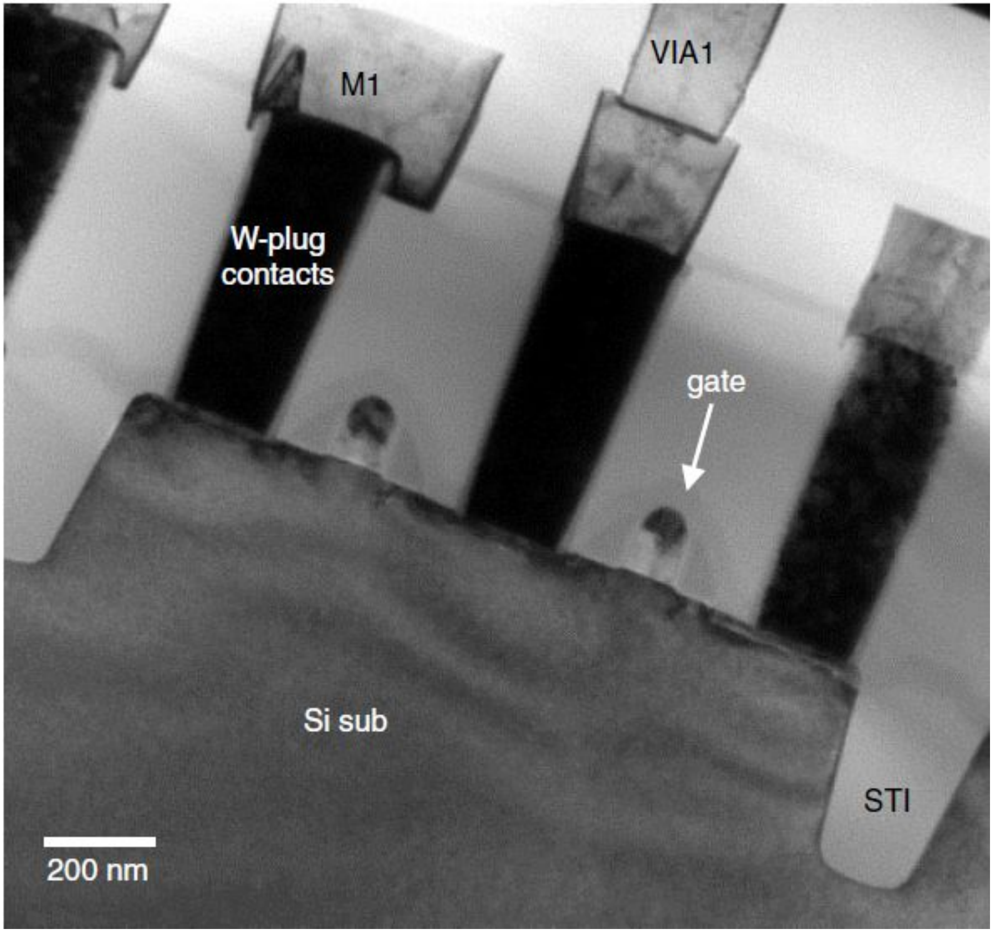


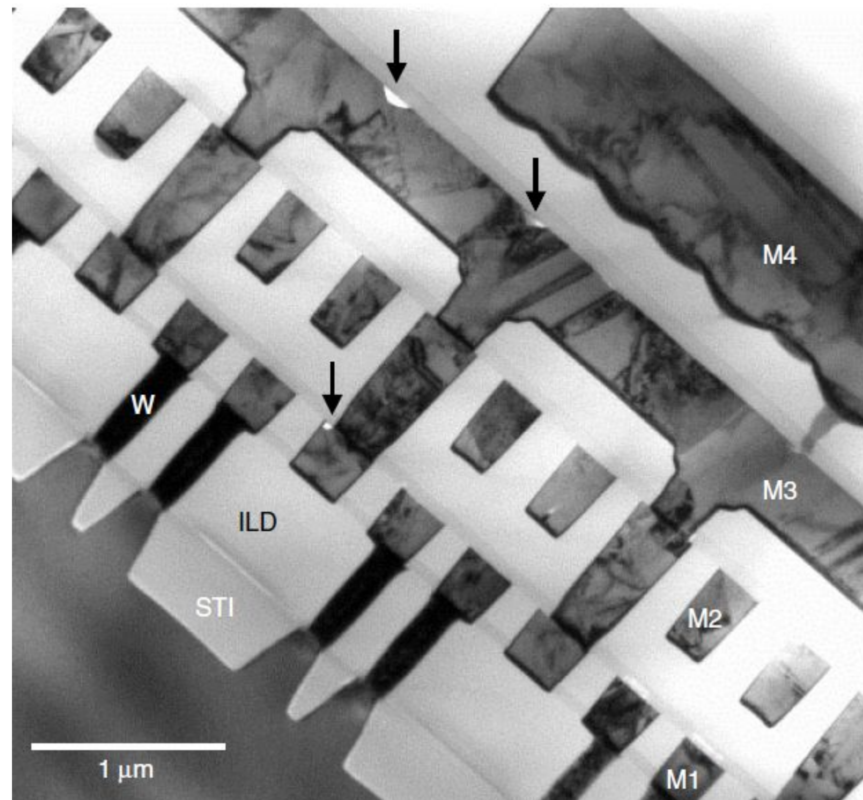
Single - Create M1 layer

Dual - It creates both vias and lines (V1 and M2) of each metal layer by etching holes in the ILD, depositing Cu in the etched features and using CMP to remove the excess copper.

CMOS Chip with 4 Metal Layers




VIA1
M1
W-plug contacts
gate
Si sub
STI
200 nm

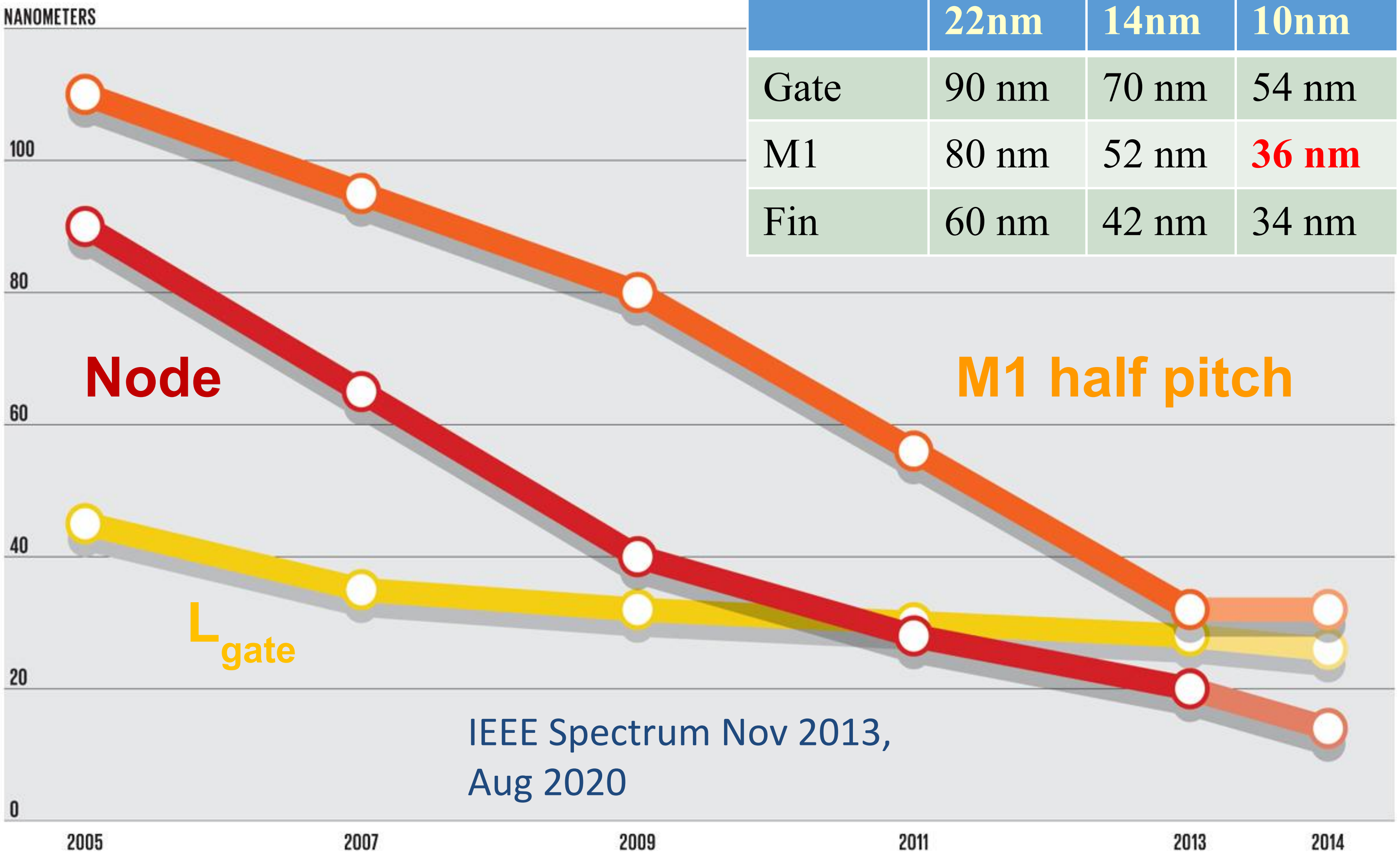

M4
W
M3
ILD
STI
M2
1 μm
M1

Points to note

- Always assume the incoming wafer as a patterned one with some pre-existing structures
- Don't put wafer with PR on hot furnace above 200 ^{0}C. Same with wafers with low MPT metals
- One cannot do selective deposition
- It is always blanket deposition followed selective etching
- Oxidation and silicidation consumes silicon, while deposition doesn't
- All BEOL processing is done below 400 ^{0}C as silicided wafer cannot handle it.
- Ion implantation cannot reach beyond 1-2 µm. Thicker layers may need epitaxy
- Dopant deactivation may happen at 500-800 ^{0}C. Higher temp leads to dopant profile changes
- Wrong process may contaminate tool, scrapping of wafers and may result in tool down time
- Process assumption – At every step specify the structural expectation with specs
- **If things go wrong, do you have electrical test structures or process monitors to debug?**

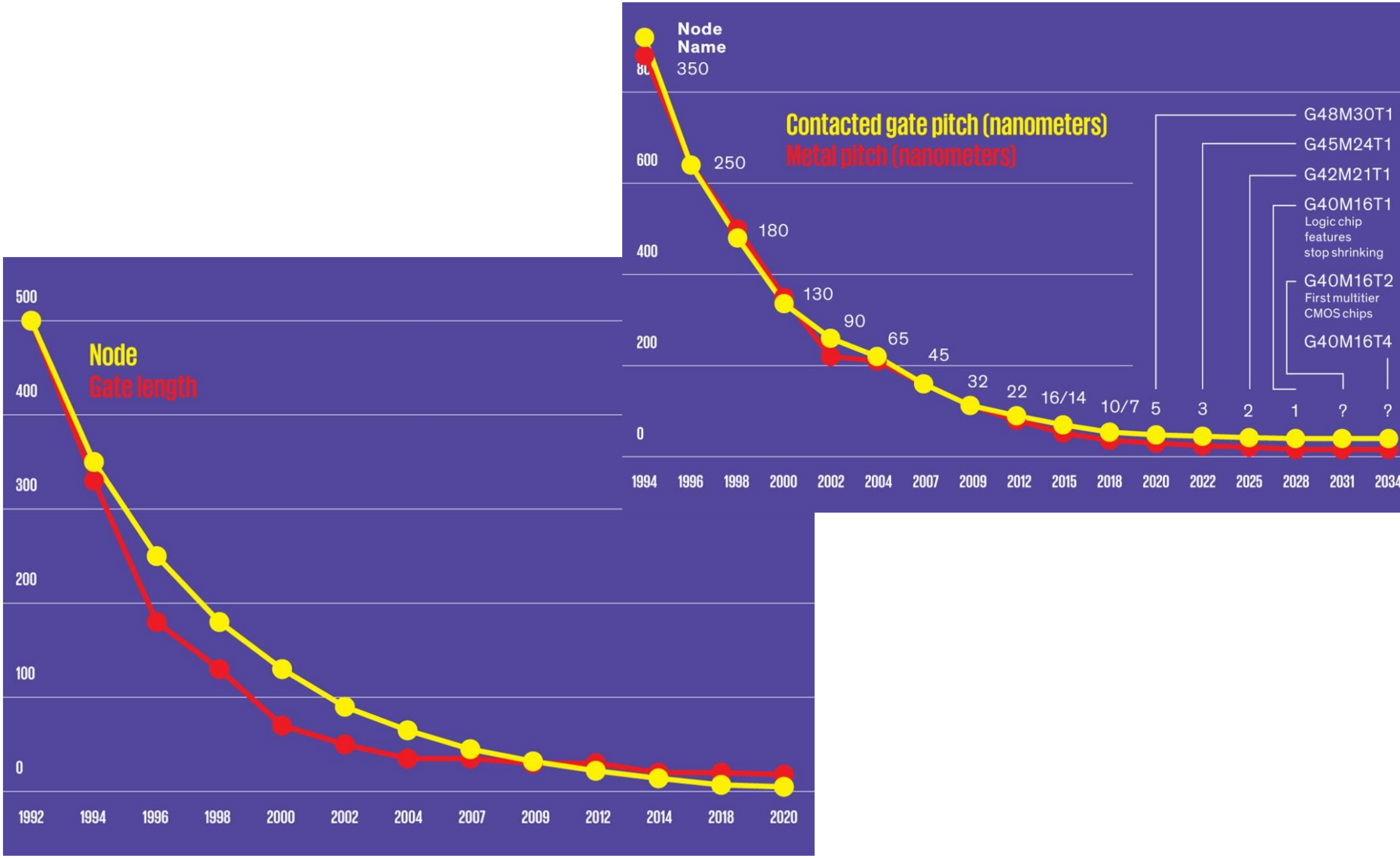
What does node name means ? (Recap)

Pitch	Intel 22nm	Intel 14nm	Intel 10nm
Gate	90 nm	70 nm	54 nm
M1	80 nm	52 nm	**36 nm**
Fin	60 nm	42 nm	34 nm



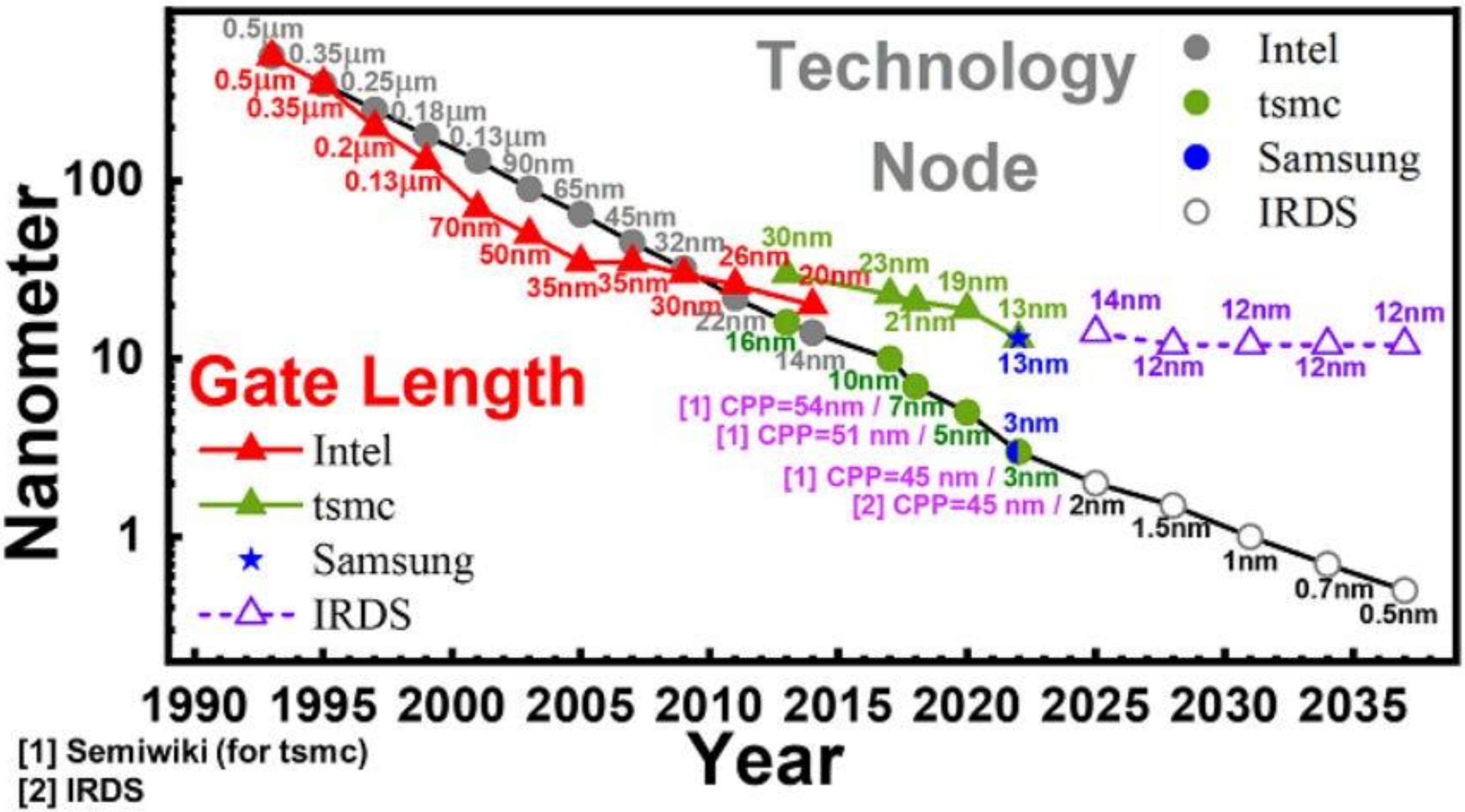
Node name doesn’t represent any feature size on the wafer or in the design rules. There are also nodelets or half nodes !!

The Node is Nonsense (Recap)

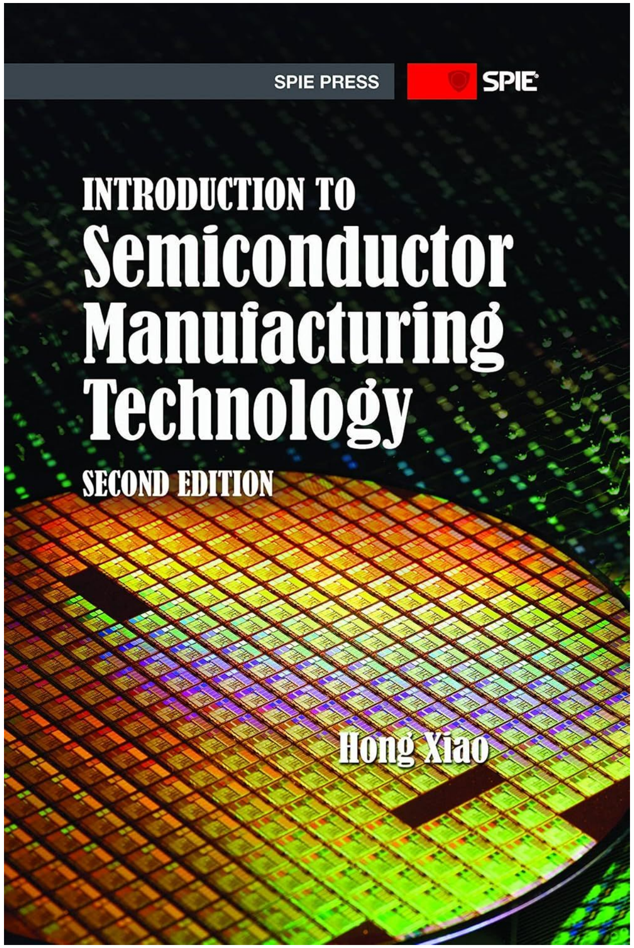


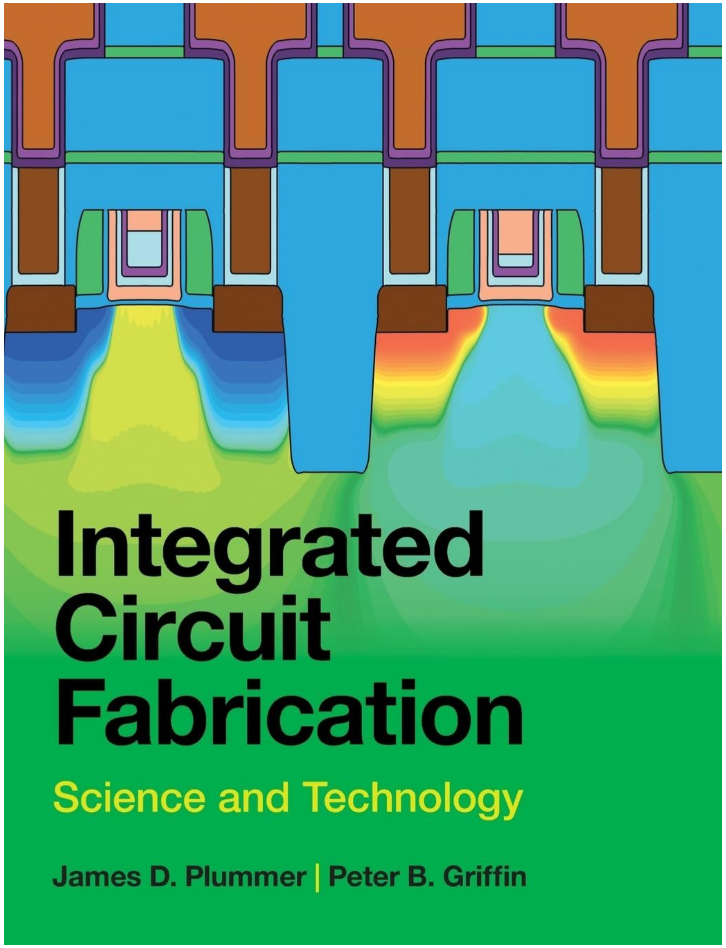
The Node is Nonsense – IEEE Spectrum August 2020

The Node is Nonsense (Recap)



For additional reading





Youtube Videos

[The Amazing, Humble Silicon Wafer by Asianometry]

[How are Microchips Made? CPU Manufacturing Process Steps by Branch Education]

[The World's Most Important Machine by Veritasium]